

Piranha HS

Camera User's Manual

HS-40-04k40

HS-80-04k40-00-R

HS-80-08k40-xx-R

HS-80-08k80-00-R

HS-82-04k80-00-R



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TELEDYNE DALSA

A Teledyne Technologies Company

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1

Introduction to the Piranha HS-xx Camera

1.1 Camera Highlights

Features

- 4096 or 8192 pixels, 7 μm x 7 μm or 14 μm x 14 μm (HS-82) pixel pitch, 90% fill factor
- 4 or 8 taps, bidirectional TDI
- Up to 160, 320, or 640 megapixels/ second throughput
- Greater than 100 kHz line rate (HS-82 model)
- 100x antiblooming
- Broadband responsivity of 1170 DN (nJ/cm^2)
- 6 independently stage-selectable Time Delay and Integration (TDI) imaging regions for remarkable user-controlled sensitivity
- CE compliant
- The following models are RoHS compliant: HS-80-04k40, HS-80-08k40, HS-80-08k80, and HS-82-04k80

Programmability

- Serial interface (ASCII, 9600 baud, adjustable to 19200, 57600, 115200), through Camera Link™
- Programmable gain, offset, frame rates, trigger mode, test pattern output, and camera diagnostics
- Mirroring and direction control
- Selectable Area or TDI Mode of operation. Area Mode facilitates camera alignment and focusing. Area mode can also be used for regular operation

- Selectable pixel size (binning)
- Flat-field correction—minimizes lens vignetting, non-uniform lighting, and sensor FPN and PRNU
- Selectable Base, Medium, or Full Camera Link configuration, depending on camera model

Description

The Piranha HS camera family represent Teledyne DALSA's latest generation of high sensitivity, TDI based cameras. The Piranha HS family maximizes system throughput and provides the largest number of pixels available in a TDI camera. All cameras are capable of bi-directionality with up to 96 stages of integration.

Applications

The Piranha HS family is ideal for applications requiring high speed, superior image quality, and high responsivity. These applications include:

- Postal sorting (flats)
- Flat panel display inspection
- Printed circuit board inspection
- High performance document scanning
- Large web inspection
- Low-light applications

Models

The Piranha HS-xx cameras are available in the following model configurations:

Table 1: Piranha HS-xx Camera Models Overview

Model Number	Description
HS-40-04k40	4k resolution, programmable 2 or 4 taps, up to 160 megapixels/ second throughput, 40/ 80MHz strobe rate, Base or Medium Camera Link configuration
HS-80-04k40	4k resolution, programmable 4 or 8 taps, up to 320 megapixels/ second throughput, 40/ 80MHz strobe rate, Medium or Full Camera Link configuration. RoHS compliant.
HS-80-08k40	8k resolution, programmable 4 or 8 taps, up to 320 megapixels/ second throughput, 40/ 80MHz strobe rate, Medium or Full Camera Link configuration. RoHS compliant.
HS-80-08k80	8k resolution, programmable 4 or 8 taps, up to 640 megapixels/ second throughput, 40/ 80MHz strobe rate, Medium or Full Camera Link configuration. RoHS compliant.
HS-82-04k80	4k resolution, programmable 4 or 8 taps, up to 640 megapixels / second throughput, 40/ 80MHz strobe rate, over 100 kHz line rate, Medium or Full Camera Link configuration. RoHS compliant.

Throughout the manual, the cameras are referred to as the Piranha HS-xx camera family unless a section is valid to a specific model only where the camera's model number is used.

1.2 Camera Performance Specifications

Features and Specifications					
Model	HS-40-04k40	HS-80-04k-40	HS-80-08k40	HS-80-08k80	HS-82-04k80
Imager Format	Bidirectional TDI — all models				
Resolution	4096 pixels	4096	8196	8196	4096 x 48
Pixel Fill Factor	90 %	100 %	90 %	100 %	100 %
Pixel Size	7x7 μm	7x7 μm	7x7 μm	7x7 μm	14 x 14 μm
Output Format (# of Camera Link taps)	2 or 4	4 or 8	4 or 8	4 or 8	4 or 8
Stage Selection	16, 32, 48, 64, 80, 96 — all models				
Antiblooming	100 x — all models				
CCD Shift Direction Change	0.2 seconds— all models				

Optical Interface					
Model	HS-40-04k40	HS-80-04k-40	HS-80-08k40	HS-80-08k80	HS-82-04k80
Back Focal Distance					
F Mount	46.50 $\pm$ 0.18 mm	—	—	—	—
M42x1 Mount	6.56 $\pm$ 0.25 mm	—	—	—	—
M72 Mount		6.56 $\pm$ 0.25	6.56 $\pm$ 0.25	6.56 $\pm$ 0.25	6.56 $\pm$ 0.25
Sensor Alignment (aligned to sides of camera)	$\pm$ 0.05 mm x $\pm$ 0.05 mm y $\pm$ 0.25 mm z $\pm$ 0.2 ° Θ_z				
Lens Mount Hole ¹	62 mm hole. M42 or F mount adapter available.	M72 x 0.75	M72 x 0.75	M72 x 0.75	M72 x 0.75

Mechanical Interface					
Model	HS-40-04k40	HS-80-04k-40	HS-80-08k40	HS-80-08k80	HS-82-04k80
Camera Size	85 (l) x 85 (h) x 55.4 (w) mm	80 (l) x 150 (h) x 65 (w)	80 (l) x 150 (h) x 65 (w)	80 (l) x 150 (h) x 65 (w)	80 (l) x 150 (h) x 65 (w)
Mass	< 500 g	< 800 g	< 800 g	< 800 g	< 800 g
Connectors	6 pin male Hirose, power MDR26 female, data				

Electrical Interface					
Model	HS-40-04k40	HS-80-04k40	HS-80-08k40	HS-80-08k80	HS-82-04k80
Input Voltage	+ 12 to + 15 $\pm$ 5 % Volts DC — all models				
Power Dissipation ²	10 W	19 W	14.4 W	19 W	< 20 W
Operating Temperature ³	0 to 50 °C — all models				
Bit Width	8 or 12 bit user selectable bits — all models				
Output Data Configuration	Base or Medium Camera Link	Medium or Full Camera Link	Medium or Full Camera Link	Medium or Full Camera Link	Medium or Full Camera Link

Operating Ranges					
Model	HS-40-04k40	HS-80-04k40	HS-80-08k40	HS-80-08k80	HS-82-04k80
Minimum Line Rate	3.5 kHz — all models				
Maximum Line Rate	36.563 kHz	68.610 kHz	34.305 kHz	68.610	110 kHz
Throughput	selectable 80 or 160 Mpix/ sec	selectable 160 or 320 Mpix/ sec	selectable 80, 160, or 320 Mpix/ sec	selectable 160, 320 or 640 Mpix/ sec	selectable 320 or 640 Mpix/ sec
Gain	-10 to +10 dB — all models				
Calibration Time	4.3 seconds — all models				

Table 2: Camera Performance Specifications

Performance	Gain -10 dB			Gain 0 dB			Gain +10 dB		
	Min	Typ	Max	Min	Typ	Max	Min	Typ	Max
Dynamic Range									
HS-40-04k40	313	1140		208	357		63	114	
HS-80-04k40	312	1190		166	357		56	119	
HS-80-08k40	312	1920		312	607		100	192	
HS-80-08k80	312	1190		166	357		56	119	
HS-82-04k80	312	1190		166	357		56	119	
Random Noise DN rms									
HS-40-04k40		0.22	0.8		0.70	1.2		2.20	4
HS-80-04k40		0.22	0.8		0.70	1.5		2.1	4.5
HS-80-08k40		0.16	0.8		0.41	0.8		1.3	2.5
HS-80-08k80		0.22	0.8		0.70	1.5		2.1	4.5
HS-82-04k80		0.22	0.8		0.7	1.5		2.1	4.5
SEE nJ/cm²									
HS-40-04k40		2.13			0.68			0.21	
HS-80-04k40		2.13			0.68			0.21	
HS-80-08k40		2.13			0.65			0.21	
HS-80-08k80		2.13			0.68			0.21	
HS-82-04k80		2.13			0.68			0.21	
NEE pJ/cm²									
HS-40-04k40		1.9	6.8		1.9	3.2		1.9	3.4
HS-80-04k40		1.9	6.8		1.9	4.1		1.8	3.8
HS-80-08k40		1.1	6.8		1.1	2.2		1.1	2.1
HS-80-08k80		1.9	6.8		1.9	4.1		1.8	3.8
HS-82-04k80		1.9	6.8		1.9	4.1		1.8	3.8
Analog Broadband Responsivity (DN/nJ/ cm²)									
HS-40-04k40		117		352	370	389		1170	
HS-80-04k40		117		352	370	389		1170	
HS-80-08k40		117		352	370	389		1170	
HS-80-08k80		117		352	370	389		1170	
HS-82-04k80		117		352	370	389		1170	
FPN DN p-p with correction									
HS-40-04k40					0.5	2			

Performance	Gain -10 dB			Gain 0 dB			Gain +10 dB		
HS-80-04k40					0.5	2			
HS-80-08k40					0.5	2			
HS-80-08k80					0.5	2			
HS-82-04k80					0.5	2			
FPN DN p-p w/o correction									
HS-40-04k40		1	3		2	6		10	18
HS-80-04k40			3		4.5	3			9
HS-80-08k40			3		3	3			9
HS-80-08k80			3		4.5	3			9
HS-82-04k80			3		4.5	3			9
PRNU DN p-p with correction									
HS-40-04k40					3	5.5			
HS-80-04k40					4.0	8			
HS-80-08k40					3.2	5.5			
HS-80-08k80					4.0	8			
HS-82-04k80					4.0	8			
PRNU % w/o correction									
HS-40-04k40		4	10		5	12		10	25
HS-80-04k40			22			22		10	25
HS-80-08k40			22			22		7	25
HS-80-08k80			22			22		10	25
HS-82-04k80			22			22		10	25
Saturation Output Amplitude DN	255 typ								
DC Offset DN	3 min			5 typ			6 max		

Test conditions for all models, unless otherwise noted:

- TDI mode of operation. These specifications are not guaranteed for area mode of operation.
 - Line Rate: 10 kHz.
 - Nominal Gain setting.
 - Light Source: Broadband Quartz Halogen, 3250 k, with 700 nm IR cutoff filter installed.
 - All Max specifications are valid over a 0-50 °C temperature range.
 - All Typ specifications are measured at 25 °C.
 - All values are referenced at 8-bit.
1. Maximum using highest Camera Link mode and maximum line rate.
 2. Measured at the front plate.

Figure 1: PRNU Uncorrected (pk-pk) Forward or Reverse @ 10 kHz line rate and 0 dB Gain over Temperature

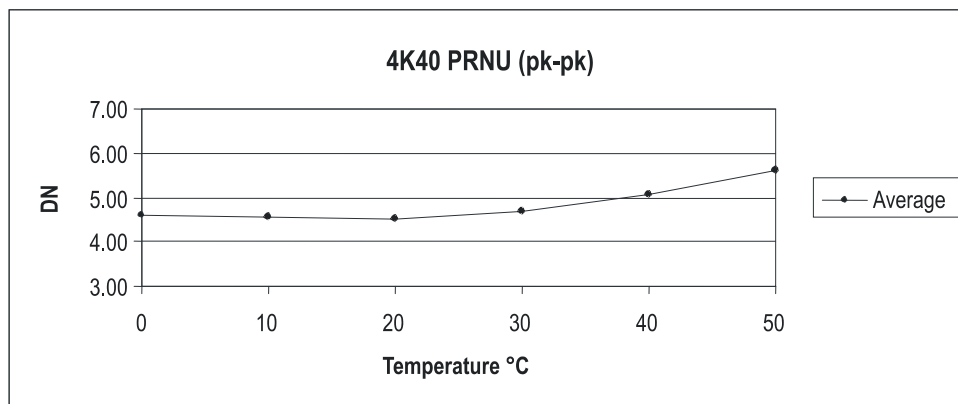
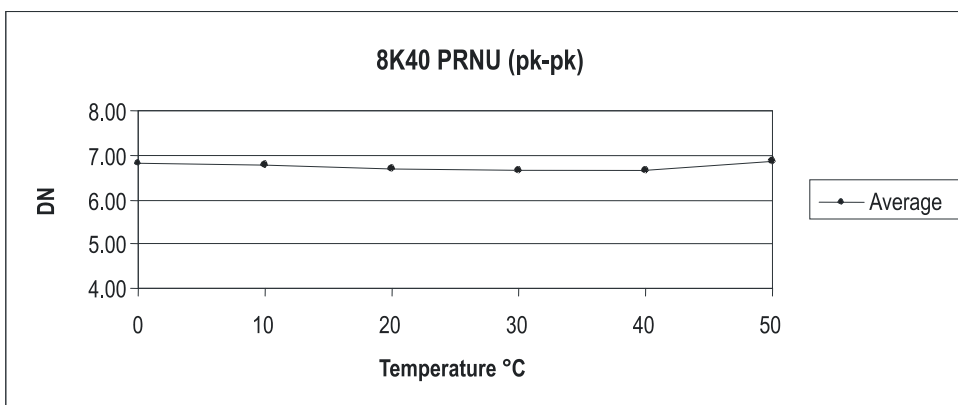
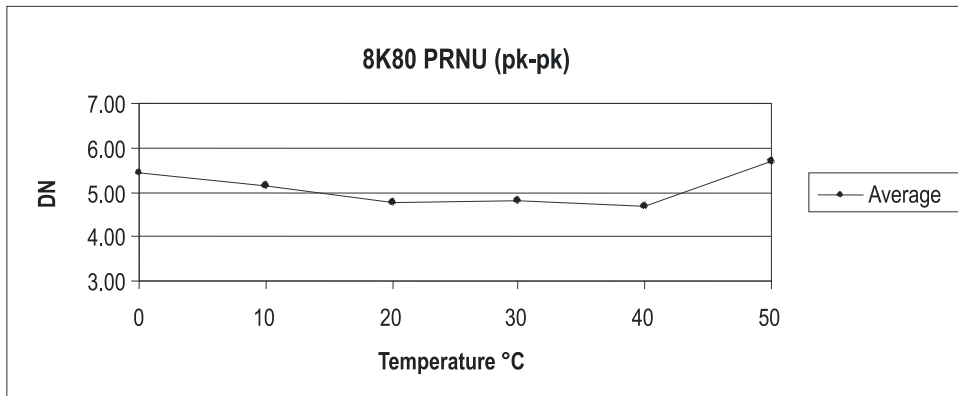


Figure 2: FPN Forward or Reverse @ 10 kHz line rate and 0 dB Gain over Temperature

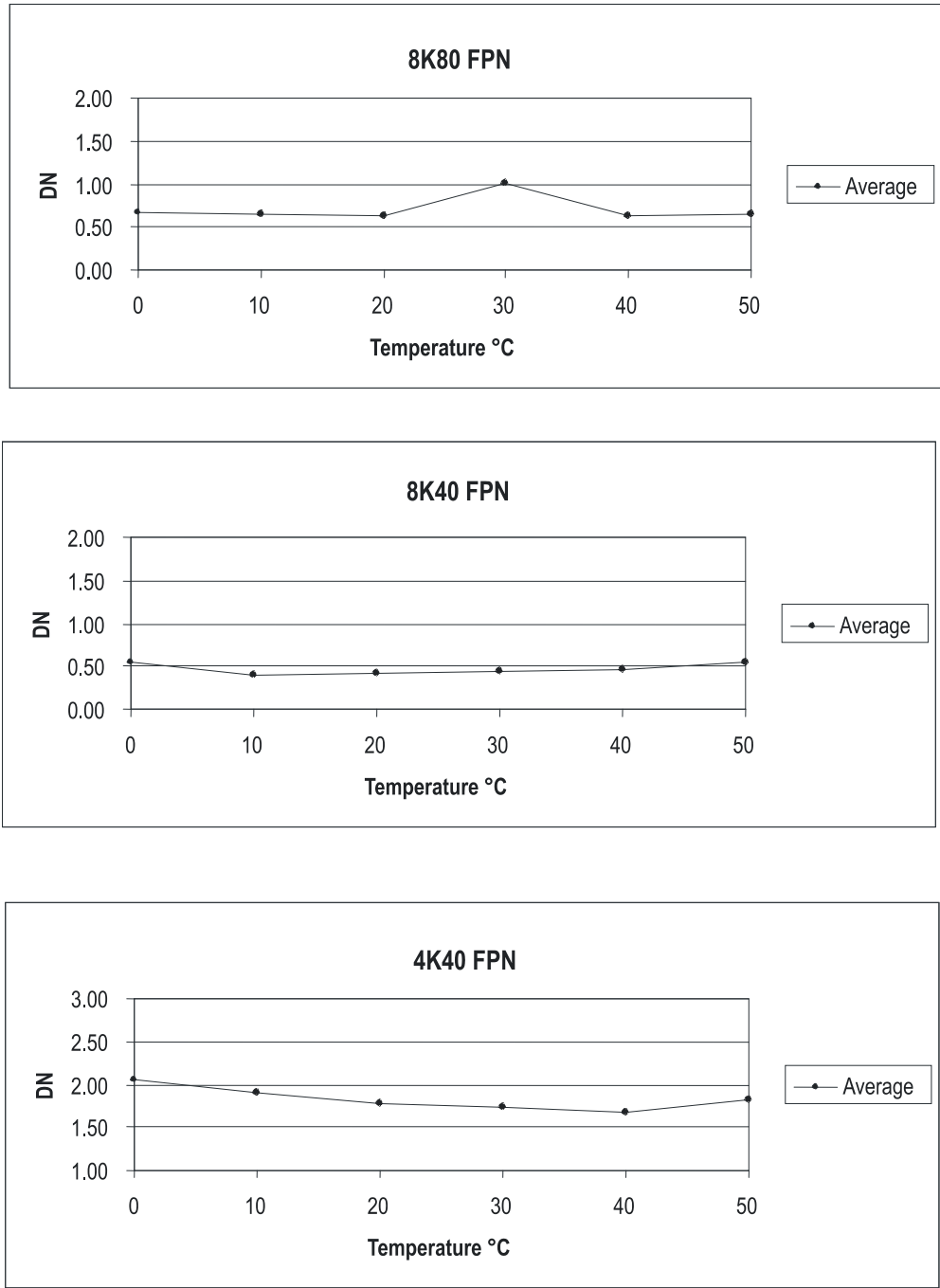


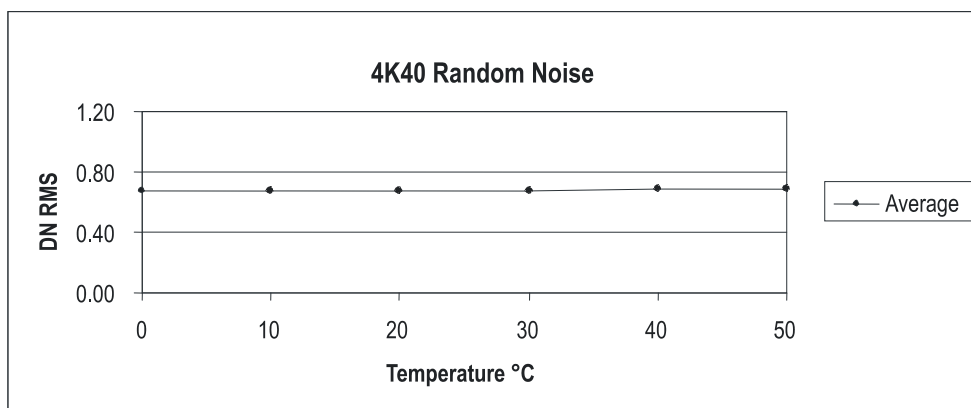
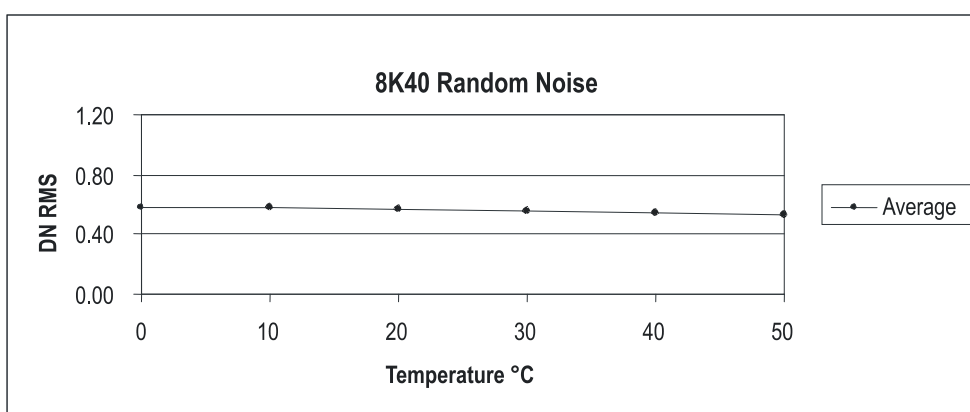
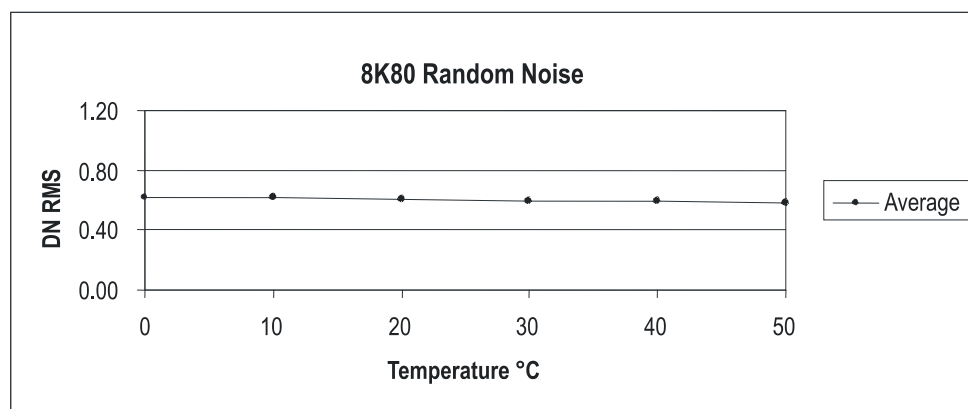
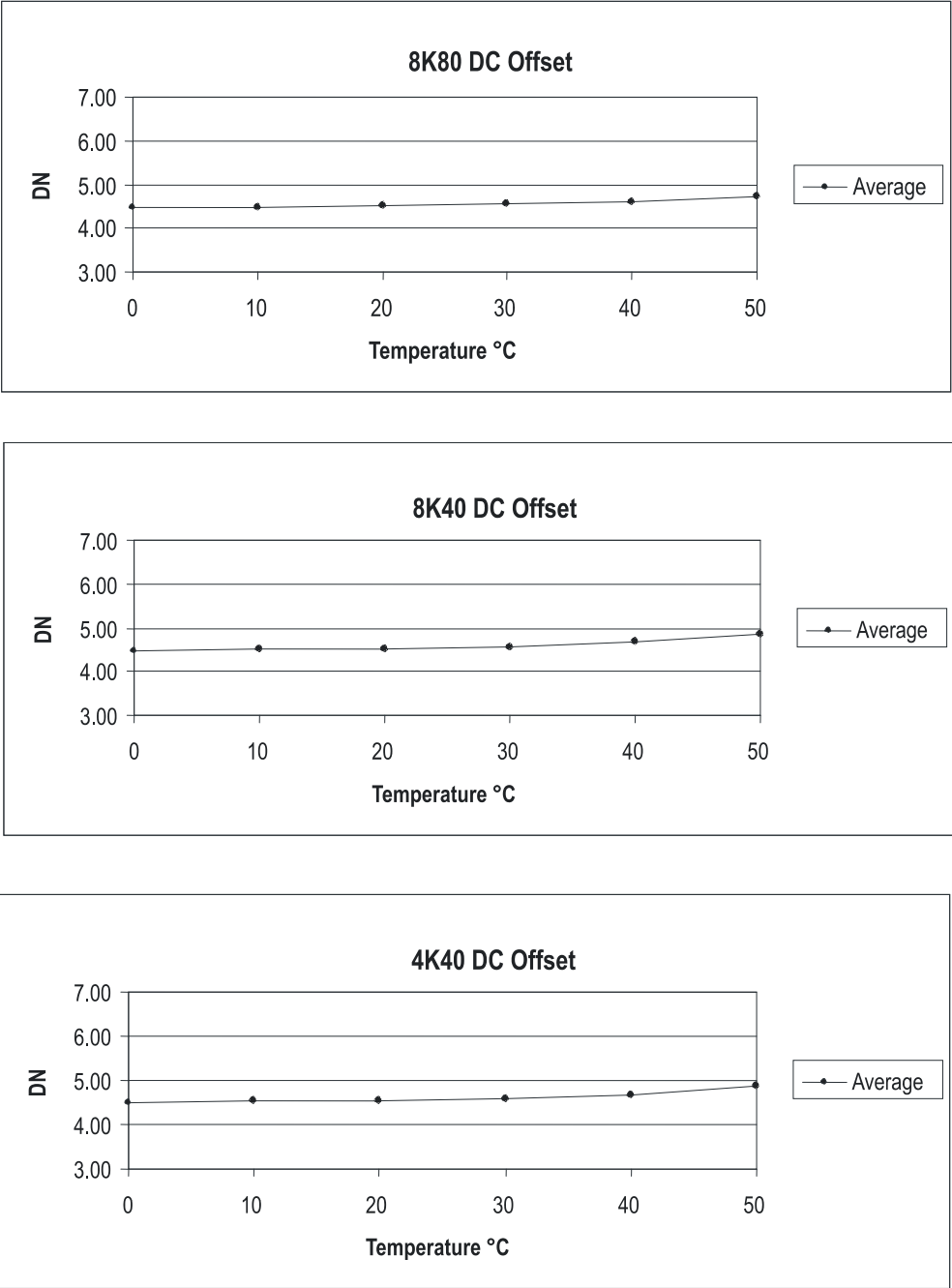
Figure 3: Random Noise Forward or Reverse @ 10 kHz line rate and 0 dB Gain over Temperature

Figure 4: DC Offset Forward or Reverse @ 10 kHz line rate and 0 dB Gain over Temperature



1.3 Image Sensor

The camera uses Teledyne DALSA's newest bidirectional TDI sensors. The camera can be configured to read out in either Forward or Reverse CCD shift direction. Readout direction is controlled by the software command **scd**.

Figure 5: 4 Tap Sensor Block Diagram (HS-40-04k40)

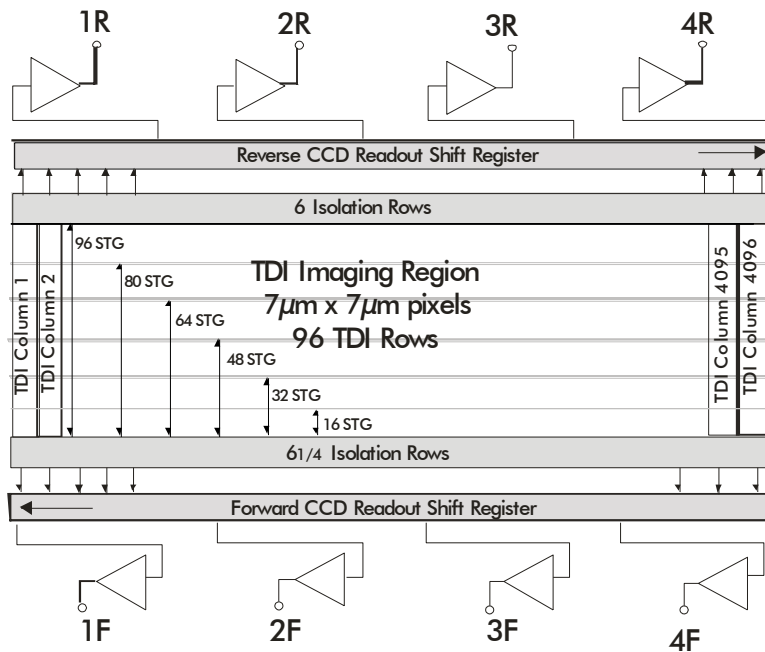
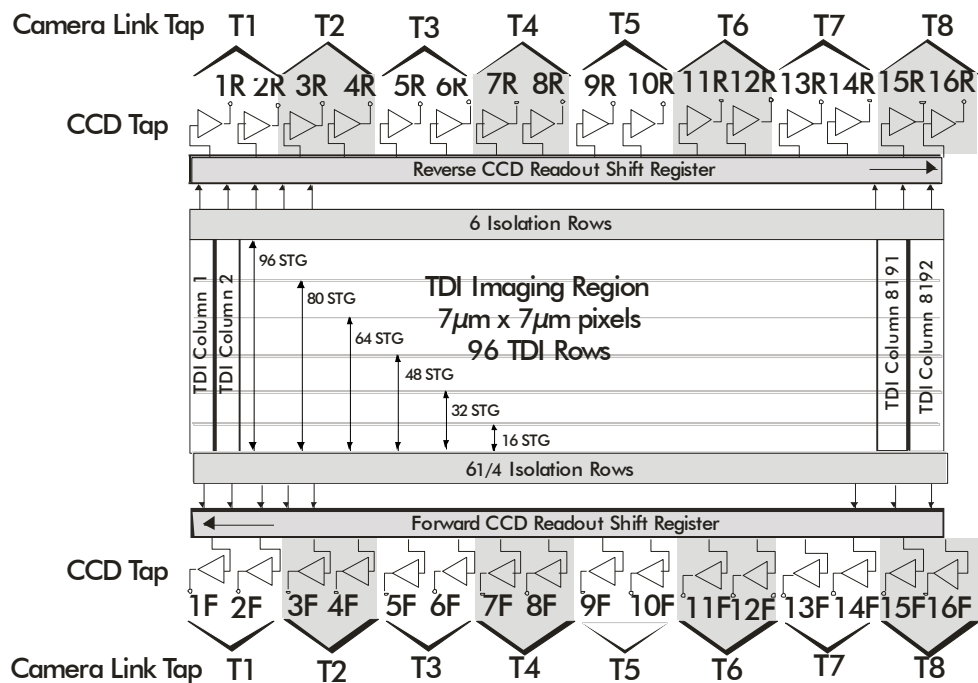
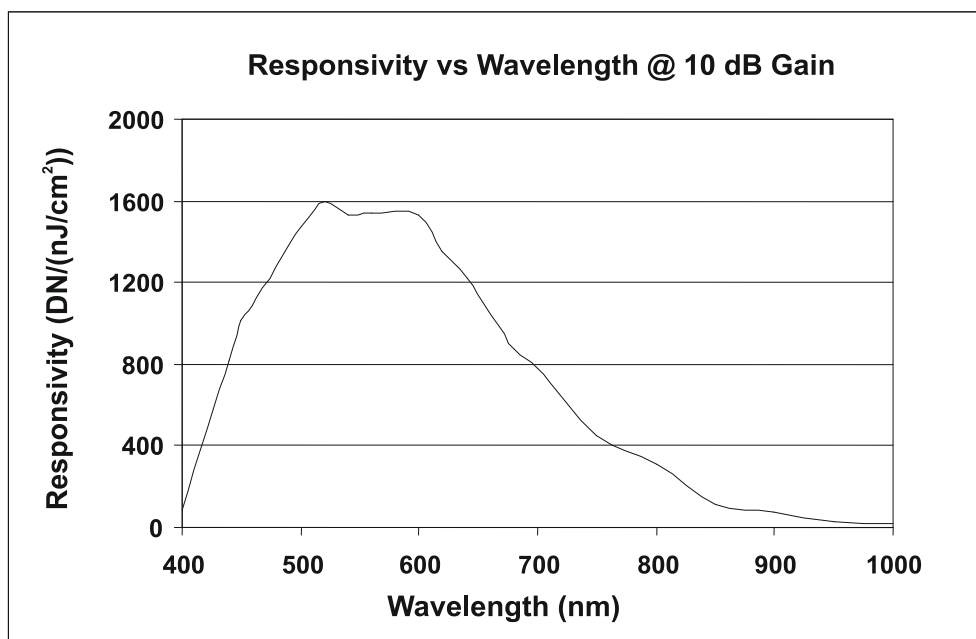


Figure 6: 16 Tap Sensor Block Diagram (HS-80-04k40, HS-80-08k80)



1.4 Responsivity

Figure 7: Piranha HS Analog Responsivity



2

Camera Hardware Interface

2.1 Installation Overview

This installation overview assumes you have not installed any system components yet.

When installing your camera, you should take these steps:

1. Power down all equipment.
2. Follow the manufacturer's instructions to install the framegrabber (if applicable). Be sure to observe all static precautions.
3. Install any necessary imaging software.
4. Before connecting power to the camera, test all power supplies. Ensure that all the correct voltages are present at the camera end of the power cable. Power supplies must meet the requirements defined in section 2.2.2 Power Connector.
5. Inspect all cables and connectors prior to installation. Do not use damaged cables or connectors or the camera may be damaged.
6. Connect Camera Link and power cables.
7. After connecting cables, apply power to the camera.
8. Check the diagnostic LED. See 2.2.1 LED Status Indicator for an LED description.

You must also set up the other components of your system, including light sources, camera mounts, host computers, optics, encoders, and so on.

2.2 Input/Output Connectors and LED

The camera uses:

- A diagnostic LED for monitoring the camera. See LED Status Indicator in section 2.2.1 LED Status Indicator for details.

- High-density 26-pin MDR26 connectors for Camera Link control signals, data signals, and serial communications. Refer to section 2.2.3 Camera Link Data Connector for details.
- One 6-pin Hirose connector for power. Refer to section 2.2.2 Power Connector for details.

Figure 8: Piranha HS-xx Input and Output Connectors (4k Models)

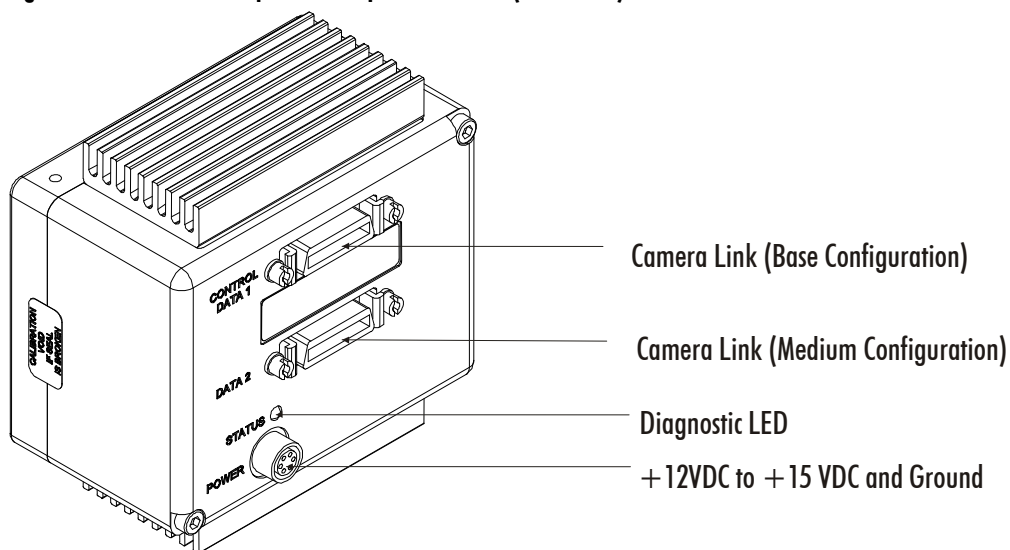
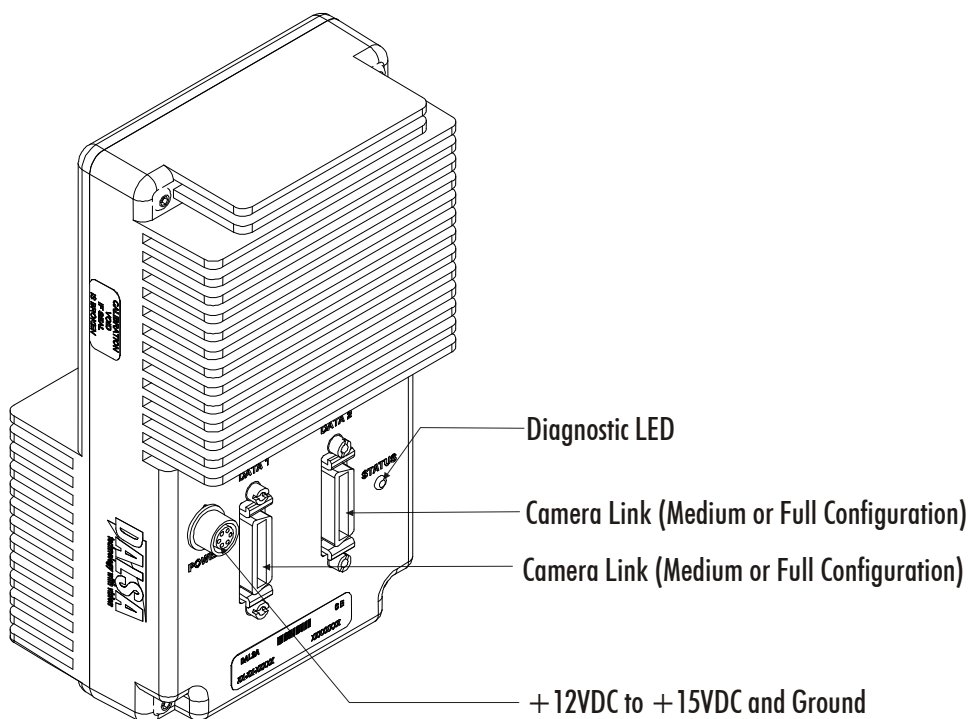


Figure 9: Piranha HS-xx Input and Output Connectors (8k Models)



WARNING: It is extremely important that you apply the appropriate voltages to your camera. Incorrect voltages will damage the camera. See 2.2.2 Power Connector for more details.

2.2.1 LED Status Indicator

The camera is equipped with a red/ green LED used to display the operational status of the camera. The table below summarizes the operating states of the camera and the corresponding LED states.

When more than one condition is active, the LED indicates the condition with the highest priority. Error and warning states are accompanied by corresponding messages further describing the current camera status.

Table 3: Diagnostic LED

Priority	Color of Status LED	Meaning
1	Flashing Red	Fatal Error. Camera temperature is too high and camera thermal shutdown has occurred or a power on failure has been detected.
2	Solid Red	Warning. Loss of functionality.
3	Flashing Green	Camera initialization or executing a long command (e.g., flat field correction commands ccp or ccf)
4	Solid Green	Camera is operational and functioning correctly.

2.2.2 Power Connector

Figure 10: Hirose 6-pin Circular Male—Power Connector

Hirose 6-pin Circular Male



Table 4: Hirose Pin Description

Pin	Description	Pin	Description
1	Min +12 to Max +15VDC	4	GND
2	Min +12 to Max +15VDC	5	GND
3	Min +12 to Max +15VDC	6	GND

The camera requires a single voltage input (+12 to +15VDC). The camera meets all performance specifications using standard switching power supplies, although well-regulated linear supplies provide optimum performance.



WARNING: When setting up the camera's power supplies follow these guidelines:

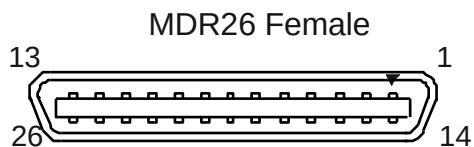
- Apply the appropriate voltages
- Protect the camera with a **fast-blow fuse** between power supply and camera.
- Do not use the shield on a multi-conductor cable for ground.
- Keep leads as short as possible to reduce voltage drop.
- Use high-quality **linear** supplies to minimize noise.
- Use an isolated type power supply to prevent LVDS common mode range violation.

Note: Camera performance specifications are not guaranteed if your power supply does not meet these requirements.

Teledyne DALSA offers a power supply with attached 6' power cable that meets the Piranha HS camera's requirements, but it should not be considered the only choice. Many high quality supplies are available from other vendors.

2.2.3 Camera Link Data Connector

Figure 11: Camera Link MDR26 Connector



Mating Part : 3M 334-31 series

Cable: 3M 14X26-SZLB-XXX-0LC**

The Camera Link interface is implemented as a Base, Medium or Full Configuration in the Piranha HS cameras depending on the model number. The following table summarizes the different configurations and lists the configurations available to each Piranha HS model number.

Table 5: Camera Link Hardware Configuration Summary for Piranha HS-xx Models

Configuration	8 Bit Ports Supported	Serializer Bit Width	Number of Chips	Number of MDR26 Connectors	Applicable Camera Models
Base	A, B, C	28	1	1	HS-40-04k40
Medium	A, B, C, D, E, F	28	2	2	HS-40-04k40 HS-80-08k40 HS-80-08k80
Full	A, B, C, D, E, F, G, H	28	3	2	HS-80-08k40 HS-80-08k80

Table 6: Camera Link Connector Pinout

Medium and Full Configurations Up to an additional 2 Channel Link Chips				Base Configuration One Channel Link Chip + Camera Control + Serial Communication		
Camera Connector	Right Angle Frame Grabber	Channel Link Signal	Cable Name	Camera Connector	Right Angle Frame Grabber	Channel Link Signal
1	1	inner shield	Inner Shield	1	1	inner shield
14	14	inner shield	Inner Shield	14	14	inner shield
2	25	Y0-	PAIR1-	2	25	X0-
15	12	Y0+	PAIR1+	15	12	X0+
3	24	Y1-	PAIR2-	3	24	X1-
16	11	Y1+	PAIR2+	16	11	X1+
4	23	Y2-	PAIR3-	4	23	X2-
17	10	Y2+	PAIR3+	17	10	X2+
5	22	Yclk-	PAIR4-	5	22	Xclk-
18	9	Yclk+	PAIR4+	18	9	Xclk+
6	21	Y3-	PAIR5-	6	21	X3-
19	8	Y3+	PAIR5+	19	8	X3+
7	20	100 ohm	PAIR6+	7	20	SerTC+
20	7	terminated	PAIR6-	20	7	SerTC-
8	19	Z0-	PAIR7-	8	19	SerTFG-
21	6	Z0+	PAIR7+	21	6	SerTFG+
9	18	Z1-	PAIR8-	9	18	CC1-
22	5	Z1+	PAIR8+	22	5	CC1+
10	17	Z2-	PAIR9+	10	17	CC2+
23	4	Z2+	PAIR9-	23	4	CC2-
11	16	Zclk-	PAIR10-	11	16	CC3-
24	3	Zclk+	PAIR10+	24	3	CC3+
12	15	Z3-	PAIR11+	12	15	CC4+
25	2	Z3+	PAIR11-	25	2	CC4-
13	13	inner shield	Inner Shield	13	13	inner shield
26	26	inner shield	Inner Shield	26	26	inner shield

Notes:

*Exterior Overshield is connected to the shells of the connectors on both ends.

**3M part 14X26-SZLB-XXX-0LC is a complete cable assembly, including connectors.

Unused pairs should be terminated in 100 ohms at both ends of the cable.

Inner shield is connected to signal ground inside camera

Table 7: Teledyne DALSA Camera Control Configuration

Signal	Configuration
CC1	EXSYNC
CC2	Spare
CC3	Forward
CC4	Spare

See Appendix B for the complete Teledyne DALSA Camera Link configuration table, and refer to the Teledyne DALSA Web site, [Knowledge Center application notes](#), for the official Camera Link documents.

Input Signals, Camera Link

The camera accepts control inputs through the Camera Link MDR26F connector.



The camera ships in internal sync, internal programmed integration (exposure mode 7) TDI Mode.



EXSYNC (Triggers Frame Readout)

Frame rate can be set internally using the serial interface. The external control signal EXSYNC is optional and enabled through the serial interface. This camera uses the **falling edge of EXSYNC** to trigger pixel readout. Section Exposure Mode and Line/ Frame Rate for details on how to set frame times, exposure times, and camera modes.

Direction Control

You control the CCD shift direction through the serial interface. With the software command, **scd**, you determine whether the direction control is set via software control or via the Camera Link control signal on CC3. Refer to section Setting the Camera's CCD Shift Direction for details.

Output Signals, Camera Link



IMPORTANT:

This camera's data should be sampled on the rising edge of STROBE.

These signals indicate when data is valid, allowing you to clock the data from the camera to your acquisition system. These signals are part of the Camera Link configuration and you should refer to the Teledyne DALSA Camera Link Implementation Road Map, available at [Knowledge Center](#), for the standard location of these signals.

Clocking Signal	Indicates
LVAL (high)	Outputting valid line
DVAL (high)	Valid data
STROBE (rising edge)	Valid data
FVAL (high)	Outputting valid frame

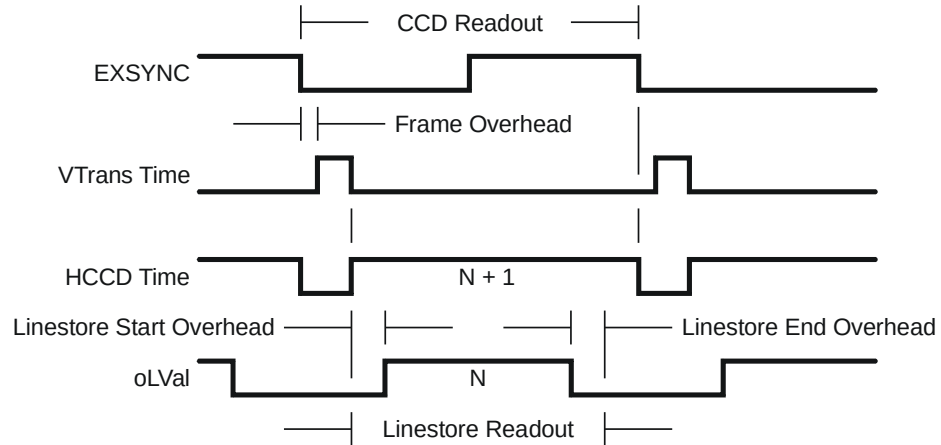
- The camera internally digitizes 12 bits and outputs 8 MSB or all 12 bits depending on the camera's Camera Link operating mode. Refer to 4.4.2 Setting the Camera Link Mode for details.
- For a Camera Link reference refer to Appendix A on page 114.

2.3 Camera Link Video Timing

The Piranha HS-xx camera has two different readout times. The first readout time is the CCD Readout where the camera pixels are read out into the camera's linestore. The second readout is the linestore readout where the linestore pixels are read out to your acquisition system. The camera's minimum readout time is dependant on which of these two readout times are greater where the greater readout time will be the camera's minimum readout time.

The figure below illustrates camera timing when the CCD readout is greater than the linestore readout.

Figure 12: Piranha HS-xx TDI Mode Timing (CCD Limited)



The following figure illustrates camera timing when the linestore readout is greater than the CCD readout.

Figure 13: Piranha HS-xx TDI Mode Timing (Linestore Limited)

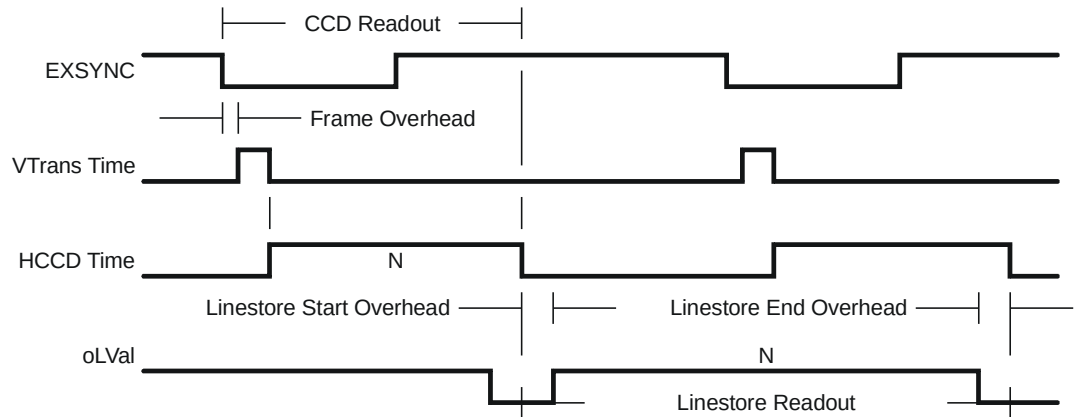


Table 8: Piranha HS-xx Timing Values

Symbol	Time
HCCD Frequency	40MHz (4K40, 8K80) 20MHz (8K40)
HCCD Pixels/Tap	1024 (4k camera) 512 (8k camera)
Frame Overhead	$\frac{4}{\text{HCCD Frequency}}$
VTransTime	$\frac{33}{\text{HCCD Frequency}}$
HCCD Read Overhead	HS 4k: 35 pixels, HS 8k: 34 pixels
HTime	$HTime = \frac{\text{HCCD Read Overhead} + \text{HCCD Pixels/Tap}}{\text{HCCD Frequency}}$
CCD Readout Time	$CCDReadoutTime = \text{FrameOverhead} + HTime + VTransTime \times \text{Vertical Binning Factor}$

Symbol	Time
HCCD Taps	4 (4K40) 16 (8Kxx)
Linestore Start Overhead	15 clocks
Linestore End Overhead	23 clocks
Linestore Readout Time	$\left[\frac{\text{HCCD Pixels/Tap} \times \text{HCCD Taps}}{\left(\# \text{ Camera Link Taps} \times \text{Horizontal Binning Factor} \right)} + \text{Linestore Start Overhead} + \text{Linestore End Overhead} \right] \times \frac{\# \text{ Camera Link Taps}}{\text{Throughput (MHz)}}$
Horizontal Binning Factor	Value set with <code>sbh</code> command
Vertical Binning Factor	Value set with <code>sbv</code> command
Vertical Readout Rows	Stage selection set with <code>stg</code> command
# of Camera Link Taps	Value set with the <code>clm</code> command
Throughput	Value set with <code>sot</code> command

Figure 14: Piranha HS-xx Area Mode Timing (CCD Limited)

**IMPORTANT:**

This camera uses the *falling* edge of EXSYNC to trigger line readout, unlike previous DALSA cameras, which used the rising edge.

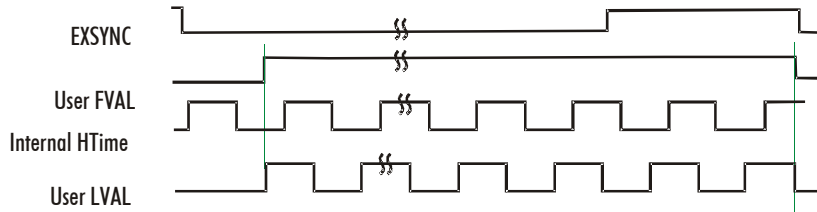


Figure 15: Piranha HS-xx Area Mode Timing (Linestore Limited)

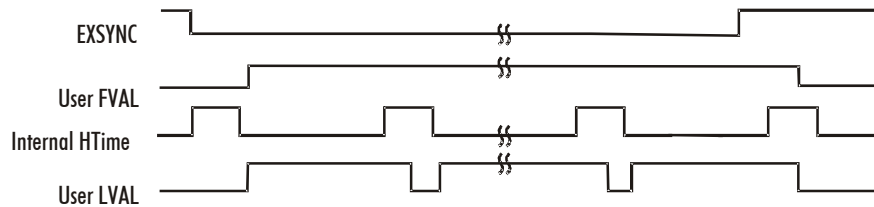


Table 9: Piranha Input and Output

Symbol	Time
CCD Readout Time (Area Mode)	$\text{CCD ReadoutTime} = \frac{\text{Frame Overhead}}{\text{HCCD Frequency}} + \left[\text{HTime} + \text{VTransTime} \times \text{Remainder} \left(\frac{\text{Vertical readout rows} + \text{ISORows}}{\text{Vertical Binning Factor}} \right) \right] +$ $\left[\left(\text{HTime} + \text{VTrans} \times \text{Vertical Binning Factor} \right) \times \text{Integer} \left(\frac{\text{Vertical readout rows} + \text{ISORows}}{\text{Vertical Binning Factor}} \right) \right]$

8k Resolution

3.2 Lens Mounts

Model Number	Lens Mount Options
HS-40-04k40	62mm hole. M42x1 and F-mount lens adapters available through Teledyne DALSA Sales.
HS-80 and HS-82 models	M72x0.75 thread.

3.3 Optical Interface

Illumination

The amount and wavelengths of light required to capture useful images depend on the particular application. Factors include the nature, speed, and spectral characteristics of objects being imaged, exposure times, light source characteristics, environmental and acquisition system specifics, and more.

It is often more important to consider exposure than illumination. The total amount of energy (which is related to the total number of photons reaching the sensor) is more important than the rate at which it arrives. For example, $5\mu\text{J}/\text{cm}^2$ can be achieved by exposing $5\text{mW}/\text{cm}^2$ for 1ms just the same as exposing an intensity of $5\text{W}/\text{cm}^2$ for $1\mu\text{s}$.

Light Sources

Keep these guidelines in mind when setting up your light source:

- LED light sources are relatively inexpensive, provide a uniform field, and longer life span compared to other light sources. However, they also require a camera with excellent sensitivity, such as the HS-xx camera.
- Halogen light sources generally provide very little blue relative to infrared light (IR).
- Fiber-optic light distribution systems generally transmit very little blue relative to IR.
- Some light sources age; over their life span they produce less light. This aging may not be uniform—a light source may produce progressively less light in some areas of the spectrum but not others.

Filters

CCD cameras are extremely responsive to infrared (IR) wavelengths of light. To prevent infrared from distorting the images you scan, use a “hot mirror” or IR cutoff filter that transmits visible wavelengths but does not transmit wavelengths over 750nm. Examples are the Schneider Optics™ B+W 489, which includes a mounting ring, the CORION™ LS-750, which does not include a mounting ring, and the CORION™ HR-750 series hot mirror.

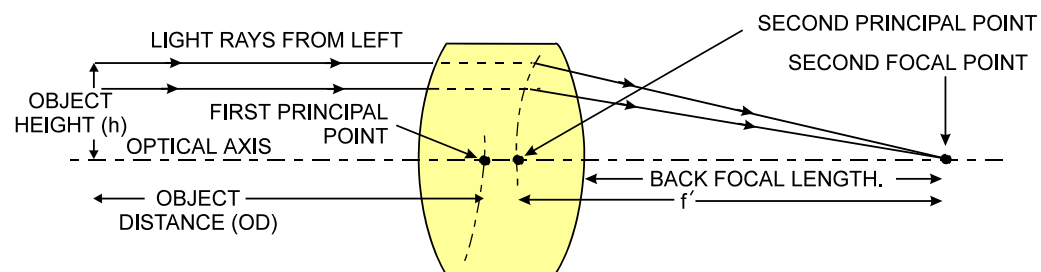
Lens Modeling

Any lens surrounded by air can be modeled for camera purposes using three primary points: the first and second principal points and the second focal point. The primary points for a lens should be available from the lens data sheet or from the lens manufacturer. Primed quantities denote characteristics of the image side of the lens. That is, h is the object height and h' is the image height.

The *focal point* is the point at which the image of an infinitely distant object is brought to focus. The *effective focal length* (f') is the distance from the second principal point to the second focal point. The *back focal length* (BFL) is the distance from the image side of the

lens surface to the second focal point. The *object distance (OD)* is the distance from the first principal point to the object.

Figure 17: Primary Points in a Lens System



4

Software Interface: How to Control the Camera



This chapter outlines the more commonly used commands. See section [Commands](#) for a list of all available commands.

All Piranha HS-xx camera features can be controlled through the serial interface. The camera can also be used without the serial interface after it has been set up correctly. Functions available include:

- Controlling basic camera functions such as gain and sync signal source.
- Flat field correction.
- Mirroring and readout control.
- Generating a test pattern for debugging.

The serial interface uses a simple ASCII-based protocol and the PC does not require any custom software.

Note: This command set has changes from previous Teledyne DALSA cameras. Do not assume that the Piranha HS commands perform similarly to older cameras.

Serial Protocol Defaults

- 8 data bits
- 1 stop bit
- No parity
- No flow control
- 9.6kbps
- Camera does not echo characters

Command Format

When entering commands, remember that:

- A carriage return <CR> ends each command.
- A space or multiple space characters separate parameters. Tabs or commas are invalid parameter separators.
- Upper and lowercase characters are accepted
- The backspace key is supported
- The camera will answer each command with either <CR><LF> "OK >" or <CR><LF>"Error xx: Error Message >" or <CR><LF>"Warning xx: Warning Message >". The ">" is used exclusively as the last character sent by the camera.

The following parameter conventions are used in the manual:

- *i* = integer value
- *f* = real number
- *m* = member of a set
- *s* = string
- *t* = tap id
- *x* = pixel column number
- *y* = pixel row number

Example: to return the current camera settings

`gcp <CR>`

Setting Baud Rate

Purpose:	Sets the speed in bps of the camera serial communication port.
Syntax:	sbr <i>m</i>
Syntax Elements:	<i>m</i>
	Baud rate. Available baud rates are: 9600 (Default), 19200 , 57600 , and 115200 .
Notes:	• Power-on rate is always 9600 baud. • The <u>rc</u> (reset camera) command will <i>not</i> reset the camera to the power-on baud rate and will reboot using the last used baud rate.
Example:	sbr 57600

Camera Help Screen

For quick help, the camera can return all available commands and parameters through the serial interface.

There are two different help screens available. One lists all of the available commands to configure camera operation. The other help screen lists all of the commands available for retrieving camera parameters (these are called “get” commands).

To view the help screen listing all of the camera configuration commands, use the command:

Syntax: `h`

To view a help screen listing all of the “get” commands, use the command:

Syntax: `gh`

Notes: For more information on the camera’s “get” commands, refer to section 4.7.6 Returning Camera Settings.

The camera configuration command help screen lists all commands available. Parameter ranges displayed are the extreme ranges available. Depending on the current camera operating conditions, you may not be able to obtain these values. If this occurs, values are clipped and the camera returns a warning message.

Some commands may not be available in your current operating mode. The help screen displays NA in this case.

Example Help Screen for HS-80-08k80 TDI Mode Operation

Command	cao	calibrate analog offset	ti	0-16:0-255
	ccf	correction calibrate fpn		
	ccg	calibrate camera gain	iti	1-4:0-16:1024-4055
	ccp	correction calibrate prnu		
	clm	camera link mode	m	15/16/21/
	cpa	calculate prnu algorithm	ii	1-4:1024-4055
	css	correction set sample	m	256/512/1024/
	dpc	display pixel coeffs	xx	1-8191:1-8192
	els	end of line sequence	i	0-1
	epc	enable pixel coefficients	ii	0-1:0-1
	gcm	get camera model		
	gcp	get camera parameters		
	gcs	get camera serial		
	gcv	get camera version		
	get	get values	s	
	get	fpn coeff	x	1-8192
	gh	get help		
	gl	get line	xx	1-8191:1-8192
	gla	get line average	xx	1-8191:1-8192
	gpc	get prnu coeff	x	1-8192
	gsf	get signal frequency	i	1-4
	h	help		
	lpc	load pixel coefficients	i	0-4
	rc	reset camera		
	rfs	restore factory settings		
	roi	region of interest	xyxy	1-8192:1-1:1-8192:1-1
	rpc	reset pixel coeffs		
	rus	restore user settings		
	sag	set analog gain	tf	0-16:-10.0-+10.0
	sao	set analog offset	ti	0-16:0-255
	sbh	set binning horizontal	m	1/2/4/8/
	sbr	set baud rate	m	9600/19200/57600/115200/
	sbv	set binning vertical	i	1-8
	scd	set ccd direction	i	0-2
	sdo	set digital offset	ti	0-16:0-255
	sem	set exposure mode	m	3/7/
	sfc	set fpn coeff	xi	1-8192:0-2048
	sfr	set fpn range	xxi	1-8192:1-8192:0-2048
	slt	set lower threshold	i	0-4095

Parameters
i = integer
f = floating point number
m = member of a set
s = string
t = tap
x = pixel column number
y = pixel row number

Parameter Range
- = range
: = multiple parameter separator
/ = member of a set separator
NA = command not available in current operating mode

smm	set mirroring mode	i	0-1
sot	set output throughput	m	160/320/640/
spc	set prnu coeff	xi	1-8192:0-28671
spr	set prnu range	xxi	1-8192:1-8192:0-28671
spt	set pretrigger	i	0-16
ssb	set subtract background	ti	0-16:0-4095
ssf	set sync frequency	f	3499.87-68610.6 [Hz]
ssg	set system gain	ti	0-16:0-65535
stg	set stage	m	16/32/48/64/80/96/
sut	set upper threshold	i	0-4095
svm	set video mode	i	0-2
tdi	set area or TDI mode	i	0-1
ugr	update gain reference		
vt	verify temperature		
vv	verify voltage		
wfc	write FPN coefficients	i	1-4
wpc	write PRNU coefficients	i	1-4
wus	write user settings		

Example Help Screen for HS-80-08k80 Area Mode Operation

cao	calibrate analog offset	ti	0-16:0-255
ccf	correction calibrate fpn		NA
ccg	calibrate camera gain	iti	1-3:0-16:1024-4055
ccp	correction calibrate prnu		NA
clm	camera link mode	m	5/16/21/
cpa	calculate prnu algorithm	ii	NA
css	correction set sample	m	NA
dpc	display pixel coeffs	xx	NA
els	end of line sequence	i	0-1
epc	enable pixel coefficients	ii	NA
gcm	get camera model		
gcp	get camera parameters		
gcs	get camera serial		
gcv	get camera version		
get	get values	s	
gfc	get fpn coeff	x	NA
gh	get help		
gl	get line	xx	NA
gla	get line average	xx	NA
gpc	get prnu coeff	x	NA
gsf	get signal frequency	i	1-4
gss	get sensor serial		
h	help		
lpc	load pixel coefficients	i	NA
rc	reset camera		
rfs	restore factory settings		
roi	region of interest	xyxy	1-8192:1-96:1-8192:1-96
rpc	reset pixel coeffs		
rus	restore user settings		
sag	set analog gain	tf	0-16:-10.0-+10.0
sao	set analog offset	ti	0-16:0-255
sbh	set binning horizontal	m	1/2/4/8/
sbr	set baud rate	m	9600/19200/57600/115200/
sbv	set binning vertical	i	1-8
scd	set_ccd_direction	i	0-1
sdo	set digital offset	ti	0-16:0-2048
sem	set exposure mode	m	3/7/
sfc	set fpn coeff	xi	NA
sfr	set fpn range	xxi	NA
slt	set lower threshold	i	0-4095
smm	set mirroring mode	i	0-1

sot	set output throughput	m	160/320/640/
spc	set prnu coeff	xi	NA
spr	set prnu range	xxi	NA
spt	set pretrigger	i	0-16
ssb	set subtract background	ti	0-16:0-4095
ssf	set sync frequency	f	1-6169.03 [Hz]
ssg	set system gain	ti	0-16:0-65535
stg	set stage	m	16/32/48/64/80/96/
sut	set upper threshold	i	0-4095
svm	set video mode	i	0-2
tdi	set TDI or area mode	i	0-1
ugr	update gain reference		
vt	verify temperature		
vv	verify voltage		
wfc	write FPN coefficients	i	NA
wpc	write PRNU coefficients	i	NA
wus	write user settings		

4.1 First Power Up Camera Settings

When the camera is powered up for the first time, it operates using the following factory settings:

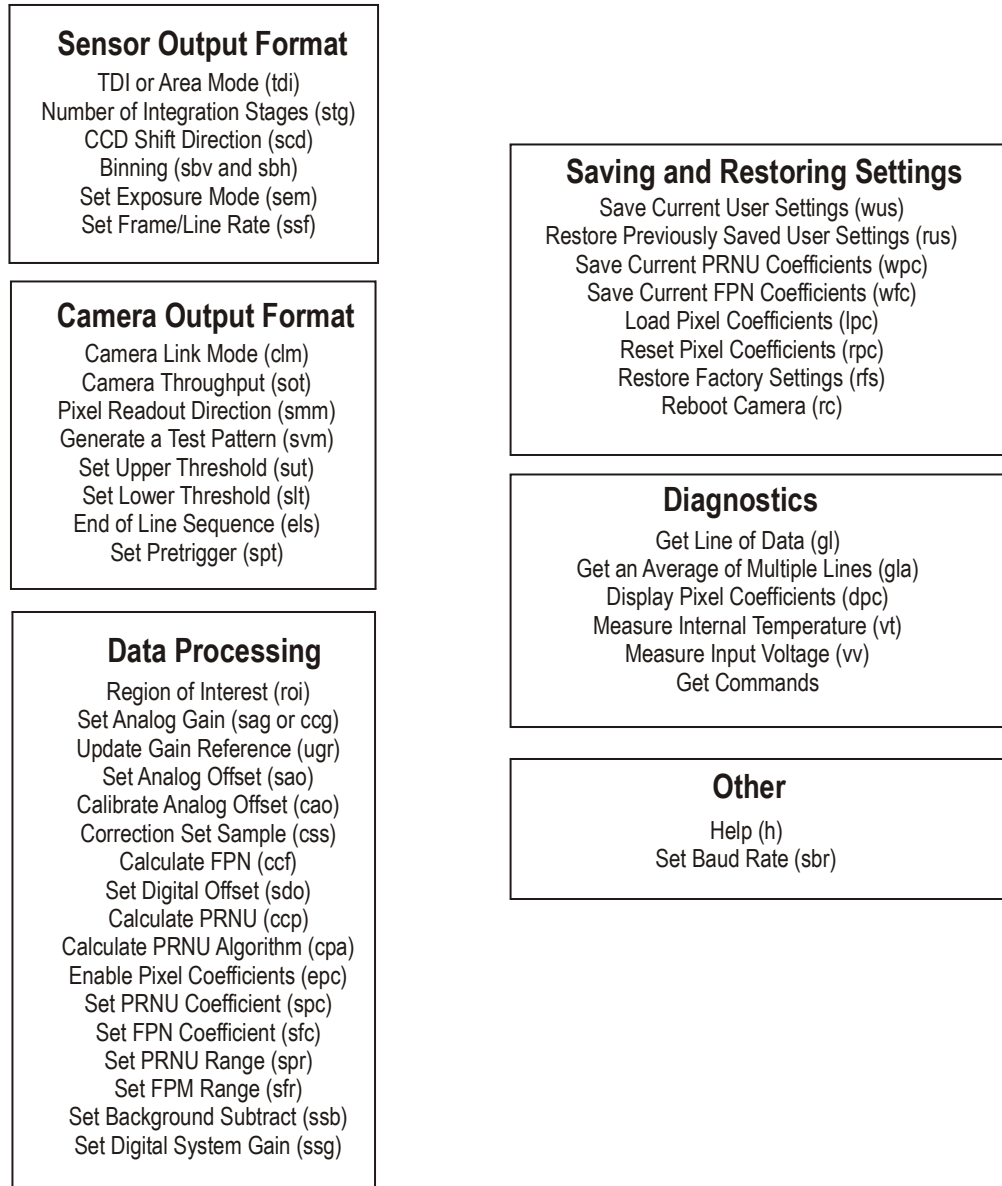
- TDI mode
- Left to right pixel readout
- Forward CCD shift direction
- 96 integration stages
- No binning
- Camera Link Mode 4k: 15 (8 bit, 4 taps, 40MHz strobe rate)
 8k: 21 (8 bit, 8 taps, 40MHz strobe rate)
- Exposure mode 7
- 10kHz line rate
- 160 (HS-4k) or 320 (HS-8k) throughput
- Factory calibrated analog gain and offset
- Factory calibrated FPN and PRNU coefficients using the following process:

ssf	10000	(line rate of 10000Hz)
ccg	2 0 3280	(analog gain calibrated to an average pixel value of 3280)
ccf		(fpn calibration)
cpa	2 3920	(calculate PRNU algorithm)
ssg	0 4096	(set system gain)

4.2 Command Categories

The following diagram categorizes and lists all of the camera's commands. This chapter is organized by command category.

Figure 18: Command Categories



4.3 Sensor Output Format

4.3.1 Selecting TDI or Area Mode Operation

The Piranha HS-xx cameras have the ability to operate in both TDI and Area Mode.

In Area Mode, the camera operates as an area array camera using a two dimensional array of pixels. Area Mode is useful for aligning the camera to your web direction or when you need a rectangular 2D image and the lighting supports a full frame imager.

In TDI Mode, the camera operates as a TDI high sensitivity line scan camera and combines multiple exposures of an object into one high-resolution result.

The camera stores user settings for Area Mode and TDI Mode separately, allowing you to switch between Area and TDI mode without losing settings specific to each mode. See section 4.6 Saving and Restoring Settings for an explanation on how user settings are stored and retrieved.

NOTE: Sensor cosmetic specifications for Area Mode of operation are neither tested nor guaranteed

Purpose:	Selects the camera's operating mode. Area Mode is useful for aligning and focusing your camera.
Syntax:	tdi i
Syntax Elements:	i 0 Area mode 1 TDI mode
Notes:	- Remember to save your user settings before changing mode. Sending the tdi command always restores your last saved user settings for the mode of operation requested even if you are already operating in the requested mode. See section 4.6 Saving and Restoring Settings for an explanation on how user settings are stored and retrieved for each mode. - Flat field correction is not available in Area Mode
Example	tdi 1

4.2 Selecting the Number of CCD Integration Stages

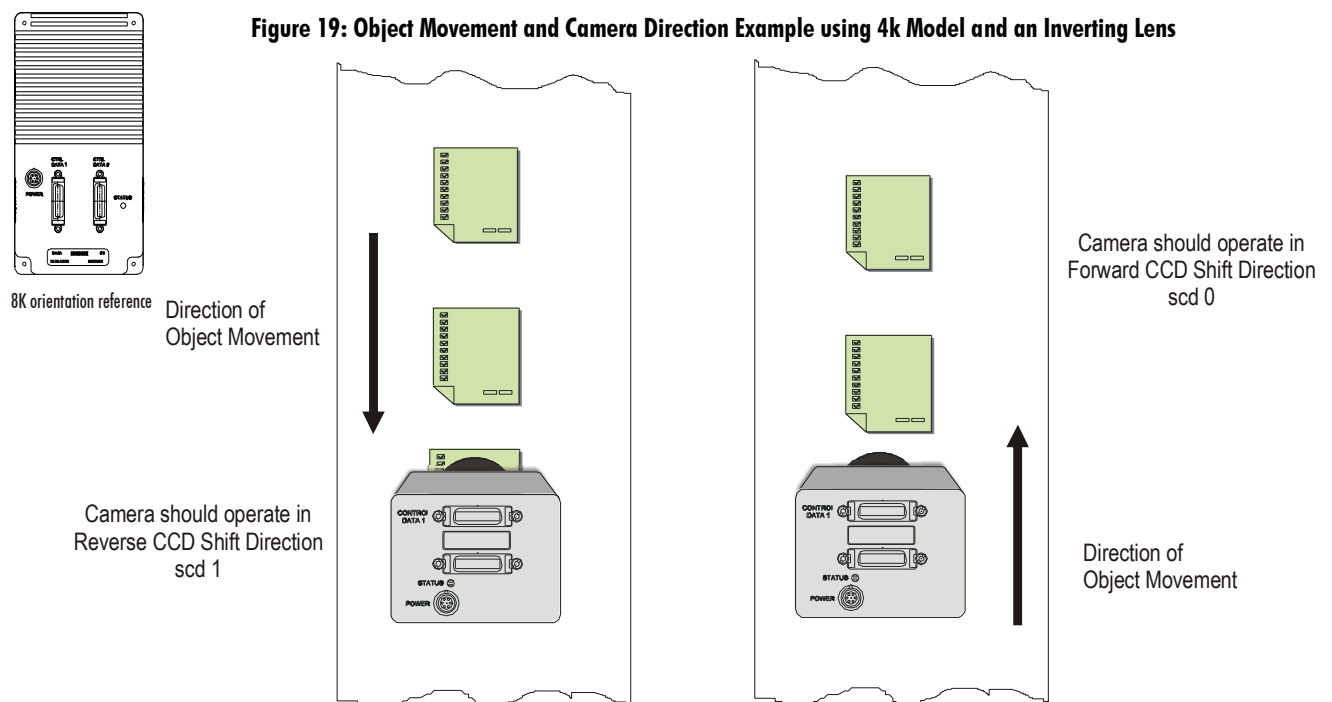
Purpose:	In TDI Mode, this command adjusts the sensitivity level in your camera by setting the number of CCD integration stages. In Area Mode, the vertical height of the image sensor is controlled by the number of stages.
Syntax:	stg m
Syntax Elements:	m Number of stages to use. Available values (not including the HS-82 model) are 16, 32, 48, 64, 80 , and 96 . Factory setting is 96 . Available values for the HS-82 model are 8, 16, 24, 32, 40 , and 48 (default).
Example	stg 64

A note on stage selection (stg) and the HS-82 model of camera:

The sensitivity level of a camera can be adjusted by setting the number of CCD integration stages. For such purpose, the standard HS-80-08K80-00-R camera has six different allowable settings: 16, 32, 48, 64, 80 and 96. With modifications to the binning setup, the HS-82-04k80-00-R camera retains this feature of the standard camera, but the number of stages is halved (8, 16, 24, 32, 40 and 48) to reflect this change.

4.3.3 Setting the Camera's CCD Shift Direction

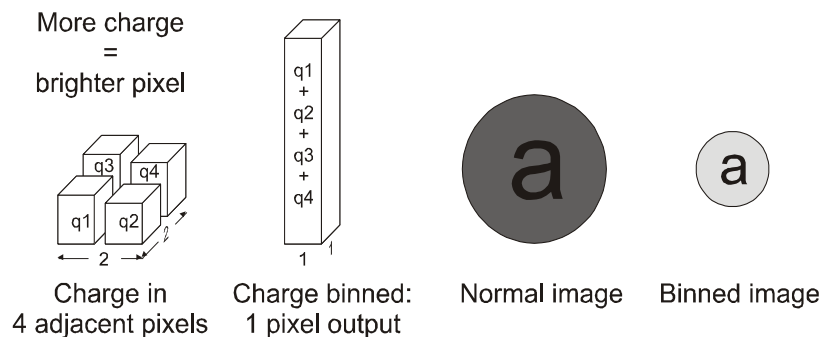
Purpose:	When in TDI Mode, selects the forward or reverse CCD shift direction or external direction control. This accommodates object direction change on a web and allows you to mount the camera “upside down”. In Area Mode, selects the vertical readout direction. This allows you to mirror the image vertically or mount the camera “upside down”.
Syntax:	scd i
Syntax Elements:	i Readout direction. Allowable values are: 0 = Forward CCD shift direction. 1 = Reverse CCD shift direction. 2 = Externally controlled direction control via Camera Link control CC3 (CC3=1 forward, CC3=0 reverse). Available only in TDI Mode.
Notes:	- The following user settings are stored separately for forward and reverse direction; analog gain, analog offset, digital gain, digital offset, background subtract, and pixel coefficients. These settings are automatically loaded when you switch direction. All other settings are common to both directions. - See the following figures for an illustration of CCD shift direction in relation to object movement. - Note that some commands that require longer processing time, like ccg, delay implementation of an external direction change.
Example	scd 1



4.3.4 Increasing Sensitivity with Binning

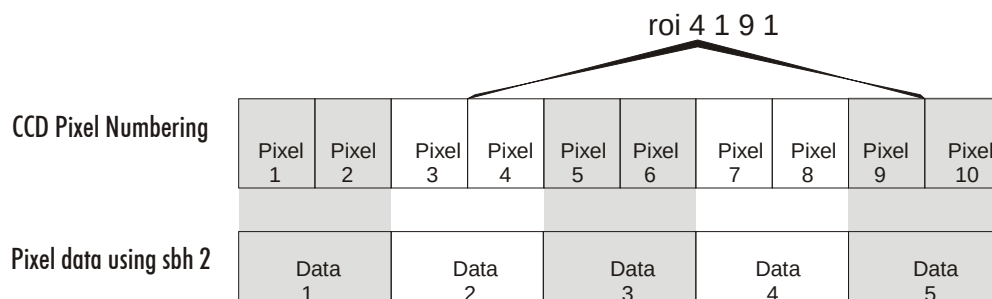
Binning increases the camera's light sensitivity by decreasing horizontal and/or vertical resolution—the charge collected by adjacent pixels is added together. Binning is also useful for increasing frame rate (vertical binning) or increasing the pixel pitch. For example, if you set your vertical binning to 2 and your horizontal binning to 2, your pixel size increases from $7\mu\text{m} \times 7\mu\text{m}$ (no binning) to $14\mu\text{m} \times 14\mu\text{m}$ (2x2 binning).

Figure 20: 2x2 Binning in Area Mode



Setting Horizontal Binning

Purpose:	Increases the horizontal pixel pitch and light sensitivity by decreasing horizontal resolution. The amount of data being sent from the camera is reduced by the horizontal binning factor. Different framegrabber files are needed for different horizontal binning factors.
Syntax:	sbh m
Syntax Elements:	m Horizontal binning value. Available values are 1 (factory setting, no binning) 2 , 4 , or 8 .
Notes:	- If you are using horizontal binning, the min, max, and mean statistics generated by the gl or gla command are for every second pixel (or valid data) only (e.g., if sbh 2, every second pixel). - For optimal flat field correction, you should rerun the ccp and ccf commands after changing binning values. - Changing binning values does not automatically alter gain, frame rate generation, or other functions of the camera. - Pixel numbering remains unchanged for the roi, gl, gla, dpc, gfc, sfc, gpc, and spc commands. Refer to Figure 21 for an explanation of pixel numbering and pixel start and stop values when using a region of interest.
Example:	sbh 2

Figure 21: Binning Start and End Values when using a Region of Interest

In this example a region of interest is set to include pixels 4 to 9 and horizontal binning is set to 2. Because pixel 3 is now included in the same data group as pixel 4, the region of interest will now include the data from pixel 3. Also, pixel 10 is included in the same data group as pixel 9, so pixel 10 is now part of the region of interest. To see how the region of interest start and stop values have been rounded, use the command **get roi**.

Setting Vertical Binning

Syntax: Increases the vertical pixel pitch and light sensitivity by decreasing vertical resolution. Vertical binning is also useful for increasing frame rate in Area Mode. Vertical binning in TDI Mode should only be used if your web's shaft encoder provides a reduced ratio of pulses to match web speed.

Syntax: **sbv i**

Syntax Elements: **i**

Vertical binning value.

Available values (not including the HS-82 model) are **1** (factory setting, no binning) to **8**.

Available values for the HS-82 model are **1** to **4**.

- Notes:**
- You will have to recalibrate the camera after changing binning values.
 - Increasing the vertical binning, decreases the maximum allowable line rate. You may have to enter a new camera frame rate after changing vertical binning values if the current value becomes invalid. The camera sends a warning message in this situation.

Example: **sbv 2**

A note on vertical binning (sbv) and the HS-82 model of camera:

Binning increases the camera's light sensitivity by decreasing horizontal and/ or vertical resolution — the charge collected by adjacent pixels is added together. Binning is also useful for increasing frame rate (vertical binning) or increasing the pixel pitch.

The default binning setting (1 x 1) in the HS-82-04k80-00-R camera is essentially equivalent to performing 2 x 2 binning in standard HS-80-08K80-00-R camera. Horizontal binning is achieved by binning two adjacent columns together during the VCCD to HCCD transfer while vertical binning is achieved by transferring two VCCD rows into the enlarged HCCD prior to horizontal transfers. This difference translates the 1x - 8x

binning capability of HS-80-08K80-00-R camera to 1x - 4x binning in the HS-82-04k80-00-R camera.

4.3.5 Exposure Mode and Line/Frame Rate

How to Set Exposure Mode and Line/Frame Rate

You have a choice of operating the camera in one of two exposure modes. Depending on your mode of operation, the camera's line/ frame rate (synchronization) can be generated internally through the software command [ssf](#) or set externally with an EXSYNC signal (CC1). When operating in TDI Mode, it is important that the line rate used matches the web speed. Failure to match the web speed will result in smearing.

To select how you want the camera's line/frame rate to be generated:

1. You must first set the camera's exposure mode using the `sem` command. Refer to section Setting the Exposure Mode below for details.
2. Next, if using mode 7, use the command [ssf](#) to set the line/ frame rate. Refer to section Setting Frame Rate for details.

Setting the Exposure Mode

Purpose:	Sets the camera's exposure mode allowing you to control your sync and line/ frame rate generation.
Syntax:	<code>sem m</code>
Syntax Elements:	<code>m</code>
Notes:	Exposure mode to use. Factory setting is 7. • Refer to Table 10: Piranha HS Exposure Modes for a quick list of available modes or to the following sections for a more detailed explanation including timing diagrams. • To obtain the current value of the exposure mode, use the command gcp or get sem. • When setting the camera to external signal modes, EXSYNC must be supplied. • Refer to section 4.5.1 for more information on how to operate your camera in TDI or Area Mode. • Exposure Modes are saved separately for TDI Mode and Area Mode. Refer to section 4.6 Saving and Restoring Settings for more information on how to save camera settings.
Related Commands:	ssf
Example:	<code>sem 3</code>

Table 10: Piranha HS Exposure Modes

Mode	SYNC	Programmable Frame Rate		Programmable Exposure Time		Description
		↓	↓	↓	↓	
3	External	No	No			Maximum exposure time with no charge reset.
7	Internal	Yes	No			Internal sync, maximum exposure time with no charge reset.

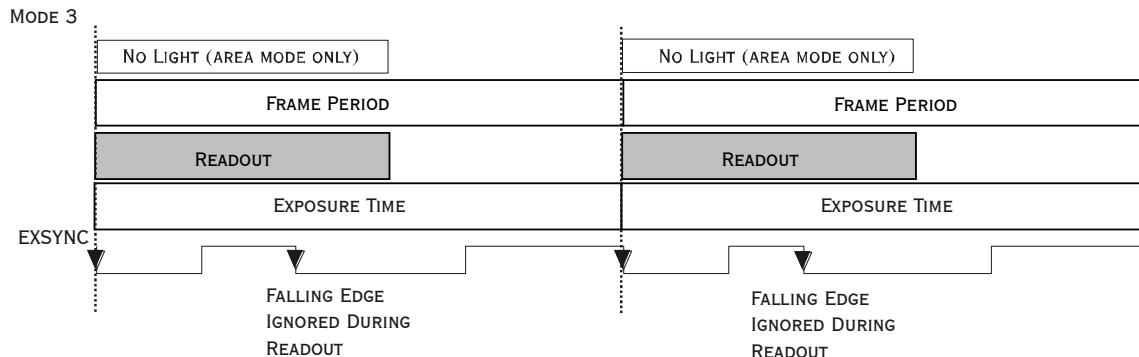
Exposure Modes in Detail

When setting the frame rate (using the `ssf` command), exposure time will be reduced, if necessary, to accommodate the new frame rate. The exposure time will always be set to the maximum time (frame period – frame transfer time – pixel reset time) for that frame rate when a new frame rate requiring reduced exposure time is entered.

When setting the exposure time (using the `set` command), frame time will be increased, if necessary, to accommodate the exposure time. Under this condition, the frame period will equal the Readout+Exposure Pulse+Exposure Time.

Frame rate is set by the period of the external trigger pulses. EXSYNC pulses faster than the read out time are ignored. The falling edge of EXSYNC marks the start of readout.

Note: In TDI mode the frame period equals the line period.

Mode 3: External Trigger, Maximum Exposure Time**Figure 23: Mode 3 Timing**

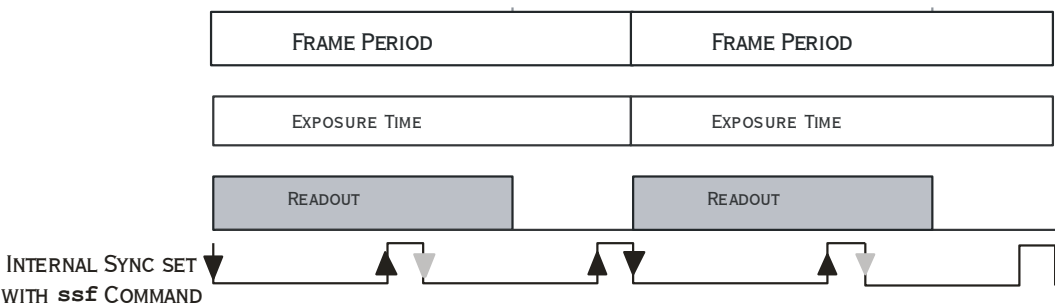
In this mode, the external trigger rate sets both the frame rate and the exposure time. The rising pulse of the external trigger marks the beginning of the exposure time. Readout begins on the falling edge of EXSYNC. A charge dump occurs at the beginning of the exposure time, clearing any accumulated charge gathered up to this point. An external shutter or strobe should be used to remove light during readout.

Mode 7: Internal Frame Rate, Maximum Exposure Time

In this mode, the frame rate is set internally using the `ssf` command with a maximum exposure time.

Note: In TDI mode the frame period equals the line period.

Figure 24: Mode 7 Camera Timing



Setting Frame Rate

Purpose:	Sets the camera’s frame rate in Hz. Camera must be operating in exposure mode 7.
Syntax:	ssf i
Syntax Elements:	i Set the frame rate to a value from: TDI HS-40-4k40: 3499.87-36563.1 HS-80-4k40: 3499.87-68610 HS-80-8k40: 3499.56-34305.3 HS-80-8k80: 3499.87-68610.6 HS-82-4k80: 3499.87-111111 Area HS-40-4k40: 1-3783.58 HS-80-4k40: 1-6169 HS-80-8k40: 1-3084.52 HS-80-8k80: 1-6169.03 HS-82-4k80: 1-9017.13 Value rounded up/ down as required. The maximum line/ frame rate is affected by horizontal and vertical binning factors, throughput setting, Camera Link mode, and number of CCD integration stages.
Notes:	• If you enter an invalid frame rate frequency the value, the camera clips the frame rate to be within the current operating range and a warning message is returned . • If you enter a frame rate frequency outside of the range displayed on the help screen, an error message is returned and the frame rate remains unchanged. • The camera does not automatically change the frame rate after you change binning or stage selection values. You may have to adjust your frame rate to avoid ignored syncs. • To return the camera’s frame rate, use the commad gcp or get ssf.
Related Commands:	sem
Example:	ssf 10000

4.4 Camera Output Format

4.4.1 How to Configure Camera Output

The Piranha HS cameras offer great flexibility when configuring your camera output. Using the [clm](#) command, you determine the camera's Camera Link configuration, number of output taps, and bit depth. Using the [sot](#) command, you determine the camera's output rate. These two commands work together to determine your final camera output configuration.

You can further configure your readout using the [smm](#) command to select the camera's pixel readout direction.

The following tables summarize the possible camera configurations for each of the HS-xx camera models. Refer to the figure below for a description on how to select your camera output.

Figure 25: How to Read the Camera Link Tables

A Find the rows in the table with a line rate greater than or equal to your desired line rate.

C From the rows determined by part B, find the Camera Link modes with your desired bit depth.

Camera Link Mode Configuration (Controlled by clm command)				Readout Direction (Controlled by smm command)	Pixel Rate Configuration (Controlled by sot command)
Command	Camera Link Configuration	Camera Link Taps	Bit Depth	smm 0 increment = 1 smm 1 increment = -1	
clm 2	Base	2 Camera Link taps where: 1 = CCD taps 1 + 2 2 = CCD taps 3 + 4	8	smm 0 = CL tap 1 (1-2048) CL tap 2 (2049-4096) smm 1 = CL tap 1 (4096-2049) CL tap 2 (2048-1)	sot 160 = 80 MHz strobe (max line rate 36kHz) sot 80 = 40 MHz strobe (Max line rate 18kHz)

D From the remaining rows determined in part C, select an acceptable Camera Link configuration. If none of the remaining configurations are acceptable for your system, you will have to reduce your bit depth or line rate.

E Set the pixel readout direction. Note: Horizontal binning reduces the number of pixels sent to the frame grabber by the binning factor.

B From the rows determined in part A, find the sot values in the table with a throughput greater than or equal to your required throughput.

Note: In the following tables, a CCD tap refers to the actual physical taps on the sensor, while the Camera Link taps refer to the way the data is configured for output over Camera Link. For a diagram illustrating sensor taps, see section 1.3 Image Sensor.

Table 11: HS-40-04k40 Data Readout Configurations

Camera Link Mode Configuration (Controlled by clm command)				Readout Direction (Controlled by smm command)	Pixel Rate Configuration (Controlled by sot command)
Command	Camera Link Configuration	Camera Link Taps	Bit Depth	smm 0 increment = 1 smm 1 increment = -1	
clm 2	Base	2 Camera Link taps where: 1 = CCD taps 1+2 2 = CCD taps 3+4	8	smm 0 = CL tap 1 (1-2048) CL tap 2 (2049-4096) smm 1 = CL tap 1 (4096-2049) CL tap 2 (2048-1)	sot 160 = 80 MHz strobe (max line rate 36563Hz) sot 80 = 40 MHz strobe (Max line rate 19166Hz)
clm 3	Base	2 Camera Link taps where: 1 = CCD taps 1+2 2 = CCD taps 3+4	12	smm 0 = CL tap 1 (1-2048) CL tap 2 (2049-4096) smm 1 = CL tap 1 (4096-2049) CL tap 2 (2048-1)	sot 160 = 80 MHz strobe (max line rate 36563Hz) sot 80 = 40 MHz strobe (Max line rate 19166Hz)
clm 15	Medium	4 Camera Link taps where: 1 = CCD tap 1 2 = CCD tap 2 3 = CCD tap 3 4 = CCD tap 4	8	smm 0 = CL tap 1 (1-1024) CL tap 2 (1025-2048) CL tap 3 (2049-3072) CL tap 4 (3073-4096) smm 1 = CL tap 1 (4096-3073) CL tap 2 (3072-2049) CL tap 3 (2048-1025) CL tap 4 (1024-1)	sot 160 = 40 MHz strobe (max line rate 36563Hz)
clm 16	Medium	4 Camera Link taps where: 1 = CCD tap 1 2 = CCD tap 2 3 = CCD tap 3 4 = CCD tap 4	12	smm 0 = CL tap 1(1-1024) CL tap 2(1025-2048) CL tap 3(2049-3072) CL tap 4(3073-4096) smm 1 = CL tap 1(4096-3073) CL tap 2(3072-2049) CL tap 3(2048-1025) CL tap 4(1024-1)	sot 160 = 40 MHz strobe (max line rate 36563Hz)

Table 12: HS-80-04k40 Configurations

Camera Link Mode Configuration (Controlled by clm command)				Readout Direction (Controlled by smm command)	Pixel Rate Configuration (Controlled by sot command)
Command	Camera Link Configuration	Camera Link Taps	Bit Depth		
clm 15	Medium	4 Camera Link taps where: 1 = CCD tap 1+2 2 = CCD tap 3+4 3 = CCD tap 5+6 4 = CCD tap 7+8	8	smm 0 = CL tap 1(1-1024) CL tap 2(1025-2048) CL tap 3(2049-3072) CL tap 4(3073-4096) smm 1 = CL tap 1(4096-3073) CL tap 2(3072-2049) CL tap 3(2048-1025) CL tap 4(1024-1)	sot 160 = 40 MHz strobe (max line rate 37629Hz) sot 320 = 80 MHz strobe (max line rate 68610Hz)
clm 16	Medium	4 Camera Link taps where: 1 = CCD tap 1+2 2 = CCD tap 3+4 3 = CCD tap 5+6 4 = CCD tap 7+8	12	smm 0 = CL tap 1(1-1024) CL tap 2(1025-2048) CL tap 3(2049-3072) CL tap 4(3073-4096) smm 1 = CL tap 1(4096-3073) CL tap 2(3072-2049) CL tap 3(2048-1025) CL tap 4(1024-1)	sot 160 = 40 MHz strobe (max line rate 37629Hz) sot 320 = 80 MHz strobe (max line rate 68610Hz)
clm 21	Full	8 Camera Link taps where: 1 = CCD tap 1 2 = CCD tap 2 3 = CCD tap 3 4 = CCD tap 4 5 = CCD tap 5 6 = CCD tap 6 7 = CCD tap 7 8 = CCD tap 8	8	smm 0 = CL tap 1(1-512) CL tap 2(513-1024) CL tap 3(1025-1536) CL tap 4(1537-2048) CL tap 5(2049-2560) CL tap 6(2561-3072) CL tap 7(3073-3584) CL tap 8(3585-4096) smm 1 = CL tap 1(4096-3585) CL tap 2(3584-3073) CL tap 3(3072-2561) CL tap 4(2560-2049) CL tap 5(2048-1537) CL tap 6(1536-1025) CL tap 7(1024-513) CL tap 8(512-1)	sot 320 = 40 MHz strobe (max line rate 68610Hz)
clm 15	Medium	4 Camera Link taps where: 1 = CCD tap 1+2 2 = CCD tap 3+4 3 = CCD tap 5+6 4 = CCD tap 7+8	8	smm 0 = CL tap 1(1-1024) CL tap 2(1025-2048) CL tap 3(2049-3072) CL tap 4(3073-4096) smm 1 = CL tap 1(4096-3073) CL tap 2(3072-2049) CL tap 3(2048-1025) CL tap 4(1024-1)	sot 160 = 40 MHz strobe (max line rate 37629Hz) sot 320 = 80 MHz strobe (max line rate 68610Hz)

Table 13: HS-80-08k40 Configurations

Camera Link Mode Configuration (Controlled by clm command)				Readout Direction (Controlled by smm command)	Pixel Rate Configuration (Controlled by sot command)
Command	Camera Link Configuration	Camera Link Taps	Bit Depth		
clm 15	Medium	4 Camera Link taps where: 1 = CCD tap 1+2+3+4 2 = CCD tap 5+6+7+8 3 = CCD tap 9+10+11+12 4 = CCD tap 13+14+15+16	8	smm 0 = CL tap 1(1-2048) CL tap 2(2049-4096) CL tap 3(4097-6144) CL tap 4(6145-8192) smm 1 = CL tap 1(8192-6145) CL tap 2(6144-4097) CL tap 3(4096-2049) CL tap 4(2048-1)	sot 80 = 20 MHz strobe (max line rate 9583Hz) sot 160 = 40 MHz strobe (max line rate 19157Hz)
clm 16	Medium	4 Camera Link taps where: 1 = CCD tap 1+2+3+4 2 = CCD tap 5+6+7+8 3 = CCD tap 9+10+11+12 4 = CCD tap 13+14+15+16	12	smm 0 = CL tap 1(1-2048) CL tap 2(2049-4096) CL tap 3(4097-6144) CL tap 4(6145-8192) smm 1 = CL tap 1(8192-6145) CL tap 2(6144-4097) CL tap 3(4096-2049) CL tap 4(2048-1)	sot 80 = 20 MHz strobe (max line rate 9583Hz) sot 160 = 40 MHz strobe (max line rate 19157Hz)
clm 21	Full	8 Camera Link taps where: 1 = CCD tap 1+2 2 = CCD tap 3+4 3 = CCD tap 5+6 4 = CCD tap 7+8 5 = CCD tap 9+10 6 = CCD tap 11+12 7 = CCD tap 13+14 8 = CCD tap 15+16	8	smm 0 = CL tap 1(1-1024) CL tap 2(1025-2048) CL tap 3(2049-3072) CL tap 4(3073-4096) CL tap 5(4097-5120) CL tap 6(5121-6144) CL tap 7(6145-7168) CL tap 8(7169-8192) smm 1 = CL tap 1(8192-7169) CL tap 2(7168-6145) CL tap 3(6144-5121) CL tap 4(5120-4097) CL tap 5(4096-3073) CL tap 6(3072-2049) CL tap 7(2048-1025) CL tap 8(1024-1)	sot 160 = 20 MHz strobe (max line rate 18814Hz) sot 320 = 40 MHz strobe (max line rate 34305Hz)
clm 15	Medium	4 Camera Link taps where: 1 = CCD tap 1+2+3+4 2 = CCD tap 5+6+7+8 3 = CCD tap 9+10+11+12 4 = CCD tap 13+14+15+16	8	smm 0 = CL tap 1(1-2048) CL tap 2(2049-4096) CL tap 3(4097-6144) CL tap 4(6145-8192) smm 1 = CL tap 1(8192-6145) CL tap 2(6144-4097) CL tap 3(4096-2049) CL tap 4(2048-1)	sot 80 = 20 MHz strobe (max line rate 9583Hz) sot 160 = 40 MHz strobe (max line rate 19157Hz)

Table 14: HS-80-08k80 Configurations

Camera Link Mode Configuration (Controlled by clm command)				Readout Direction (Controlled by smm command)	Pixel Rate Configuration (Controlled by sot command)
Command	Camera Link Configuration	Camera Link Taps	Bit Depth		
clm 15	Medium	4 Camera Link taps where: 1 = CCD tap 1+2+3+4 2 = CCD tap 5+6+7+8 3 = CCD tap 9+10+11+12 4 = CCD tap 13+14+15+16	8	smm 0 = CL tap 1(1-2048) CL tap 2(2049-4096) CL tap 3(4097-6144) CL tap 4(6145-8192) smm 1 = CL tap 1(8192-6145) CL tap 2(6144-4097) CL tap 3(4096-2049) CL tap 4(2048-1)	sot 320 = 80 MHz strobe (max rate 38314Hz) sot 160 = 40 MHz strobe (max line rate 19166Hz)
clm 16	Medium	4 Camera Link taps where: 1 = CCD tap 1+2+3+4 2 = CCD tap 5+6+7+8 3 = CCD tap 9+10+11+12 4 = CCD tap 13+14+15+16	12	smm 0 = CL tap 1(1-2048) CL tap 2(2049-4096) CL tap 3(4097-6144) CL tap 4(6145-8192) smm 1 = CL tap 1(8192-6145) CL tap 2(6144-4097) CL tap 3(4096-2049) CL tap 4(2048-1)	sot 320 = 80 MHz strobe (max line rate 38314Hz) sot 160 = 40 MHz strobe (max line rate 19166Hz)
clm 21	Full	8 Camera Link taps where: 1 = CCD tap 1+2 2 = CCD tap 3+4 3 = CCD tap 5+6 4 = CCD tap 7+8 5 = CCD tap 9+10 6 = CCD tap 11+12 7 = CCD tap 13+14 8 = CCD tap 15+16	8	smm 0 = CL tap 1(1-1024) CL tap 2(1025-2048) CL tap 3(2049-3072) CL tap 4(3073-4096) CL tap 5(4097-5120) CL tap 6(5121-6144) CL tap 7(6145-7168) CL tap 8(7169-8192) smm 1 = CL tap 1(8192-7169) CL tap 2(7168-6145) CL tap 3(6144-5121) CL tap 4(5120-4097) CL tap 5(4096-3073) CL tap 6(3072-2049) CL tap 7(2048-1025) CL tap 8(1024-1)	sot 640 = 80 MHz strobe (HS-80-08k80 only) (max line rate 68610Hz) sot 320 = 40 MHz strobe (max line rate 37629Hz)
clm 15	Medium	4 Camera Link taps where: 1 = CCD tap 1+2+3+4 2 = CCD tap 5+6+7+8 3 = CCD tap 9+10+11+12 4 = CCD tap 13+14+15+16	8	smm 0 = CL tap 1(1-2048) CL tap 2(2049-4096) CL tap 3(4097-6144) CL tap 4(6145-8192) smm 1 = CL tap 1(8192-6145) CL tap 2(6144-4097) CL tap 3(4096-2049) CL tap 4(2048-1)	sot 320 = 80 MHz strobe (max rate 38314Hz) sot 160 = 40 MHz strobe (max line rate 19166Hz)

4.4.2 Setting the Camera Link Mode

Purpose:	Sets the camera's Camera Link configuration, number of Camera Link taps and data bit depth. Refer to the tables on the previous pages to determine which configurations are valid for your camera model and how this command relates to other camera configuration commands.
Syntax:	clm <i>m</i>
Syntax Elements:	<i>m</i> Output mode to use: 2 : Base configuration, 2 taps, 8 bit output 3 : Base configuration, 2 taps, 12 bit output 15 : Medium configuration, 4 taps, 8 bit output 16 : Medium configuration, 4 taps, 12 bit output 21 : Full configuration, 8 taps, 8 bit output
Notes:	- When you change the Camera Link mode (clm command), the camera attempts to maintain the current sot throughput (pixels/ sec). If the current throughput is too slow or too fast for the current Camera Link mode, the camera will automatically adjust the throughput value and will return a warning message that a related parameter was adjusted. - To obtain the current Camera Link mode, use the command gcp or get clm. - The bit patterns are defined by the Teledyne DALSA Camera Link Roadmap available at Teledyne DALSA application notes.
Related Commands	sot
Example:	clm 15

4.4.3 Setting the Camera Throughput

Purpose:	Works in conjunction with the clm command (see previous) and determines the throughput of the camera. Refer to the tables in section 4.4.1 How to Configure Camera Output to determine which configurations are valid for your camera model and how this command relates to other camera configuration commands.
Syntax:	sot <i>m</i>
Syntax Elements:	<i>m</i> Output throughput. Allowable values are: 80 = 4 taps at 20MHz or 2 taps at 40MHz 160 = 2 taps at 80MHz or 4 taps at 40MHz 320 = 4 taps at 80MHz or 8 taps at 40MHz 640 = 8 taps at 80MHz
Notes:	Throughput is calculated as: Throughput= (Number of Camera Link Taps) x (Camera Link Pixel Rate in MHz) - To obtain the throughput setting, use the command gcp or get clm. - Throughput values are clipped if the camera is unable to maintain the current throughput setting and a warning message is displayed. - Refer to the tables in section 4.4.1 How to Configure Camera Output to determine which configurations are valid for your camera model.
Related Commands	clm
Example:	sot 160

4.4.4 Setting the Pixel Readout Direction

Purpose:	Sets the tap readout from left to right or from right to left. This command is useful if the camera must be mounted upside down.
Syntax:	smm <i>i</i>
Syntax Elements:	<i>i</i> Readout direction. Allowable values are: 0 = All pixels are read out from left to right. 1 = All pixels are read out from right to left.
Notes:	- To obtain the current readout direction, use the command gcp or get smm. - This command is available in both TDI and Area Mode. - Refer to the following figures and tables for an explanation of pixel readout and mirror direction. - Refer to section 1.3 Image Sensor for sensor architecture diagrams that illustrate sensor readout direction.

Table 16: HS-80-04k40 Forward or Reverse Pixel Readout

Readout Direction	Tap 1	Tap 2	Tap 3	Tap 4	Tap 5	Tap 6	Tap 7	Tap 8
Left to Right Software command smm 0	1-512	513-1024	1025-1536	1537-2048	2049-2560	2561-3072	3073-3584	3585-4096
Right to Left Software command smm 1	4096-3585	3584-3073	3072-2561	2560-2049	2048-1537	1536-1025	1024-513	512-1

Table 17: HS-80-08k40, HS-80-08k80 Forward or Reverse Pixel Readout

Readout Direction	Tap 1	Tap 2	Tap 3	Tap 4	Tap 5	Tap 6	Tap 7	Tap 8
Left to Right Software command smm 0	1-1024	1025-2048	2049-3072	3073-4096	4097-5120	5121-6144	6145-7168	7169-8192
Right to Left Software command smm 1	8192-7169	7168-6145	6144-5121	5120-4097	4096-3073	3072-2049	2048-1025	1024-1

4.4.5 Setting a Pretrigger

Purpose: A pretrigger may be required for some frame grabbers.

Syntax: **spt i**

Syntax Elements: **i**

Pretrigger in a range from 0 to 16.

4.5 Data Processing

4.5.1 Setting a Region of Interest

Purpose: Sets the pixel range used to collect the end-of-line statistics and sets the region of pixels used in the [ccg](#), [cao](#), [cpa](#), [gl](#), [gla](#), [ccf](#), and [ccp](#) commands.

In most applications, the field of view exceeds the required object size and these extraneous areas should be ignored. It is recommended that you set the region of interest a few pixels inside the actual useable image.

Syntax: **roi x1 y1 x2 y2**

Syntax Elements: **x1**

Column start number. Must be less than or equal to the column end number in a range from 1 to (column resolution – 1).

y1

Row start number. Must be less than or equal to the row end number in a range from 1 to (row end number – 1) except in TDI Mode where **y1** must be 1.

x2

Column end number. Must be greater than or equal to the

column start number in a range from 2 to column resolution.

y2

Row end number. Must be greater than or equal to the row start number in a range from 2 to number of stages except in TDI Mode where **y2** must be 1.

Notes:

- If you are using binning, the start pixel is rounded down to the beginning of binned area and end pixel is rounded up to the end of the binned area.
- In Area Mode, the roi must be within the stage. If the requested roi is above the stage, the roi rows will be clipped. The start and end rows will be clipped to the stage selection if necessary. A “clipped to max” warning message is returned.

Related Commands

[ccg](#), [cao](#), [cpa](#), [gl](#), [gla](#), [ccf](#), [ccp](#), [els](#)

Example:

roi 10 1 50 1 (TDI Mode)

4.5.2 Analog and Digital Signal Processing Chain

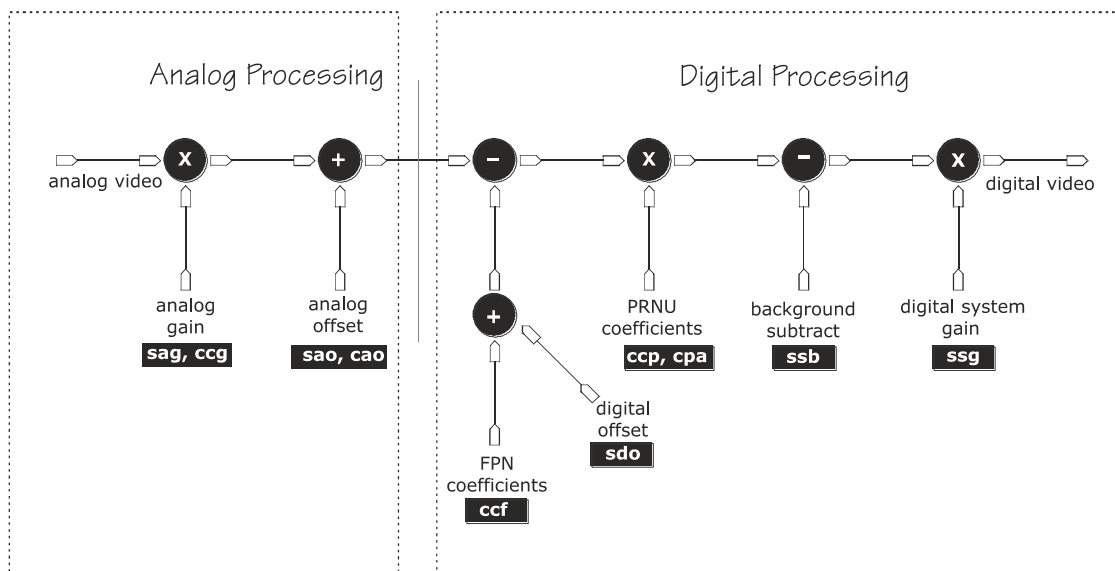
Processing Chain Overview and Description

The following diagram shows a simplified block diagram of the camera’s analog and digital processing chain. The analog processing chain begins with an analog gain adjustment, followed by an analog offset adjustment. These adjustments are applied to the video analog signal prior to its digitization by an A/ D converter.

The digital processing chain contains the FPN correction, the PRNU correction, the background subtract, and the digital gain and offset. All of these elements are user programmable.

Notes:

- FPN and PRNU correction is not available when operating the camera in Area Mode. For details on how to switch camera operation modes, refer to section 4.5.1 .
- The following user settings are stored separately for forward and reverse direction; analog gain, analog offset, digital gain, digital offset, and background subtract. They are saved using the [wus](#) command.
- FPN and PRNU coefficients are stored separately for forward and reverse direction. To save the current PRNU coefficients, use the command [wpc](#). To save the current FPN coefficients, use the command **wfc**. Settings are saved for the current direction only.

Figure 29: Signal Processing Chain

Analog Processing

Optimizing offset performance and gain in the analog domain allows you to achieve a better signal-to-noise ratio and dynamic range than you would achieve by trying to optimize the offset in the digital domain. As a result, perform all analog adjustments prior to any digital adjustments.

1. Analog gain is multiplied by the analog signal to increase the signal strength before the A/D conversion. It is used to take advantage of the full dynamic range of the A/D converter. For example, in a low light situation the brightest part of the image may be consistently coming in at only 50% of the DN. An analog gain of 6 dB (2x) will ensure full use of the dynamic range of the A/D converter. Of course the noise is also increased.
2. The analog offset or black level is an “artificial” offset introduced into the video path to ensure that the A/D will function properly. The analog offset should be set so that it is at least 3 times the rms noise value at the current gain.

Digital Processing

To optimize camera performance, digital signal processing should be completed after any analog adjustments.

1. Fixed pattern noise (FPN) calibration (calculated using the [ccf](#) command) is used to subtract away individual pixel dark current.
2. The digital offset ([sdo](#) command) enables the subtraction of the “artificial” A/D offset (the analog offset) so that application of the PRNU coefficient doesn’t result in artifacts at low light levels due to the offset value. You may want to set the [sdo](#) value if you are not using FPN correction but want to perform PRNU correction.
3. Photo-Response Non-Uniformity (PRNU) coefficients are used to correct the difference in responsivity of individual pixels (i.e. given the same amount of light different pixels will charge up at different rates) and the change in light intensity across the image either because of the light source or due to optical aberrations (e.g. there may be more light in the center of the image). PRNU coefficients are

multipliers and are defined to be of a value greater than or equal to 1. This ensures that all pixels will saturate together. When using PRNU correction, it is important that the A/ D offset and Fixed Pattern Noise (FPN) or per pixel offsets are subtracted prior to the multiplication by the PRNU coefficient. The subtraction of these 2 components ensure that the video supplied to the PRNU multiplier is nominally zero and zero multiplied by anything is still zero resulting in no PRNU coefficient induced FPN. If the offset is not subtracted from the video then there will be artifacts in the video at low light caused by the multiplication of the offset value by the PRNU coefficients.

4. Background subtract ([ssb](#) command) and system (digital) gain ([ssg](#) command) are used to increase image contrast after FPN and PRNU calibration. It is useful for systems that process 8-bit data but want to take advantage of the camera's 12-bit digital processing chain. For example, if you find that your image is consistently between 128 and 255DN(8-bit), you can subtract off 128 (**ssb 2048**) and then multiply by 2 (**ssg 0 8192**) to get an output range from 0 to 255.

The following sections are organized as follows:

1. Section Analog Signal Processing provides a detailed description of all analog processing chain commands.
2. Section Calibrating the Camera to Remove Non-Uniformity (Flat Field Correction) provides an overview of how to perform flat field calibration.
3. Section Digital Signal Processing provides a detailed description of all digital processing chain commands.

Analog Signal Processing: Setting Analog Gain and Offset

All analog signal processing chain commands should be performed prior to FPN and PRNU calibration and prior to digital signal processing commands.

Setting Analog Gain

Purpose:	Sets the camera's analog gain value. Analog gain is multiplied by the analog signal to increase the signal strength before the A/ D conversion. It is used to take advantage of the full dynamic range of the A/ D converter.
Syntax:	sag t f
Syntax Elements:	t Tap selection. Use 0 for all taps or 1 to number of CCD taps for individual tap selection. f Gain value in a range from -10 to +10dB.
Example:	sag 0 5.2

Calibrating Camera Gain

Purpose: Instead of manually setting the analog gain to a specific value, the camera can determine appropriate gain values. This command calculates and sets the analog gain according to the algorithm determined by the first parameter.

Syntax: `ccg i t i`

Syntax Elements: `i`

Calibration algorithm to use.

1 = This algorithm adjusts analog gain so that 8% to 13% of tap ROI pixels are above the specified target value.

2 = This algorithm adjusts analog gain so that the average pixel value in tap's ROI is equal to the specified target value.

3 = This algorithm adjusts digital gain so that the average pixel value in tap's ROI is equal to the specified target.

4 = This algorithm adjusts the analog gain so that the peak tap ROI pixels are adjusted to the specified target.

`t`

Tap value. Use **0** for all taps or **1** to number of CCD taps for individual tap selection.

`i`

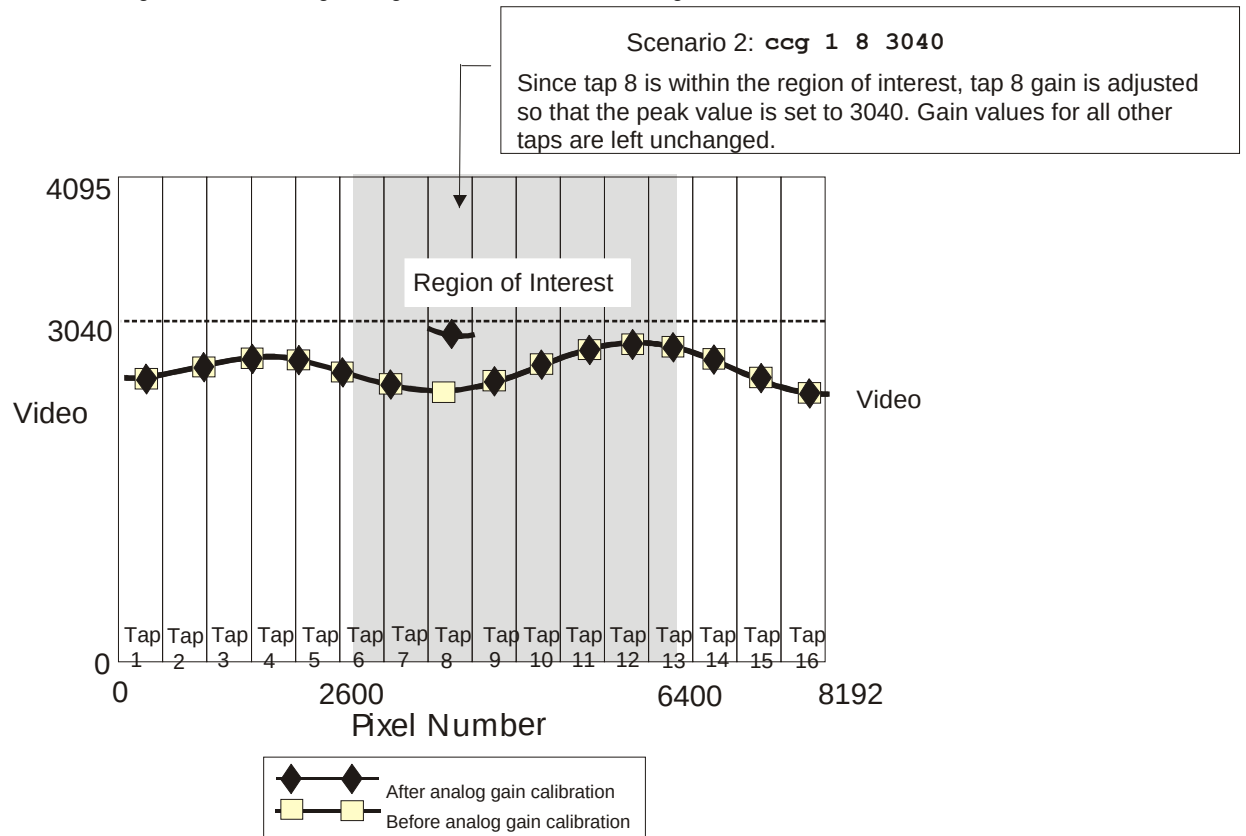
Calculation target value in a range from **1024** to **4055DN** (12 bit LSB).

- Notes:**
- All digital settings (digital offset, FPN and PRNU coefficients), digital gain, background subtract) should be turned off before calibrating analog gain to avoid unpredictable results.
 - This function requires constant light input while executing.
 - To use this command, the CCD shift direction ([scd](#)) should be set to forward (**0**) or reverse (**1**).
 - If very few tap pixels are within the ROI, gain calculation may not be optimal.
 - When all taps are selected, taps outside of the ROI are set to the average gain of the taps that are within the ROI.
 - Perform analog gain algorithms before performing FPN and PRNU calibration.

Example: `ccg 2 0 3040`

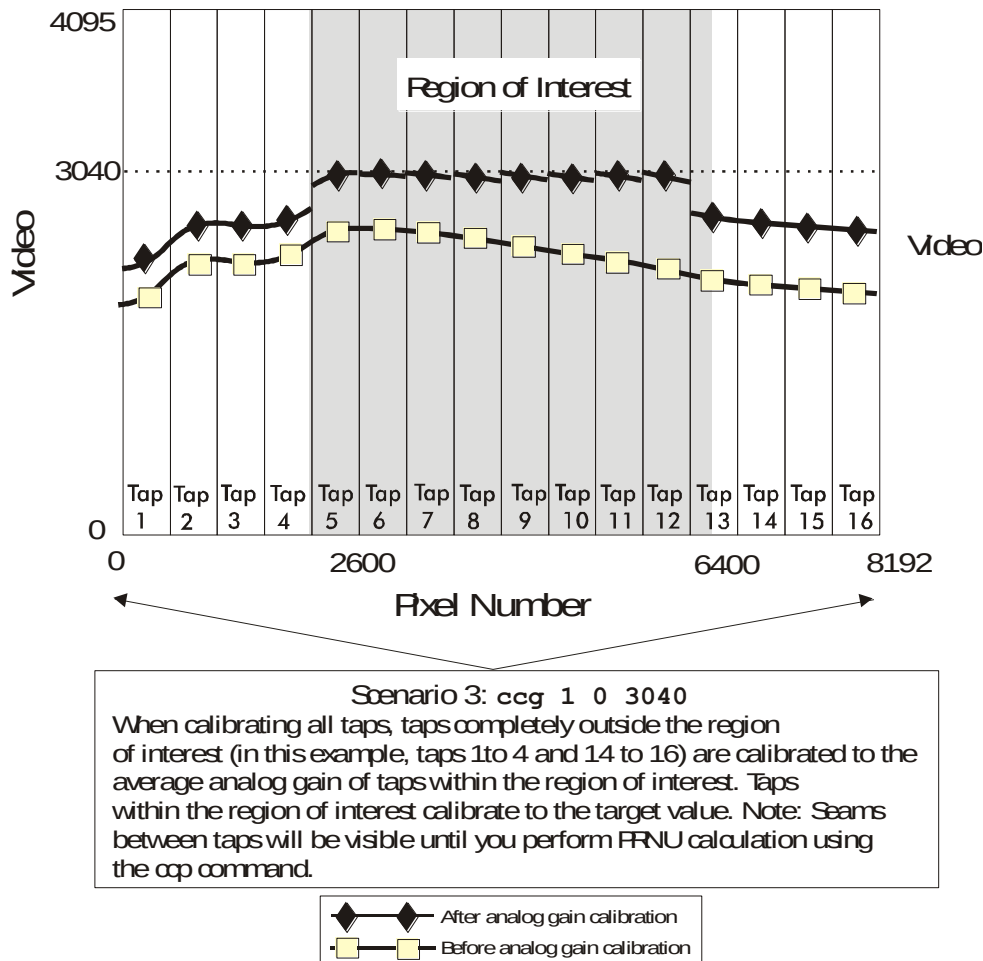
The following diagrams summarize and provide an example of how analog gain is calibrated when using a region of interest.

In the following example, analog gain is being set for a tap inside the region of interest. The peak value of the tap is calibrated to the specified target value and all other taps remain unchanged.

Figure 30: Calculating Analog Gain for a Tap inside the Region of Interest

In the following example, analog gain is set for all taps. The peak value of each tap within the region of interest is calibrated to the specified target value. All taps completely outside the region of interest are calibrated to the average analog gain value of the taps inside the region of interest.

Figure 31: Calculating Analog Gain for all Taps



Setting Analog Offset

Purpose:

Sets the analog offset. The analog offset should be set so that it is at least 3 times the rms noise value at the current gain. Teledyne DALSA configures the analog offset for the noise at the maximum specified gain and as a result you should not need to adjust the analog offset.

Syntax:

sao t i

Syntax Elements:

t

Tap selection. Use 0 for all taps or 1 to number of CCD taps for individual tap selection.

i

Offset value in a range from 0 to 255DN (12 bit LSB).

Example:

sao 3 35

Calibrating Analog Offset

Purpose:	Instead of manually setting the analog offset to a specific value, the camera can determine appropriate offset values. This command calculates and averages each tap's pixels within the ROI and sets the offset to achieve the specified average target value.
Syntax:	cao t i
Syntax Elements:	t Tap selection. Use 0 for all taps or 1 to number of CCD taps for individual tap selection. i Average target value in a range from 1 to 255DN (12 bit LSB).
Notes:	- Perform analog offset calibration before performing FPN and PRNU coefficients. - To use this command, CCD shift direction should be controlled internally, either scd 0 or 1
Example:	cao 1 50

Updating the Gain Reference

To update the analog gain reference:

Purpose:	Sets the current analog gain setting to be the 0dB point. This is useful after tap gain matching allowing you to change the gain on all taps by the same amount.
Syntax:	ugr

Calibrating the Camera to Remove Non-Uniformity (Flat Field Correction)

Flat Field Correction Overview

This camera has the ability to calculate correction coefficients in order to remove non-uniformity in the image when operating in TDI Mode. This video correction operates on a pixel-by-pixel basis and implements a two point correction for each pixel. This correction can reduce or eliminate image distortion caused by the following factors:

- Fixed Pattern Noise (FPN)
- Photo Response Non Uniformity (PRNU)
- Lens and light source non-uniformity

Correction is implemented such that for each pixel:

$$V_{\text{output}} = [(V_{\text{input}} - \text{FPN}(\text{pixel}) - \text{digital offset}) * \text{PRNU}(\text{pixel}) - \text{Background Subtract}] \times \text{System Gain}$$

where V_{output} = digital output pixel value

V_{input} = digital input pixel value from the CCD

PRNU(pixel)	=	PRNU correction coefficient for this pixel
FPN(pixel)	=	FPN correction coefficient for this pixel
Background Subtract	=	background subtract value
System Gain	=	digital gain value

The algorithm is performed in two steps. The fixed offset (FPN) is determined first by performing a calculation without any light. This calibration determines exactly how much offset to subtract per pixel in order to obtain flat output when the CCD is not exposed.

The white light calibration is performed next to determine the multiplication factors required to bring each pixel to the required value (target) for flat, white output. Video output is set slightly above the brightest pixel (depending on offset subtracted).

Flat Field Correction Restrictions

It is important to do the FPN correction first. Results of the FPN correction are used in the PRNU procedure. We recommend that you repeat the correction when a temperature change greater than 10°C occurs or if you change the analog gain, integration time, binning, or number of integration stages.

Note: If your illumination or white reference does not extend the full field of view of the camera, the camera will send a warning.

PRNU correction requires a clean, white reference. The quality of this reference is important for proper calibration. White paper is often not sufficient because the grain in the white paper will distort the correction. White plastic or white ceramic will lead to better balancing.

For best results, ensure that:

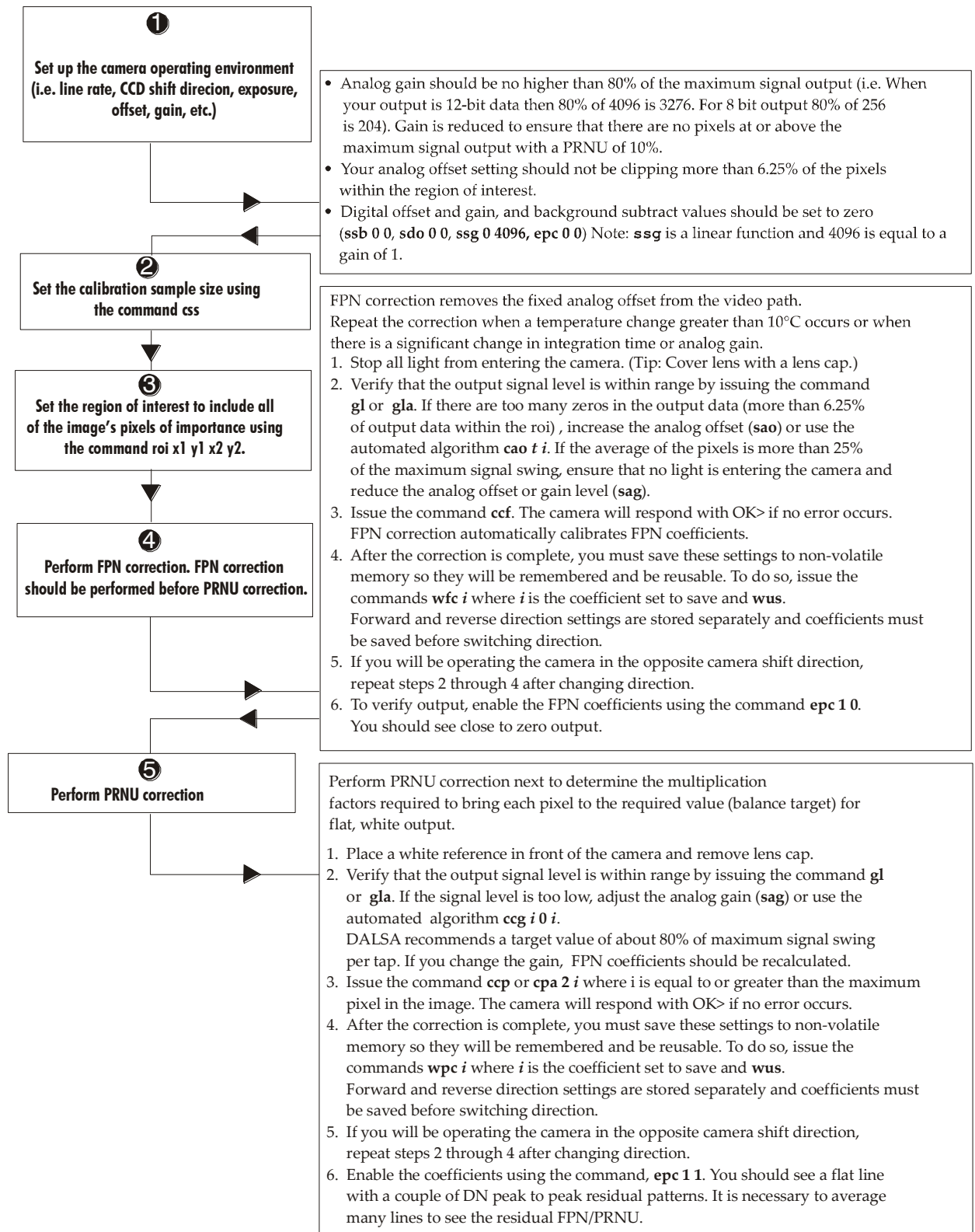
1. 60 Hz ambient light flicker is sufficiently low not to affect camera performance and calibration results.
2. For best results, the analog gain should be adjusted for the expected operating conditions and the ratio of the brightest to darkest pixel in a tap should be less than 3 to 1 where:

$$3 > \frac{\text{Brightest Pixel (per tap)}}{\text{Darkest Pixel (per tap)}}$$

The camera is capable of operating under a range of 8 to 1, but will clip values larger than this ratio.

3. The brightest pixel should be slightly below the target output.
4. When 6.25% of pixels from a single row within the region of interest are clipped, flat field correction results may be inaccurate.
5. Correction results are valid only for the current stage selection. If you change the number of stages, it is recommended that you recalculate your coefficients.
6. Correction results are valid only for the current analog gain and offset values. If you change these values, it is recommended that you recalculate your coefficients.

How to Perform Flat Field Correction



Digital Signal Processing

Subtracting Background

Purpose:	Use the background subtract command after performing flat field correction if you want to improve your image in a low contrast scene. You should try to make your darkest pixel in the scene equal to zero.
Syntax	ssb t i
Syntax Elements:	t Tap selection. Allowable range is 1 to number of CCD taps, or 0 for all taps. i Subtracted value in a range in DN from 0 to 4095 (12 bit LSB).
Notes:	See the following section for details on the ssg command.
Related Commands	ssg
Example	ssb 0 500

Setting Digital Gain

Purpose:	Improves signal output swing after a background subtract. When subtracting a digital value from the digital video signal, using the ssb command, the output can no longer reach its maximum. Use the this command to correct for this where: $\text{ssg value} = \frac{\text{max output value}}{\text{max output value} - \text{ssb value}}$
Syntax:	ssg t i
Syntax Elements:	t Tap selection. Allowable range is 1 to number of CCD taps, or 0 for all taps. i Gain setting. The gain ranges are 0 to 65535. The digital video values are multiplied by this value where: $\text{Digital Gain} = \frac{i}{4096}$
Notes:	- Use this command in conjunction with the ssb command (described above). - Teledyne DALSA recommends that i is never set below 4096. Setting i to 0 will result in only 0 output data. - Digital offset is set to zero after sending the ccf command
Related Commands:	ssb , sdo
Example:	ssg 1 4500

Setting Digital Offset

Purpose:	Sets the digital offset. Digital offset is set to zero when you perform FPN correction (ccf command). If you are unable to perform FPN correction, you can partially remove FPN by adjusting the digital offset.
Syntax:	sdo t i
Syntax Elements:	t Tap selection. Allowable range is 1 to number of CCD taps, or 0 for all taps. i Subtracted offset value in a range from 0 to 511 (12-bit LSB).
Notes:	When subtracting a digital value from the digital video signal, the output can no longer reach its maximum unless you apply digital gain using the ssg command. See the previous section for details on the ssg command.
Related Commands:	ssg , ccf
Example:	sdo 0 100

Performing FPN Correction

Syntax:	Performs FPN correction and eliminates FPN noise by subtracting away individual pixel dark current. For a complete description on how to use this command, see the Flat Field Correction Overview on page 57.
Syntax:	ccf
Notes:	- Before performing this command, stop all light from entering the camera. (Tip: cover lens with a lens cap.) - Perform all analog and digital adjustments before performing FPN correction. - Perform FPN correction before PRNU correction. - The ccf command is not available when the CCD direction is externally controlled (scd 2) (see Direction Control on page 21). Direction control must be stable while the camera is calculating coefficients. - Available in TDI Mode only. - Save coefficients before changing directions, changing operating mode, or powering off.
Related Commands:	ccp , cpa
Example:	ccf

Setting a Pixel's FPN Coefficient

Purpose:	Sets an individual pixel's FPN coefficient.
Syntax	sfc x i
Syntax Elements:	x The pixel number from 1 to sensor pixel count. i Coefficient value in a range from 0-511 (12-bit LSB).
Notes:	Available in TDI Mode only.
Example:	sfc 10 50

Returning FPN Coefficients

Purpose:	Returns a pixel's FPN coefficient value in DN (12-bit LSB)
Syntax:	gfc i
Syntax Elements:	i The pixel number to read in a range from 1 to sensor pixel count.
Notes:	Available in TDI Mode only.
Example:	gfc 10

Setting a Range of FPN Coefficients

Purpose:	Sets a range of pixel FPN coefficients.
Syntax	sfr x1 x2 i
Syntax Elements:	x1 The first pixel number of the range. x2 The last pixel number of the range. i Coefficient value in a range from 0-2048.
Notes:	The first pixel of the range must be less than the last.
Example:	sfr 1 100 80

Performing PRNU Correction

Purpose:	Performs PRNU correction to a camera calibrated peak value and eliminates the difference in responsivity between the most and least sensitive pixel creating a uniform response to light. For a complete description on how to use this command, see the Flat Field Correction Overview on page 57.
Syntax	ccp
Notes:	• Perform all analog adjustments before calculating PRNU. • Perform FPN correction before PRNU correction. • If FPN cannot be calibrated, use the rpc command to reset all coefficients to zero, and save them to memory with the wpc command. You can then adjust the digital offset (sdo command) to remove some of the FPN. • The ccp command is not available when the camera shift direction is externally controlled. Direction control must be stable while the camera is calculating coefficients (see Direction Control on page 21). • Ensure camera is operating at its expected analog gain, integration time, and temperature. • To avoid losing your current direction coefficients, you must save the PRNU coefficients using the command wpc before changing camera shift direction or changing from TDI to Area Mode. • Available in TDI Mode only. • Executing these algorithms causes the ssb command to be set to 0 (no background subtraction) and the ssg command to 4096 (unity digital gain). The pixel coefficients are disabled (epc 0 0) during the algorithm execution but returned to the state they were prior to command execution.

Performing PRNU to a user entered value

Purpose:	Performs PRNU calibration to user entered value and eliminates the difference in responsivity between the most and least sensitive pixel creating a uniform response to light. Using this command, you must provide a calibration target. Executing these algorithms causes the ssb command to be set to 0 (no background subtraction) and the ssg command to 4096 (unity digital gain). The pixel coefficients are disabled (epc 0 0) during the algorithm execution but returned to the state they were prior to command execution.
Syntax:	cpa i1 i2
Syntax Elements:	i1

PRNU calibration algorithm to use:

1 = This algorithm first adjusts each tap's analog gain so that 8-13% of pixels within a tap are above the value specified in the target value parameter. PRNU calibration then occurs using the peak pixel in the region of interest.

This algorithm is recommended for use only when FPN is negligible and FPN coefficients are set to zero. Since this

algorithm adjusts the analog gain, it also affects FPN. If FPN is calibrated prior to running this algorithm, FPN will be observable in dark conditions and an incorrect FPN value will be used during PRNU calibration resulting in incorrect PRNU coefficients.

2 = Calculates the PRNU coefficients using the entered target value as shown below:

$$\text{PRNU Coefficient}_i = \frac{\text{Target}}{(\text{AVG Pixel Value}_i) - (\text{FPN}_i + \text{sdo value})}$$

The calculation is performed for all sensor pixels but warnings are only applied to pixels in the region of interest. This algorithm is useful for achieving uniform output across multiple cameras. It is important that the target value (set with the next parameter) is set to be at least equal to the highest pixel across all cameras so that all pixels can reach the highest pixel value during calibration.

3 = This algorithm includes an analog gain adjustment prior to PRNU calibration. Analog gain is first adjusted so that the peak pixel value in tap's ROI is within 97 to 99% of the specified target value. It then calculates the PRNU coefficients using the target value as shown below:

$$\text{PRNU Coefficient}_i = \frac{\text{Target}}{(\text{AVG Pixel Value}_i) - (\text{FPN}_i + \text{sdo value})}$$

The calculation is performed for all sensor pixels but warnings are only applied to pixels in the region of interest. This algorithm is useful for achieving uniform output across multiple cameras.

This algorithm is useful for achieving uniform output across multiple cameras by first adjusting analog gain and then performing PRNU calibration. This algorithm is recommended for use only when FPN is negligible and FPN coefficients are set to zero. Since this algorithm adjusts the analog gain, it also affects FPN. If FPN is calibrated prior to running this algorithm, FPN will be observable in dark conditions and an incorrect FPN value will be used during PRNU calibration resulting in incorrect PRNU coefficients.

This algorithm is more robust and repeatable than algorithm 1 because it uses an average pixel value rather than a number above target. However, this algorithm is slower.

4 = Calculates the PRNU coefficient in the same way as **cpa 2** with the exception that this command only calculates PRNU for pixels within the current Region of Interest (ROI).

i2

Peak target value in a range from 1024 to 4055DN. The target value must be greater than the current peak output value.

Notes:

- Perform all analog adjustments before calibrating PRNU.
- This command performs the same function as the **cpp** command but forces you to enter a target value.
- Calibrate FPN before calibrating PRNU. If you are not performing FPN calibration then issue the **rpc** (reset pixel coefficients) command and set the **sdo** (set digital offset) value

so that the output is near zero under dark.

Example: **cpa 1 600**

Setting a Pixel's PRNU Coefficient

Purpose: Sets an individual pixel's PRNU coefficient.

Syntax: **spc i1 i2**

Syntax Elements: **i1**

The pixel number from 1 to sensor pixel count.

i2

Coefficient value in a range from 0 to 28671 where:

$$\text{prnu coefficient} = 1 + \frac{i}{4096}$$

Setting a range of Pixel PRNU Coefficients

Purpose: Sets a range of pixel PRNU coefficients.

Syntax: **spr i1 i2 x**

Syntax Elements: **i1**

The first pixel number of the range.

i2

The last pixel number of the range.

x

Coefficient value in a range from 0 to 28671 where:

$$\text{prnu coefficient} = 1 + \frac{i}{4096}$$

Notes:

- The first pixel of the range must be less than the last.

Example: **spr 4001 4096 0**

Returning FPN and PRNU Coefficients

Purpose: Returns all the current pixel coefficients in the order FPN, PRNU, FPN, PRNU... for the range specified by **x1** and **x2**. The camera also returns the pixel number with every fifth coefficient.

Syntax: **dpc x1 x2**

Syntax Elements: **x1**

Start pixel to display in a range from 1 to (sensor pixel count-1).

x2

End pixel to display in a range from **x1** +1 to sensor pixel count.

Notes:

- If **x2 < x1** then **x2** is forced to be **x1**.

Example: `dpc 10 20`

Enabling and Disabling Pixel Coefficients

Purpose: Enables and disables FPN and PRNU coefficients.

Syntax: `epc i1 i2`

Syntax Elements: `i1`

FPN coefficients.

0 = FPN coefficients disabled

1 = FPN coefficients enabled

`i2`

PRNU coefficients.

0 = PRNU coefficients disabled

1 = PRNU coefficients enabled

Example: `epc 0 1`

4.5.3 End-of-line Sequence

Purpose: Produces an end-of-line sequence that provides basic calculations including "frame counter", "line sum", "pixels above threshold", "pixels below threshold", and "derivative line sum" within the region of interest. These basic calculations are used to calibrate analog offset ([cao](#)) and calibrate analog gain ([ccg](#)). To further aid in debugging and cable/ data path integrity, the first three pixels after Line Valid are "aa", "55", "aa". Refer to the following table. These statistics refer only to pixels within the region of interest.

Syntax: `els i`

Syntax Elements: `i`

0 Disable end-of-line sequence

1 Enable end-of-line sequence

Notes:

- LVAL is not high during the end-of-line statistics.

Example: `els 1`

Table 18: End-of-Line Sequence Description

Location	Value	Description
1	A's	By ensuring these values consistently toggle between "aa" and "55", you can verify cabling (i.e. no stuck bits)
2	S's	
3	A's	
4	4 bit counter LSB justified (Upper 4 bits are row counter in Area Mode)	Counter increments by 1. Use this value to verify that every line is output
5	Line sum (7...0)	LSB justified. Use these values to help calculate line average and gain
6	Line sum (15...8)	

Location	Value	Description
7	Line sum (23...16)	
8	Line sum (31...24)	
9	Line sum (39...32)	
10	Pixels above threshold (7...0)	Monitor these values (either above or below threshold) and adjust camera digital gain and background subtract to maximize scene contrast. This provides a basis for automatic gain control (AGC)
11	Pixels above threshold (15...8)	
12	Pixels above threshold (23...16)	
13	Pixels below threshold (7...0)	
14	Pixels below threshold (15...8)	
15	Pixels below threshold (23...16)	
16	Differential line sum (7..0)	Use these values to focus the camera. Generally, the greater the sum the greater the image contrast and better the focus.
17	Differential line sum (15...8)	
18	Differential line sum (23...16)	
19	Differential line sum (31...24)	
20	Differential line sum (39...32)	

Setting Thresholds

Setting an Upper Threshold

Purpose:	Sets the upper threshold limit to report in the end-of-line sequence.
Syntax:	sut i
Syntax Elements:	i Upper threshold limit in range from 0 to 4095 .
Notes:	LVAL is not high during the end-of-line statistics.
Related Commands:	els , slt
Example:	sut 1024

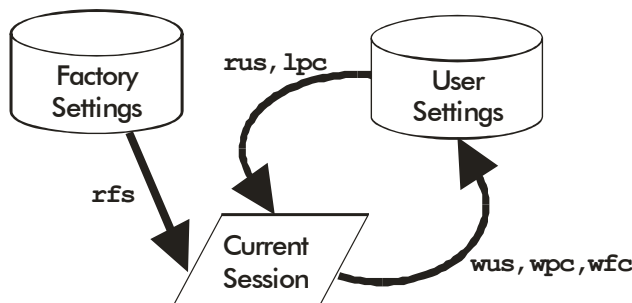
Setting a Lower Threshold

Purpose:	Sets the lower threshold limit to report in the end-of-line sequence.
Syntax:	slt i
Syntax Elements:	i
Notes:	Upper threshold limit in range from 0 to 4095. LVAL is not high during the end-of-line statistics.
Related Commands:	els , sut
Example:	slt 1024

4.6 Saving and Restoring Settings

4.6.1 Saving and Restoring Factory and User Settings

Figure 32: Saving and Restoring Overview

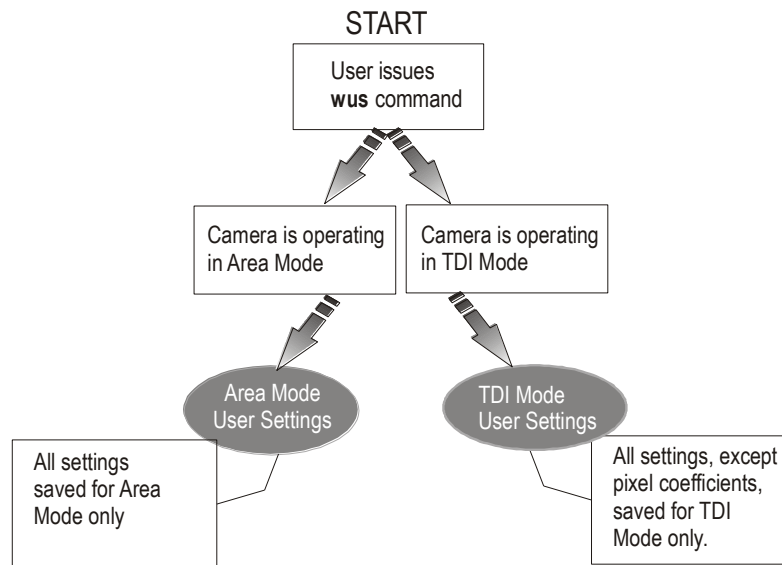


Factory Settings

You can restore the original factory settings, including the factory calibrated pixel coefficient set, at any time using the command **rfs**.

User Settings

There are two main sets of user settings: Area Mode user settings and TDI Mode user settings. After issuing the user settings save command, **wus**, settings are saved depending on which mode the camera is operating in when the command is issued. Also, when operating in TDI Mode, analog gain and offset, digital gain and offset, and background subtract values are saved as distinct values for Forward and Reverse directions. In other words, you can program the camera to operate with an analog gain value of +5db in Forward direction and an analog gain value of +3db in Reverse direction. Forward and Reverse direction settings are saved simultaneously with the **wus** command. Note that when you switch directions, the settings saved for that direction are automatically loaded.

Figure 33: How User Settings are Stored in the HS-xx Cameras after issuing the `wus` Command

You can save or restore your user settings to non-volatile memory using the following commands.

- To save all current user settings to EEPROM for the current mode for both TDI shift directions, use the command `wus`. The camera will automatically restore the saved user settings when powered up.

WARNING: While settings are being written to nonvolatile memory, do not power down camera or camera memory may be corrupted.

- To restore the last saved user settings, including the last used pixel coefficient set, for the current mode, use the command `rus`.

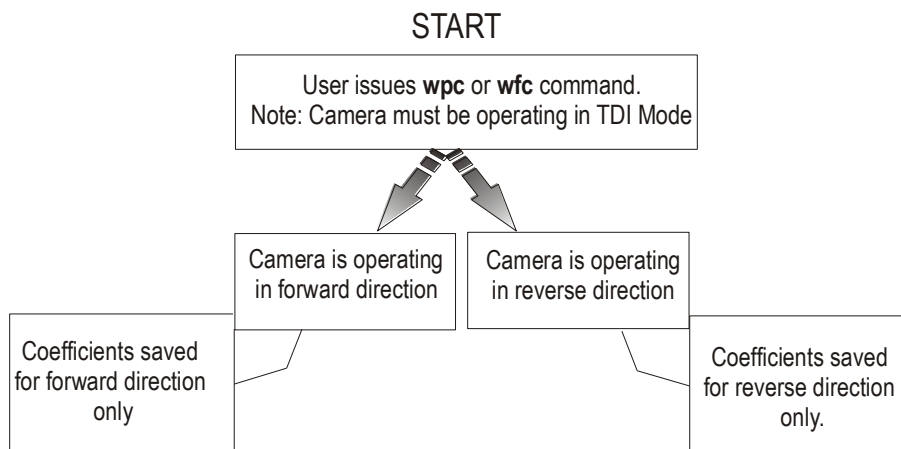
Current Session Settings

These are the current operating settings of your camera. These settings are stored in the camera's volatile memory and will not be restored once you power down your camera. To save these settings for reuse at power up, use the command `wus`. Settings are saved for the current operating mode (TDI or Area) only.

4.6.2 Saving and Restoring PRNU and FPN Coefficients

Note: Available in TDI Mode only.

Pixel coefficient sets are saved separately for Forward and Reverse direction, depending on which direction the camera is operating in when the `wpc` or `wfc` command is issued. It is important that you save pixel coefficients before switching CCD shift direction or current coefficient values will be lost.

Figure 34: How Pixel Coefficients are saved in the HS-xx Cameras after issuing the wpc or wfc Command**Saving the Current PRNU Coefficients**

Purpose: Saves the current PRNU coefficients for the current direction.

Syntax: **wpc i**

Syntax Elements: **i**

PRNU coefficients set to save.

1 = Coefficient set one

2 = Coefficient set two

3 = Coefficient set three

4 = Coefficient set four

Notes:

- Available in TDI mode only.
- Available only when operating the camera in internal direction control ([scd 0](#) or [1](#))

Example: **wpc 2**

Saving the Current FPN Coefficients

Purpose: Saves the current FPN coefficients for the current direction.

Syntax: **wfc i**

Syntax Elements: **i**

FPN coefficients set to save.

1 = Coefficient set one

2 = Coefficient set two

3 = Coefficient set three

4 = Coefficient set four

Notes:

- Available in TDI mode only.
- Available only when operating the camera in internal direction control ([scd 0](#) or [1](#))

Example: **wfc 2**

Loading a Saved Set of Coefficients

Purpose: Loads a saved set of pixel coefficients for the current direction. A factory calibrated set of coefficients is available.

Syntax: `lpc i`

Syntax Elements: `i`

FPN coefficients set to save.

0 = Factory calibrated pixel coefficients.

1 = Coefficient set one

2 = Coefficient set two

3 = Coefficient set three

4 = Coefficient set four

Notes:

- Available in TDI mode only.
- Available only when operating the camera in internal direction control ([scd 0](#) or [1](#)). When operating in external direction control, you must switch to internal direction control, load the coefficient set, and then return to external direction control.

Example: `lpc 0`

Resetting the Current Pixel Coefficients

Purpose: Resets the current pixel coefficients to zero. This command does not reset saved coefficients.

Syntax: `rpc`

Notes: The digital offset is not reset.

4.6.3 Rebooting the Camera

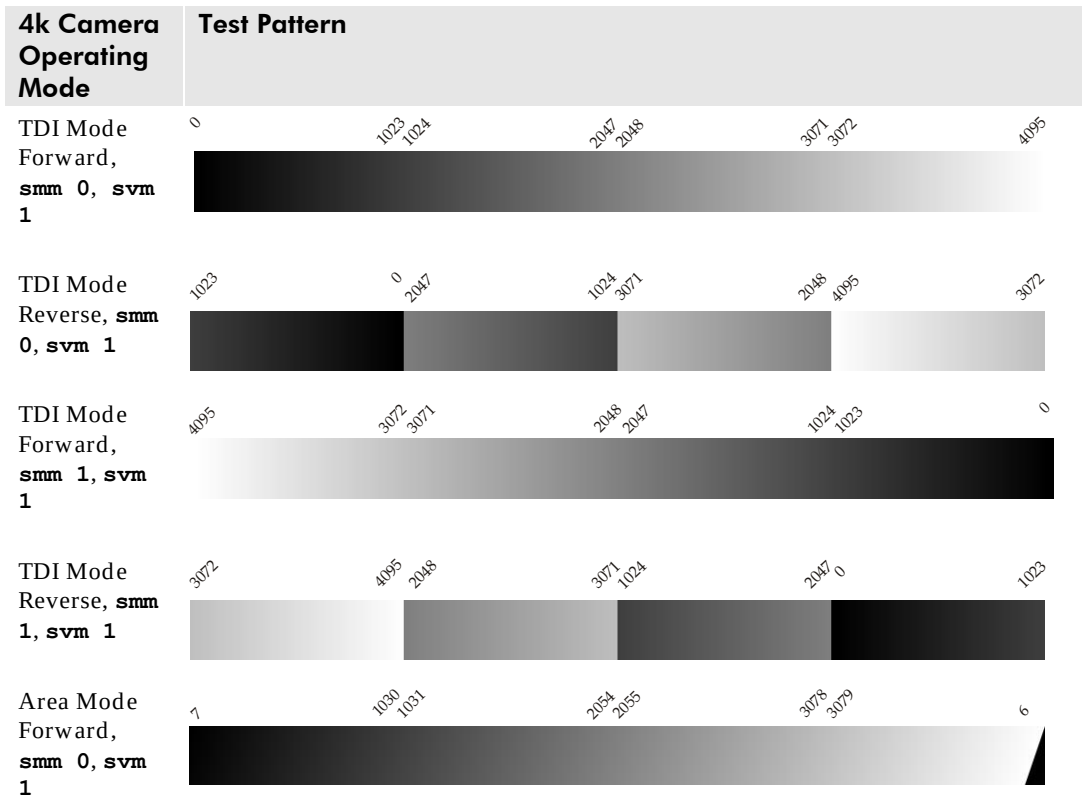
The command `rc` reboots the camera. The camera starts up with the last saved settings and the baud rate used before reboot. Previously saved pixel coefficients are also restored.

4.7 Diagnostics

4.7.1 Generating a Test Pattern

Purpose:	Generates a test pattern to aid in system debugging. The test patterns are useful for verifying proper timing and connections between the camera and the frame grabber. The following tables show each available test pattern.
Syntax:	svm i
Syntax Elements:	i 0 Video. 1 12 bit test pattern 1 (ramp) 2 8 bit test pattern 2 (step)
Notes:	- When returning to video (svm 0) after viewing a test pattern, the camera restores the saved user settings for digital offset (sdo), enable pixel coefficients (epc), set subtract background (ssb), and set system digital gain (ssg). - The following diagrams show 12-bit pixel values. When operating in 8-bit mode, pixel values will be 1/ 16th of pixel values in the diagram.
Example:	svm 2

Table 19: 4k Test Patterns



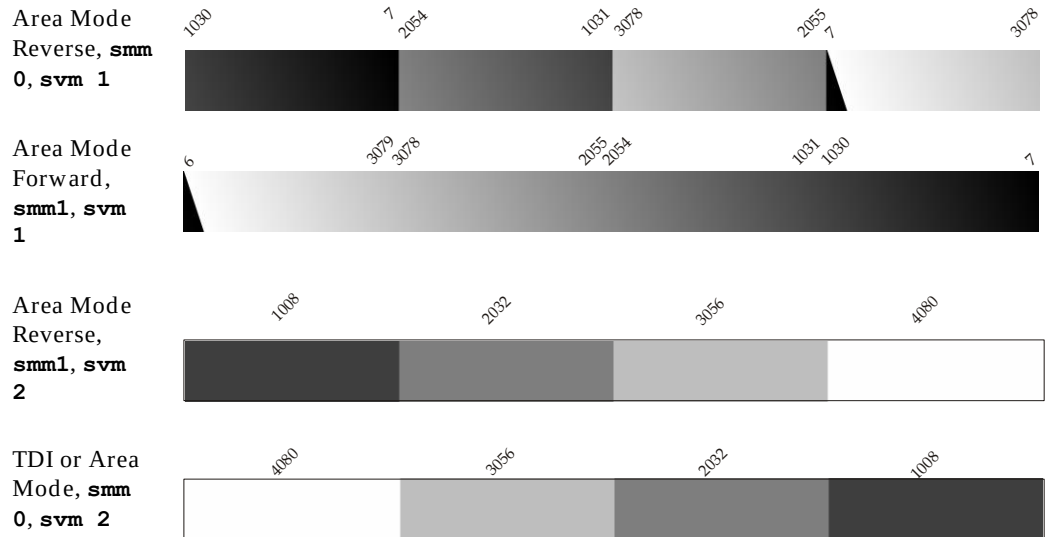
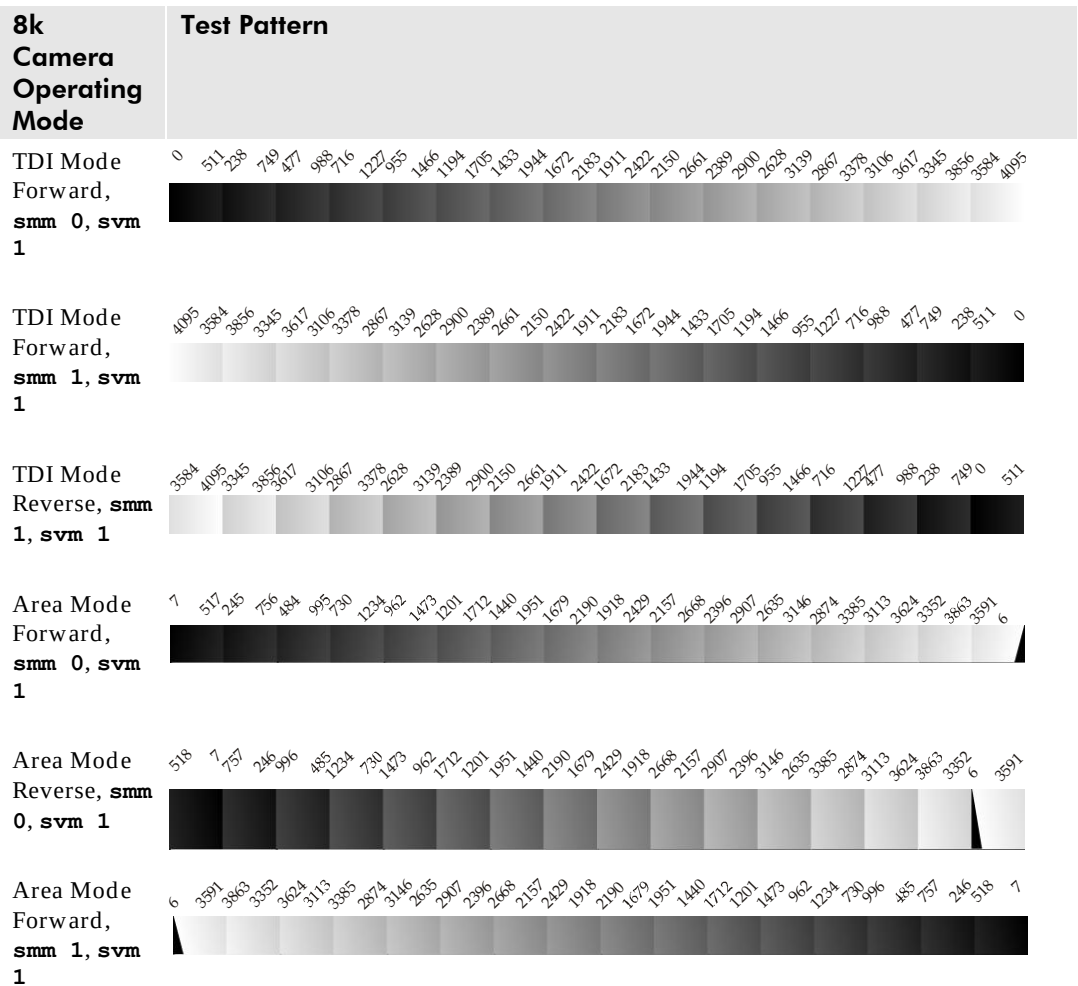
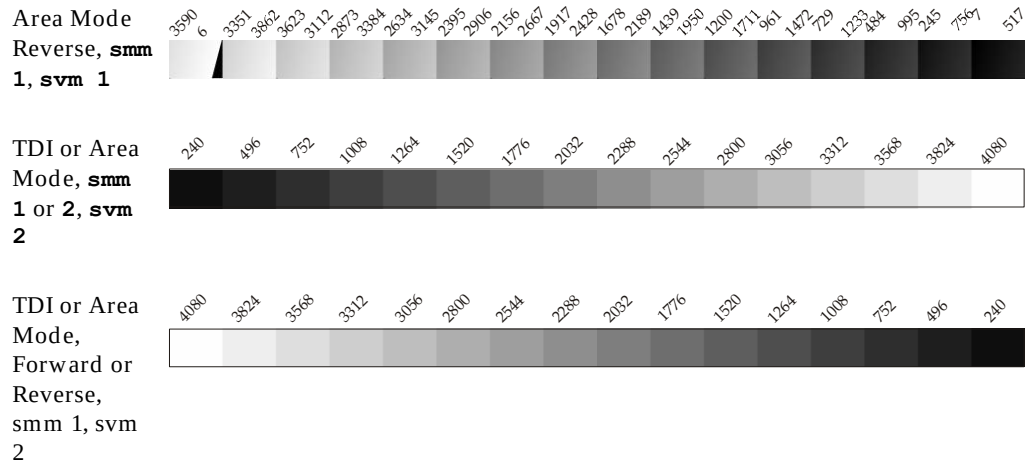


Table 20: 8k Test Patterns





4.7.2 Returning Video Information

The camera's microcontroller has the ability to read video data when operating the camera in TDI Mode. This functionality can be used to verify camera operation and to perform basic testing without having to connect the camera to a frame grabber. This information is also used for collecting line statistics for calibrating the camera.

Returning a Single Line of Video

Purpose: Returns a complete line of video (without pixel coefficients or test pattern) displaying one pixel value after another. It also displays the minimum, maximum, and mean value of the line sampled within the region of interest (the region of interest command is explained in section Setting a Region of Interest).

Use the **g1** command, or the following **g1a** command, to ensure the proper video input range into the processing chain before executing any pixel calibration commands.

Syntax: **g1 x1 x2**

Syntax Elements: **x1**

Column start number. Must be less than the column end number in a range from 1 to (**column resolution** - 1).

x2

Column end number. Must be greater than the column start number in a range from 2 to sensor resolution.

Notes:

- If $x2 \leq x1$ then **x2** is forced to be **x1**.
- Analog gain, analog offset, digital offset, background subtract, and digital system gain are applied to the data. FPN and PRNU coefficients are not included in the data.
- Values returned are in 12 bit DN.
- Available in TDI Mode only.

Related Commands [roi](#)

Example: **g1 10 20**

Returning Averaged Lines of Video

Setting the Number of Lines to Sample

Purpose:	Sets the number of lines to sample when using the gla command or for pixel coefficient calculations.
Syntax:	css i
Syntax Elements:	i Number of lines to sample. Allowable values are 256 , 512 , or 1024 (factory setting).
Notes:	To return the current setting, use the gcp command.
Related Commands:	gla
Example:	css 1024

Returning the Average of Multiple Lines of Video

Purpose:	Returns the average for multiple lines of video data (without pixel coefficients or test pattern). The number of lines to sample is set and adjusted by the css command. The camera displays the Min., Max., and Mean statistics for the pixels in the region of interest (the region of interest command is explained in section Setting a Region of Interest).
Syntax:	gla x1 x2
Syntax Elements:	x1 Column start number. Must be less than the column end number in a range from 1 to (column resolution - 1). x2 Column end number. Must be greater than the column start number in a range from 2 to column resolution.
Notes:	- If $x2 \leq x1$ then x2 is forced to be x1. - Analog gain, analog offset, digital offset, background subtract, and digital system gain are applied to the data. FPN and PRNU coefficients are not included in the data. - Values returned are in 12 bit DN. - Available in TDI Mode only.
Related Commands:	css , roi
Example:	gla 10 20

4.7.3 Temperature Measurement

The temperature of the camera can be determined by using the **vt** command. This command will return the internal chip temperature in degrees Celsius. For proper operation, this value should not exceed 75°C.

Note: If the camera reaches 75°C, the camera **will shutdown and the LED will flash red**. If this occurs, the camera **must be rebooted** using the command, [rc](#) or can be powered down manually. You will have to correct the temperature problem or the camera will shutdown again.

4.7.4 Voltage Measurement

The command **vv** displays the camera's input voltage. Note that the voltage measurement feature of the camera provides only approximate results (typically within 10%). The measurement should not be used to set the applied voltage to the camera but only used as a test to isolate gross problems with the supply voltage.

4.7.5 Camera Frequency Measurement

Purpose: Returns the frequency for the requested Camera Link control signal

Syntax: **gsf i**

Syntax Elements: **i**

Camera Link control signal to measure:

- 1: CC1 (EXSYNC)
- 2: CC2 (Spare)
- 3: CC3 (Forward)
- 4: CC4 (Spare)

Example: **gsf 1**

4.7.6 Returning Camera Settings

Returning All Camera Settings with the Camera Parameter Screen

The camera parameter (**gcp**) screen returns all of the camera's current settings. Figure 35 below describes the **gcp** screen for the Piranha HS-8k camera.

To read all current camera settings, use the command:

Syntax: **gcp**

Figure 35: Example GCP Screen for HS-40-04k40 TDI Mode Operation

GCP Screen		Description
C A M E R A S E T T I N G S		
Camera Model No.:	HS-xx-xxxxx-xx-R	Camera model number.
Sensor Serial No.:	xxxxxxxxxx	Sensor serial number.
Firmware Design Rev.:	xx-xx-xxxxx-xx	Firmware design revision number.
CCI Version:	xx-xxx-xxxxx-xx	Camera control information.
FPGA Version:	xx-xx-xxxx-xx	DSP design revision number.
UART Baud Rate:	9600	Serial communication connection speed set with the sbr command. See section Setting Baud Rate for details.

GCP Screen		Description
TDI Mode:	tdi	Current operating mode, either TDI or Area set with the tdi command. See section 4.5.1 for details.
Exposure Mode:	7	Current exposure mode value set with the sem command.
SYNC Frequency:	5000.00 Hz	Current line rate. Value is set with the ssf command.
CCD Direction:	internal/forward	CCD shift direction set with the scd command.
Stage Selection	96	Number of integration stages set with the stg command. See section 4.2 Selecting the Number of CCD Integration Stages for details.
Horizontal Binning	1	Horizontal binning value set with the sbh command. See section 4.3.4 Increasing Sensitivity with Binning for details.
Vertical Binning	1	Vertical binning value set with the sbv command. See section 4.3.4 Increasing Sensitivity with Binning for details.
Video Mode:	video	Current video mode value set with the svm command. See section 4.7.1 Generating a Test Pattern for details.
Region of Interest:	(1,1) to (8192,1)	Region of interest size set with the roi command. See section Setting a Region of Interest for details.
End-Of-Line Sequence:	on	States whether an end-of-line sequence is turned on or off. Set using the eol command. See section 4.5.3 End-of-line Sequence for details.
FFC Coefficient Set:	0	Current pixel coefficient set loaded. Refer to section 4.6.2 Saving and Restoring PRNU and FPN Coefficients for details.
FPN Coefficients:	off	States whether FPN coefficients are on or off. Set with the epc command. Refer to section Digital Signal Processing for details.
PRNU Coefficients:	off	States whether PRNU coefficients are on or off.

GCP Screen		Description
		Set with the epc command. Refer to section Digital Signal Processing for details.
Number of Line Samples:	1024	Number of lines sample with the gla command set with the css command. See section 4.7.2 Returning Video Information for details.
Lower Threshold:	0	Lower threshold value set with the slt command. See section 4.5.3 End-of-line Sequence for details.
Upper Threshold:	4095	Upper threshold value set with the sut command. See section 4.5.3 End-of-line Sequence for details.
Camera Link Mode:	21, Full, 8 taps, 8 bits, no time MUX	Camera Link configuration set with the clm command. See section 4.4.2 Setting the Camera Link Mode for details.
Output Throughput:	320	Camera throughput value set with the sot command. See section 4.4.3 Setting the Camera Throughput for details.
Pretrigger	0	Pretrigger set with the spt command. See section 4.4.5 Setting a Pretrigger.
Mirroring Mode:	0, left to right	Readout direction set with the smm command. See section 4.4.4 Setting the Pixel Readout Direction for details.
Analog Gain (dB):	0.0 0.0 0.0 0.0	Analog gain settings set with the sag command. See section Analog Signal Processing for details.
Analog Reference Gain (dB):	0.0 0.0 0.0 0.0	Analog reference gain set with the ugr command. See section Analog Signal Processing for details.
Total Analog Gain (dB):	0.0 0.0 0.0 0.0	This is the sum of the analog gain and analog gain reference values and is the total analog gain being used by the camera.

GCP Screen					Description
Analog Offset:	80	80	80	80	Analog offset settings set with the sao command. See section Analog Signal Processing for details.
Digital Offset:	80	80	80	80	Digital offset settings set with the sdo command. See section Digital Signal Processing for details.
Background Subtract:	0	0	0	0	Background subtract settings set with the ssb command. See section Digital Signal Processing for details.
System Gain:	4096 4096	4096	4096	4096	Digital gain settings set with the ssg command. See section Digital Signal Processing for details.

Returning Camera Settings with Get Commands

You can also return individual camera settings by inserting a “**get**” in front of the command that you want to query. If the command has a tap or pixel number parameter, you must also insert the tap number or pixel number that you want to query. Refer to Table 21 below for a list of available commands. To view a help screen listing the following get commands, use the command **gh**.

Table 21: Get Commands

Syntax	Parameters	Description
get cao	t	Returns the analog offset for the tap indicated t = tap selection, either 1 to number of CCD taps , or 0 for all taps
get ccf	x1 x2	Returns the FPN pixel coefficients for the pixel range indicated. x1 = Pixel start number x2 = Pixel end number
get ccp	x1 x2	Returns the PRNU pixel coefficients for the pixel range indicated. x1 = Pixel start number x2 = Pixel end number
get clm		Returns the current Camera Link mode.
get css		Returns the number of line samples averaged for pixel coefficient calculations or for output of gla command.
get dpc	x1 x2	Returns the pixel coefficients in the order FPN, PRNU, FPN, PRNU, ... x1 = Pixel start number x2 = Pixel end number in a range from 1 to sensor pixel count .

Syntax	Parameters	Description
get els		Returns whether the end-of-line statistics are turned off or on. 0 : Off 1 : On
get epc		Returns whether pixel coefficients are enabled or disabled. The first parameter returns the FPN coefficients setting where: 0 = FPN coefficients disabled 1 = FPN coefficients enabled The second parameter returns the PRNU coefficients setting where: 0 = PRNU coefficients disabled 1 = PRNU coefficients enabled
get gcm		Returns the camera's model number
get gcs		Returns the camera's serial number
get gcv		Returns the camera's software version.
get gfc	x	Returns the FPN pixel coefficient for the pixel indicated.
get gl	x1 x2	Returns pixel values for the pixel range specified.
get gla	x1 x2	Returns the average of the pixel range indicated.
get gpc	x	Returns PRNU coefficient. x = pixel number to read in a range from 1 – sensor pixel count.
get gsf	i	Returns the frequency of the Camera Link control signal indicated, either 1 , 2 , 3 , or 4 .
get h		Returns the onscreen help menu.
get lpc		Returns the current coefficient set number.
get rc		
get rfs		Returns whether factory settings have been saved. The camera always returns 1 (factory settings have been saved).
get roi		Returns the current region of interest.
get rus		Returns whether user settings have been saved. 0 = No user settings saved 1 = User settings have been saved
get sag	t	Returns the analog gain in dB for the tap indicated t = Tap value. 0 for all taps or 1-number of CCD taps for individual tap selection.
get sao	t	Returns the analog offset for the tap indicated. t = 0 for all taps or 1 to number of CCD taps for individual tap selection.
get sbh		Returns the horizontal binning value.
get sbr		Returns the speed of camera serial communication port.

Syntax	Parameters	Description
get sbv		Returns the vertical binning value.
get scd		Returns the CCD shift direction where: 0 = Forward TDI shift direction. 1 = Reverse TDI shift direction. 2 = Externally controlled direction control via Camera Link control CC3.
get sdo	t	Returns the digital offset value in DN for the tap indicated. t = Tap value. 0 for all taps or 1-number of CCD taps for individual tap selection.
get sem		Returns the current exposure mode: 3 = External SYNC, maximum exposure time 7 = Internal programmable SYNC, maximum exposure time. Factory setting.
get sfc	x	Returns the FPN coefficient for the pixel number indicated. x = pixel number within the range 1 to sensor pixel count .
get sfr	xx	Returns a range of pixel FPN coefficients x1 =first pixel number of the range. x2 =last pixel number of the range i =coefficient value in a range from 0 to 2048 .
get slt		Returns the current lower threshold value.
get smm		Returns the camera's mirror mode: 0 : Pixels readout left to right (1 to 4096 or 8192) 1 : Pixels readout right to left (8092 or 4096 to 1)
get sot		Returns the Camera Link strobe rate.
get spc	x	Returns the PRNU coefficient for the specified pixel number. x =pixel number within the range 1 to sensor pixel count .
get spr	xx	Returns a range of pixel PRNU coefficients i1 =first pixel number of the range i2 =last pixel number of the range x =coefficient value in a range from 0 to 28671 .
get spt		Returns the current pretrigger setting.
get ssb	t	Returns the current background subtract value. t = Tap value. 0 for all taps or 1 to number of CCD taps for individual tap selection.
get ssf		Returns the current line/ frame rate in Hz.
get ssg	t	Returns the current digital gain setting. t = tap selection, either 1 to number of CCD taps , or 0 for all taps
get stg		Returns the current number of CCD integration stages.
get sut		Returns the current upper threshold value.

Syntax	Parameters	Description
get svm		Returns the current video mode. 0: Normal video mode 1: Test pattern 2: Test pattern 3: Test pattern
get tdi		Returns the camera's current operating mode. 0: Area Mode 1: TDI Mode
get ugr	t	Returns the gain reference value
get vt		Returns the camera's internal chip temperature in degrees Celcius.
get vv		Returns the camera's supply voltage.
get wfc		Returns whether FPN coefficients have been saved. 0 = No FPN coefficients saved 1 = Pixel coefficients have been saved
get wpc		Returns whether PRNU coefficients have been saved. 0 = No PRNU coefficients saved 1 = Pixel coefficients have been saved
get wus		Returns whether user settings have been saved. 0 = No user settings saved 1 = User settings have been saved

Appendix A

Error Handling and Command List

Error Handling

The following table lists warning and error messages and provides a description and possible cause. Warning messages are returned when the camera cannot meet the full value of the request; error messages are returned when the camera is unable to complete the request.

Table 22: Warning and Error Messages

Warning Messages	
Camera Response	Comment
OK>	Camera executed command
Warning 01: Outside of specification>	Parameter accepted was outside of specified operating range (e.g. gain greater than ± 10 dB of factory setting, or SSF below specification).
Warning 02: Clipped to min>	Parameter was clipped to the current operating range. Use GCP or GET to see value used.
Warning 03: Clipped to max>	Parameter was clipped to the current operating range. Use GCP or GET to see value used.
Warning 04: Related parameters adjusted>	Internal operating condition is adjusted to accommodate the entered command. E.g. requesting exposure time longer than line time automatically adjusts the line time to meet the exposure time requirement.
Warning 07: Coefficient may be inaccurate A/ D clipping has occurred>	In the region of interest (ROI) greater than 6.251% single or 1% of averaged pixel values were zero or saturated.
Warning 08: Greater than 1% of coefficients have been clipped	Greater than 1% of FPN or PRNU coefficients have been calculated to be greater than the maximum allowable and so were clipped.
Warning 09: Internal line rate inconsistent with read out time>	Changing this parameter (e.g. vertical binning) has changed read out time and that is greater than the <i>internal SYNC</i>

Error Messages	
Camera Response	Comment
Error 01: Internal error xx>	Where xx is a code list below. Only output during power up. Customer should contact Teledyne DALSA customer support.
Error 02: Unrecognized command>	Command is not valid.
Error 03: Incorrect number of parameters>	Too many or too few parameters.
Error 04: Incorrect parameter value>	This response returned for: Alpha received for numeric or visa versa Float where integer expected Not an element of the set of possible values. E.g., Baud Rate Outside the range limit
Error 05: Command unavailable in this mode>	E.g. SSF when in SEM 3
Error 06: Timeout>	Command not completed in time. E.g. CCF or CCP in SEM 3 when no external EXSYNC is present.
Error 07: Camera settings not saved>	Indicates that user settings have been corrupted by turning off the power while executing the WUS command. Must build up new settings from factory and re-save with WUS.
Error 08: Unable to calibrate - tap outside ROI>	Cannot calibrate a tap that is not part of the end of line statistics.
Error 09: The camera's temperature exceeds the specified operating range>	Indicates that the camera has shut itself down to prevent damage from further overheating. (flashing red) Shuts down at internal temperature of 75°C and will not restart until below 65°C (equivalent to 50°C at front plate).
Error 10: FPGA Flash Program Failed	FCS failed either because of communication error or a bad file was sent.

Commands: Quick Reference

Parameters:

t = tap id
i = integer value
f = float
m = member of a set
s = string
x = pixel column number
y = pixel row number

As a quick reference, the following table lists all of the camera configuration commands available to the camera user. For detailed information on using these commands, refer to Chapter 3. Note: This table does not list “get” commands. Refer to section 4.7.6 Returning Camera Settings for a list of these commands.

Table 23: Command Quick Reference

Mnemonic	Syntax	Parameters	Description
calibrate analog offset	cao	t i	Calibrates the analog gain and averages each tap’s pixels within the ROI to the specified average target value. t = tap selection, either 1 to number of CCD taps, or 0 for all taps i = target value in a range from 0 to 255DN (12-bit LSB) Refer to Analog Signal Processing: Setting Analog Gain and Offset for details.
correction calibrate fpn	ccf		Performs FPN calibration and eliminates FPN noise by subtracting away individual pixel dark current. Refer to Digital Signal Processing and Processing Chain Overview and Description for details.
calculate camera gain	ccg	i1 t i2	Calculates the camera gain according to the selected algorithm. i1 = Calibration algorithm to use. 1 = This algorithm adjusts analog gain so that 8% to 13% of tap ROI pixels are above the specified target value. 2 = This algorithm adjusts analog gain so that the average pixel value in tap’s ROI is equal to the specified target value. 3 = This algorithm adjusts digital gain so that the average pixel value in tap’s ROI is equal to the specified target. 4 = This algorithm adjusts the analog gain so that the peak tap ROI pixels are adjusted to the specified target. t = Tap value. Use 0 for all taps or 1 to number of CCD taps for individual tap selection. i2 = Calibration target value in a range from: 1024 to 4055DN (12 bit LSB).

Mnemonic	Syntax	Parameters	Description
correction calibrate prnu	ccp		Performs PRNU calibration and eliminates the difference in responsivity between the most and least sensitive pixel creating a uniform response to light. Refer to Digital Signal Processing and Processing Chain Overview and Description for details.
camera link mode	clm	m	Sets the Camera Link configuration, number of Camera Link taps, and data bit depth. 2: Base configuration, 2 taps, 8 bit output 3: Base configuration, 2 taps, 12 bit output 15: Medium configuration, 4 taps, 8 bit output 16: Medium configuration, 4 taps, 12 bit output 21: Full configuration, 8 taps, 8 bit output Refer to section 4.4.2 Setting the Camera Link Mode for details.

Mnemonic	Syntax	Parameters	Description
calculate PRNU algorithm	cpa	i1 i2	Performs PRNU calibration according to the selected algorithm. The first parameter is the algorithm where i1 is: 1 = This algorithm first adjusts each tap's analog gain so that 8-13% of pixels within a tap are above the value specified in the target value parameter. PRNU calibration then occurs using the peak pixel in the region of interest. (Identical to ccp) 2 = Calculates the PRNU coefficients using the entered target value as shown below: $\text{PRNU Coefficient} = \frac{\text{Target}}{(\text{AVG Pixel Value}) - (\text{FPN} + \text{sdo value})}$ The calculation is performed for all sensor pixels but warnings are only applied to pixels in the region of interest. This algorithm is useful for achieving uniform output across multiple cameras. 3 = This algorithm includes an analog gain adjustment prior to PRNU calibration. Analog gain is first adjusted so that the peak pixel value in tap's ROI is within 97 to 99% of the specified target value. It then calculates the PRNU coefficients using the target value as shown below: $\text{PRNU Coefficient}_i = \frac{\text{Target}}{(\text{AVG Pixel Value}_i) - (\text{FPN}_i + \text{sdo value})}$ The calculation is performed for all sensor pixels but warnings are only applied to pixels in the region of interest. This algorithm is useful for achieving uniform output across multiple cameras. 4 = This algorithm is the same as 2 with the exception that it only calculates PRNU for the pixels within the current Region of Interest (ROI). The second parameter, i2, is the target value to use in a range from 1024 to 4055DN.
correction set sample	css	m	Set number of line samples averaged for pixel coefficient calculations or for output of gla command. Values: 256, 512, 1024. Factory setting: 1024 Refer to Returning Averaged Lines of Video on page 75 for details.

Mnemonic	Syntax	Parameters	Description
display pixel coeffs	dpc	x1 x2	Displays the pixel coefficients in the order FPN, PRNU, FPN, PRNU, ... x1 = Pixel start number x2 = Pixel end number in a range from 1 to sensor pixel count . Refer to Setting a range of Pixel PRNU Coefficients for details.
end of line sequence	els	i	Sets the end-of-line sequence: 0 : Off 1 : On Refer to Refer to 4.5.3 End-of-line Sequence for details.
enable pixel coefficients	epc	i i	Sets whether pixel coefficients are enabled or disabled. The first parameter sets the FPN coefficients where i is: 0 = FPN coefficients disabled 1 = FPN coefficients enabled The second parameter sets the PRNU coefficients where i is: 0 = PRNU coefficients disabled 1 = PRNU coefficients enabled Refer to section Enabling and Disabling Pixel Coefficients on page 66 for details.
get camera model	gcm		Reads the camera model number.
get camera parameters	gcp		Reads all of the camera parameters.
get camera serial	gcs		Read the camera serial number.
get camera version	gcv		Displays the firmware version and FPGA version.
get values	get	s	Retrieves camera values.
get fpn coeff	gfc	x	Read the FPN coefficient x = pixel number to read in a range from 1 – sensor pixel count . Refer to Returning FPN Coefficients on page 61 for details.
get help	gh		Returns the help screen.
get line	gl	x1 x2	Gets a line of raw video (no digital processing or test pattern) displaying one pixel value after another and the minimum, maximum, and mean value of the sampled line. x1 = Pixel start number x2 = Pixel end number in a range from 1 to sensor pixel count . Refer to Returning a Single Line of Video on page 74 for details.

Mnemonic	Syntax	Parameters	Description
get line average	gla	x1 x2	Read the average of line samples. x1 = Pixel start number x2 = Pixel end number in a range from 1 to sensor pixel count . Refer to Returning Averaged Lines of Video on page 75 for details.
get prnu coeff	gpc	x	Read the PRNU coefficient. x = pixel number to read in a range from 1 – sensor pixel count . Refer to Setting a range of Pixel PRNU Coefficients for details.
get signal frequency	gsf	i	Reads the requested Camera Link control frequency. 1 = EXSYNC frequency 2 = Spare 3 = Direction (1 = forward, 2 = reverse) 4 = Spare
help	h		Display the online help. Refer to section for details.
load pixel coefficients	lpc		Loads the previously saved pixel coefficients from non-volatile memory where i is: 0 = Factory calibrated coefficients 1 = Coefficient set one 2 = Coefficient set two 3 = Coefficient set three 4 = Coefficient set four Refer to section 4.6.2 Saving and Restoring PRNU and FPN Coefficients for details.
reset camera	rc		Reset the entire camera (reboot). Baud rate is not reset and reboots with the value last used.
restore factory settings	rfs		Restore the camera's factory settings. FPN and PRNU coefficients reset to 0. Refer to section 4.6 Saving and Restoring Settings for details.
region of interest	roi	x1 y1 x2 y2	Sets the pixel range affected by the ccg , cao , gl , gla , ccf , cpa and ccp commands. The parameters are the pixel start and end values (x1 and x2) and the column start and end values (y1 and y2) in a range from 1 to sensor pixel count . Refer to section 4.5.1 Setting a Region of Interest for details.
reset pixel coeffs	rpc		Reset the pixel coefficients to 0. Refer to section Enabling and Disabling Pixel Coefficients on page 66 for details.

Mnemonic	Syntax	Parameters	Description
restore user settings	rus		Restore the camera's last saved user settings and FPN and PRNU coefficients. Refer to section 4.6 Saving and Restoring Settings for details.
set analog gain	sag	t f	Sets the analog gain in dB. t = Tap value. 0 for all taps or 1-number of CCD taps for individual tap selection. f = gain value specified from -10 to +10 Refer to Analog Signal Processing: Setting Analog Gain and Offset on page 53 for details.
set analog offset	sao	t i	Sets the analog offset. t = 0 for all taps or 1 to number of CCD taps for individual tap selection. i = Offset value in a range from 0 to 255 (12-bit LSB). Offset increases with higher values. Refer to Analog Signal Processing: Setting Analog Gain and Offset for details.
set binning horizontal	sbh	m	Sets the horizontal binning value. Available values are 1, 2, 4, or 8 . Refer to section 4.3.4 Increasing Sensitivity with Binning on page 37 for details.
set baud rate	sbr	i	Set the speed of camera serial communication port. Baud rates: 9600, 19200, 57600, and 115200 . Default: 9600. Refer to section Setting Baud Rate on page 29 for details.
set binning vertical	sbv	m	Sets the vertical binning value. Available values (not including the HS-82 model) are 1 to 8 . Available values for the HS-82 model are 1 to 4 . Refer to section 4.3.4 Increasing Sensitivity with Binning on page 37 for details.
set ccd direction	scd	i	Sets the CCD shift direction where: 0 = Forward TDI shift direction. 1 = Reverse TDI shift direction. 2 = Externally controlled direction control via Camera Link control CC3. Available only in TDI Mode

Mnemonic	Syntax	Parameters	Description
set digital offset	sdo	t i	Subtracts the input value from the video signal prior to FPN correction. t = Tap value. 0 for all taps or 1-number of CCD taps for individual tap selection. i = Offset in a range from 0 to 511DN . Refer to Setting Digital Offset on page 61 for details.
set exposure mode	sem	m	Set the exposure mode: 3 = External SYNC, maximum exposure time 7 = Internal programmable SYNC, maximum exposure time. Factory setting.
set fpn coeff	sfc	x i	Set the FPN coefficient. x = pixel number within the range 1 to sensor pixel count . i = FPN value within the range 0 to 2048 (12-bit LSB). Refer to Performing FPN Correction on page 61 for details.
set fpn range	sfr	x1 x2 i	Set a range of pixel FPN coefficients x1 = first pixel number of the range. x2 = last pixel number of the range i = coefficient value in a range from 0 to 2048 . Refer to Setting a Range of FPN Coefficients on page 62 for details.
set lower threshold	slt	i	The pixels below the lower threshold are checked for and reported in the end-of-line sequence in a range from 0-4095 . Refer to section 4.5.3 End-of-line Sequence for details.
set mirror mode	smm	i	Set the camera's mirror mode: 0 : Pixels readout left to right (1 to 4096 or 8192) 1 : Pixels readout right to left (8092 or 4096 to 1) Refer to 4.4.4 Setting the Pixel Readout Direction for details.

Mnemonic	Syntax	Parameters	Description
set output throughput	sot	m	This command works in conjunction with the clm command and determines the Camera Link strobe rate. 80 = 4 taps at 20MHz or 2 taps at 40MHz 160 = 2 taps at 80MHz or 4 taps at 40MHz 320 = 4 taps at 80MHz or 8 taps at 40MHz 640 = 8 taps at 80MHz Refer to section 4.4.3 Setting the Camera Throughput for details on using this command. Available settings are dependent on your camera model.
set prnu coeff	spc	x i	Set the PRNU coefficient. x=pixel number within the range 1 to sensor pixel count. i= PRNU value within the range 0 to 28671. Refer to Setting a range of Pixel PRNU Coefficients for details.
set prnu range	spr	i1 i2 x	Set a range of pixel PRNU coefficients i1=first pixel number of the range i2=last pixel number of the range x=coefficient value in a range from 0 to 28671. Refer to Setting a Range of PRNU Coefficients on page 65 for details.
set pretrigger	spt	i	Set the pretrigger to a value from 0 to 16.
set subtract background	ssb	t i	Subtract the input value from the output signal. t = Tap value. 0 for all taps or 1 to number of CCD taps for individual tap selection. i = Subtracted value in a range from 0 to 4095. Refer to Subtracting Background on page 60 for details.

Mnemonic	Syntax	Parameters	Description
set sync frequency	ssf	i	Set the frame rate to a value from: TDI HS-40-4k40: 3499.87-36563.1 HS-80-8k40: 3499.56-34305.3 HS-80-8k80: 3499.87-68610.6 HS-82-4k80: 3499.87-111111 Area HS-40-4k40: 1-3783.58 HS-80-8k40: 1-3084.52 HS-80-8k80: 1-6169.03 HS-82-4k80: 1-9017.13 Value rounded up/ down as required. Refer to Setting Frame Rate on page 41 for details.
set system gain	ssg	t i	Set the digital gain. t = tap selection, either 1 to number of CCD taps , or 0 for all taps i = Digital gain in a range from 0 to 65535 . The digital video values are multiplied by this number. Refer to Setting Digital Gain on page 60 for details.
stage select	stg	i	Sets the number of TDI stages. Allowable values (not including the HS-82 model) are: 16, 32, 48, 64, 80, 96 (default) Allowable HS-82 values are: 8, 16, 24, 32, 40, 48 (default) Refer to 4.2 Selecting the Number of CCD Integration Stages for details.
set upper threshold	sut	i	The pixels equal to or greater than the upper threshold are checked for and reported in the end-of-line sequence in a range from 0-4095 . Refer to 4.5.3 End-of-line Sequence for details.
set video mode	svm	i	Switch between normal video mode and test patterns: 0 : Normal video mode 1 : Test pattern 2 : Test pattern 3 : Test pattern Refer to section 4.7.1 Generating a Test Pattern for details.
set TDI mode	tdi	i	Set the camera's operating mode. 0 : Area Mode 1 : TDI Mode Refer to section 4.5.1 for details.

Mnemonic	Syntax	Parameters	Description
update gain reference	ugr		Changes 0dB gain to equal the current analog gain value set with the sag command. Refer to section Analog Signal Processing: Setting Analog Gain and Offset for details.
verify temperature	vt		Check the internal temperature of the camera
verify voltage	vv		Check the camera's input voltages and return OK or fail
write FPN coefficients	wfc	i	Write all current FPN coefficients to EEROM where i is: 1 = FPN coefficient set one 2 = FPN coefficient set two 3 = FPN coefficient set one 4 = FPN coefficient set two Refer to section 4.6.2 Saving and Restoring PRNU and FPN Coefficients for details.
write PRNU coeffs	wpc	i	Write all current PRNU coefficients to EEROM where i is: 1 = PRNU coefficient set one 2 = PRNU coefficient set two 3 = PRNU coefficient set one 4 = PRNU coefficient set two Refer to section 4.6.2 Saving and Restoring PRNU and FPN Coefficients for details.
write user settings	wus		Write all of the user settings to EEROM. Refer to section 4.6.1 Saving and Restoring Factory and User Settings for details.

Appendix B

Camera Link™ Reference, Timing, and Configuration Table

Camera Link is a communication interface for vision applications. It provides a connectivity standard between cameras and frame grabbers. A standard cable connection will reduce manufacturers' support time and greatly reduce the level of complexity and time needed for customers to successfully integrate high speed cameras with frame grabbers. This is particularly relevant as signal and data transmissions increase both in complexity and throughput. A standard cable/ connector assembly will also enable customers to take advantage of volume pricing, thus reducing costs.

The camera link standard is intended to be extremely flexible in order to meet the needs of different camera and frame grabber manufacturers.

The Teledyne DALSA Camera Link Implementation Road Map (available from the [Knowledge Center](#) on Teledyne DALSA website) details how Teledyne DALSA standardizes its use of the Camera Link interface.

LVDS Technical Description

Low Voltage Differential Signaling (LVDS) is a high speed, low power, general purpose interface standard. The standard, known as ANSI/ TIA/ EIA-644, was approved in March 1996. LVDS uses differential signaling, with a nominal signal swing of 350mV differential. The low signal swing decreases rise and fall times to achieve a theoretical maximum transmission rate of 1.923 Gbps into a loss-less medium. The low signal swing also means that the standard is not dependent on a particular supply voltage. LVDS uses current-mode drivers, which limit power consumption. The differential signals are immune to ± 1 V common volt noise.

Camera Signal Requirements

This section provides definitions for the signals used in the Camera Link interface. The standard Camera Link cable provides camera control signals, serial communication, and video data.

Video Data

The Channel Link technology is integral to the transmission of video data. Image data and image enable signals are transmitted on the Channel Link bus. Four enable signals are defined as:

- FVAL—Frame Valid (FVAL) is defined HIGH for valid lines.
- LVAL—Line Valid (LVAL) is defined HIGH for valid pixels.

- DVAL—Data Valid (DVAL) is defined HIGH when data is valid.
- Spare— A spare has been defined for future use.

All four enable signals must be provided by the camera on each Channel Link chip. All unused data bits must be tied to a known value by the camera. For more information on image data bit allocations, refer to the official Camera Link specification located [here](#), on the Teledyne DALSA website.

Camera Control Signals

Four LVDS pairs are reserved for general purpose camera control. They are defined as camera inputs and frame grabber outputs. Camera manufacturers can define these signals to meet their needs for a particular product. The signals are:

- Camera Control 1 (CC1)
- Camera Control 2 (CC2)
- Camera Control 3 (CC3)
- Camera Control 4 (CC4)

The HS-xx uses the following camera control signals:

Table 24: Teledyne DALSA Camera Control Configuration

CC1	EXSYNC, negative edge active
CC2	Not Used
CC3	Forward
CC4	Not Used

Communication

Two LVDS pairs have been allocated for asynchronous serial communication to and from the camera and frame grabber. Cameras and frame grabbers should support at least 9600 baud. These signals are

- SerTFG—Differential pair with serial communications to the frame grabber.
- SerTC—Differential pair with serial communications to the camera.

The serial interface will have the following characteristics: one start bit, one stop bit, no parity, and no handshaking. It is recommended that frame grabber manufacturers supply both a user interface and a software application programming interface (API) for using the asynchronous serial communication port. The user interface will consist of a terminal program with minimal capabilities of sending and receiving a character string and sending a file of bytes. The software API will provide functions to enumerate boards and send or receive a character string.

Power

Power will not be provided on the Camera Link connector. The camera will receive power through a separate cable. Camera manufacturers will define their own power connector, current, and voltage requirements.

Camera Link Bit Definitions

BASE Configuration	T0			T1 (Note: Entries imply double frequency pixel rate data transmission)				
Mode	Port A Bits 0 thru 7	Port B Bits 0 thru 7	Port C Bits 0 thru 7	Port A Bits 0 thru 7	Port B Bits 0 thru 7	Port C Bits 0 thru 7		
Mode 2 2 Tap 8 bit	Tap 1 LSB..Bit 7	Tap 2 LSB..Bit7	xxxxxxx	N/ A	N/ A	N/ A		
Mode 3 2 Tap n bit Where n=10,12	Tap 1 LSB.. Bit 7	Tap 1 Bits 8,9,10,11, Tap 2 Bits 8,9,10,11	Tap 2 LSB..Bit 7	N/ A	N/ A	N/ A		
Medium Configuration	T0							
Mode	Port A Bits 0 thru 7	Port B Bits 0 thru 7	Port C Bits 0 thru 7	Port D Bits 0 thru 7	Port E Bits 0 thru 7	Port F Bits 0 thru 7		
Mode 15 4 Tap 8 bit	Tap 1 LSB..Bit 7	Tap 2 LSB..Bit 7	Tap 3 LSB..Bit 7	Tap 4 LSB...Bit 7	xxxxxxxx	Xxxxxxxxx		
Mode 16 4 Tap 10/ 12 bit	Tap 1 LSB.. Bit 7	Tap 1 Bits 8,9,10,11, Tap 2 Bits 8,9,10,11	Tap 2 LSB..Bit 7	Tap 4 LSB...Bit 7	Tap 3 LSB...Bit 7	Tap 3 Bit 8,9,10,11 Tap 4 Bit 8,9,10,11		
Full Configuration	T0							
Mode	Port A LSB... Bit 7	Port B LSB...Bi t 8	Port C LSB... Bit 8	Port D LSB... Bit 8	Port E LSB... Bit 8	Port F LSB... Bit 8	Port G LSB... Bit 8	Port H LSB... Bit 8
Mode 21 8 Tap 8 bit	Tap 1 LSB... Bit 7	Tap 2 LSB...Bit 7	Tap 3 LSB... Bit 7	Tap 4 LSB... Bit 7	Tap 5 LSB... Bit 7	Tap 6 LSB... Bit 7	Tap 7 LSB... Bit 7	Tap 8 LSB... Bit 7

Camera Link Configuration Tables

The following table provides tap reconstruction information. Teledyne DALSA is working with the machine vision industry to use this table as the basis for auto configuration. Visit the [Knowledge Center](#) Web site and view the Teledyne DALSA Camera Link Implementation Road Map document, 03-32-00450, for further details.

Interface Parameters

Table 25: Framegrabber Interface Parameters

Item (when programmable configuration the options are separated with a)	HS-40-04k40
Imager Dimension <1,2 or 1 2>	1 2
Imager Columns<number of active columns, X>	4096
Imager Rows<number of active rows, Y> Line Scan/ TDI are defined as 1	1 96
Number of CCD Taps <1,2,3.....>	4
Sensor Tap Clock Rate <xx MHz>	40
Camera Standard <NTSC, PAL, VS, VW, MW>	VS
Variable Window <Column Start, Column End, Row Start, Row End>	(0,0,0,0) All zeros indicates an unsupported feature
Multiple Window Number of Windows (Column Start 1, Column End 1, Row Start 1, Row End 1) (Column Start 2, Column End 2,...)	0, (0,0,0,0) All zeros indicates an unsupported feature
Number of Camera Configurations<1,2,3,...>	8
Configuration Definition Cx= HDW, Number of Output Taps, Bit Width, Number of Processing Nodes where Cx is the configuration ID x is <1,2,3...> HDW is <Base, Medium, Full> Number of Output Taps is <1,2,3...> Bit width is <8, 10, 12...> Number Processing Nodes is <1 or 2>	TDI Mode C1 = Medium, 4, 8, 1 C2 = Medium, 4, 12, 1 C3 = Base, 2, 8, 1 C4 = Base, 2, 12, 1 Area Mode C7 = Medium, 4, 8, 1 C8 = Medium, 4, 12, 1 C9 = Base, 2, 8, 1 C10 = Base, 2, 12, 1
Tap Reconstruction In some configurations the reconstruction may change. C0 is the default output format and must be listed. Output configurations that don't conform are listed separately. <Cx,Tn (Column Start, Column End, Column Increment, Row Start, Row End, Row Increment>	Horizontal mirroring is supported. Mirror right to left changes the following unmirrored values to: - The sign of the column increment is inverted. - Column Start becomes the Column End value - Column End becomes the Column Start value TDI Mode Left to Right Readout (smm 0) C1/ 2, T1 (1, 1024, 1, 1, 1, 1) C1/ 2, T2 (1025, 2048, 1, 1, 1, 1) C1/ 2, T3 (2049, 3072, 1, 1, 1, 1) C1/ 2, T4 (3073, 4096, 1, 1, 1, 1) C3/ 4, T1 (1, 2048, 1, 1, 1, 1) C3/ 4, T2 (2049, 4096, 1, 1, 1, 1) TDI Mode Right to Left Readout (smm 1) C1/ 2, T1 (4096, 3073, -1, 1, 1, 1) C1/ 2, T2 (3072, 2049, -1, 1, 1, 1) C1/ 2, T3 (2048, 1025, -1, 1, 1, 1) C1/ 2, T4 (1024, 1, -1, 1, 1, 1) Area Mode Left to Right Readout (smm 0) C7/ 8, T1 (1, 1024, 1, 1, 96, 1) C7/ 8, T2 (1025, 2048, 1, 1, 96, 1) C7/ 8, T3 (2049, 3072, 1, 1, 96, 1) C7/ 8, T4 (3073, 4096, 1, 1, 96, 1)

Item (when programmable configuration the options are separated with a)	HS-40-04k40
	Area Mode Right to Left Readout (smm 1) C7/ 8, T1 (4096, 3073, -1, 1, 96, -1) C7/ 8, T2 (3072, 2049, -1, 1, 96, -1) C7/ 8, T3 (2048, 1025, -1, 1, 96, -1) C7/ 8, T4 (1024, 1, -1, 1, 96, -1) C9/ 10, T1 (2049, 4096, 1, 1, 96, 1) C9/ 10, T2 (1, 2048, 1, 1, 96, 1)
Camera Color <Hybrid, Mono, Pattern, Solid>	Mono
RGB Pattern Size < (T1, Columns*Rows) (T2, Columns*Rows) (T3, Columns*Rows....>	(T0, 1*1) where 0 is reserved for the default case and individual taps don't need to be articulated
Color Definition (Column, Row, Color) Where color is R,G,B	T0 = (1, 1, M) where 0 is reserved for the default case and individual taps don't need to be defined
Row Color Offset <0,1,2,3...>	0
Column Color Offset <0,1,2,3...>	0
Row Binning Factor <1,2,3 or 1 2 3>	1...8
Column Binning Factor <1,2,3 or 1 2 3>	1 2 4 8
Pretrigger Pixels <0,1,2...or 0..15>	0
Pretrigger Lines <0,1,2.. or 0..15>	0
Frame Time Minimum <xx µs>	TDI Mode: TBD Area Mode: TBD
Frame Time Maximum <xx µs>	TDI Mode: TBD Area Mode: TBD
Internal Line/ Frame Time Resolution <xx ns> 0 if not applicable	25
Pixel Reset Pulse Minimum Width <xx ns> 0 if not applicable	TDI Mode: 0 Area Mode: TBD
Internal Pixel Reset Time Resolution <xx ns> 0 if not applicable	25
Pixel Reset to Exsync Hold time <xx ns>	TBD
BAUD Rate <9600....>	9600, 19200, 57600, 115200
CC1 <Exsync>	EXSYNC
CC2 <Spare>	PRIN (Area Mode only)
CC3 <Forward, Spare>	Forward/ ReverseB
CC4 <Spare>	Spare
DVAL out <Strobe Valid, Alternate>	Strobe Valid
Spare out <Spare>	Spare
TDI Stages	TDI Mode: 16 32 48 64 80 96 Area Mode: 16 32 48 64 80 96

Item (when programmable configuration the options are separated with a)	HS-80-04k40
Imager Dimension <1,2 or 1 2>	1 2
Imager Columns<number of active columns, X>	4096
Imager Rows<number of active rows, Y> Line Scan/ TDI are defined as 1	16 32 48 64 80 96
Number of CCD Taps <1,2,3.....>	8
Sensor Tap Clock Rate <xx MHz>	40
Camera Standard <NTSC, PAL, VS, VW, MW>	VS
Variable Window <Column Start, Column End, Row Start, Row End>	(0,0,0,0) All zeros indicates an unsupported feature
Multiple Window Number of Windows (Column Start 1, Column End 1, Row Start 1, Row End 1) (Column Start 2, Column End 2,...)	0, (0,0,0,0) All zeros indicates an unsupported feature
Number of Camera Configurations<1,2,3,...>	6
Configuration Definition Cx= HDW, Number of Output Taps, Bit Width, Number of Processing Nodes where Cx is the configuration ID x is <1,2,3...> HDW is <Base, Medium, Full> Number of Output Taps is <1,2,3...> Bit width is <8, 10, 12...> Number Processing Nodes is <1 or 2>	TDI Mode C1 = Full, 8, 8, 1 C2 = Medium, 8, 8, 1 C3 = Medium, 8, 12, 1 Area Mode C4 = Full, 8, 8, 1 C5 = Medium, 8, 8, 1 C6 = Medium, 8, 12, 1
Tap Reconstruction In some configurations the reconstruction may change. C0 is the default output format and must be listed. Output configurations that don't conform are listed separately. <Cx,Tn (Column Start, Column End, Column Increment, Row Start, Row End, Row Increment>	Horizontal mirroring is supported. Mirror "on" changes the following unmirrored values to: - The sign of the column increment is inverted. - Column Start becomes the Column End value - Column End becomes the Column Start value TDI Mode Forward Direction left to right readout C1, T1 (1, 512, 1, 1, 1, 1) C1, T2 (513, 1024, 1, 1, 1, 1) C1, T3 (1023, 1536, 1, 1, 1, 1) C1, T4 (1537, 2048, 1, 1, 1, 1) C1, T5 (2049, 2560, 1, 1, 1, 1) C1, T6 (2561, 3072, 1, 1, 1, 1) C1, T7 (3073, 3584, 1, 1, 1, 1) C1, T8 (3585, 4096, 1, 1, 1, 1) C2/ 3, T1 (1, 1024, 1, 1, 1, 1) C2/ 3, T2 (1025, 2048, 1, 1, 1, 1) C2/ 3, T3 (2049, 3072, 1, 1, 1, 1) C2/ 3, T4 (3073, 4096, 1, 1, 1, 1) TDI Mode Reverse Direction right to left readout C1, T1 (4096, 3585, -1, 1, 1, 1) C1, T2 (3584, 3073, -1, 1, 1, 1) C1, T3 (3072, 2561, -1, 1, 1, 1) C1, T4 (2560, 2049, -1, 1, 1, 1) C1, T5 (2048, 1537, -1, 1, 1, 1) C1, T6 (1536, 1023, -1, 1, 1, 1) C1, T7 (1024, 513, -1, 1, 1, 1)

Item (when programmable configuration the options are separated with a)	HS-80-04k40
	C1, T8 (512, 1, -1, 1, 1, 1) C2/ 3, T1 (4096, 3073, 1, 1, 1, 1) C2/ 3, T2 (3072, 2049, 1, 1, 1, 1) C2/ 3, T3 (2048, 1025, 1, 1, 1, 1) C2/ 3, T4 (1024, 1, 1, 1, 1, 1) Area Mode Forward Direction left to right readout C4, T1 (1, 512, 1, 1, 96, 1) C4, T2 (513, 1024, 1, 1, 96, 1) C4, T3 (1023, 1536, 1, 1, 96, 1) C4, T4 (1537, 2048, 1, 1, 96, 1) C4, T5 (2049, 2560, 1, 1, 96, 1) C4, T6 (2561, 3072, 1, 1, 96, 1) C4, T7 (3073, 3584, 1, 1, 96, 1) C4, T8 (3585, 4096, 1, 1, 96, 1) C5/ 6, T1 (1, 1024, 1, 1, 96, 1) C5/ 6, T2 (1025, 2048, 1, 1, 96, 1) C5/ 6, T3 (2049, 3072, 1, 1, 96, 1) C5/ 6, T4 (3073, 4096, 1, 1, 96, 1) Area Mode Reverse Direction right to left readout C4, T1 (4096, 3585, -1, 96, 1, -1) C4, T2 (3584, 3073, -1, 96, 1, -1) C4, T3 (3072, 2561, -1, 96, 1, -1) C4, T4 (2560, 2049, -1, 96, 1, -1) C4, T5 (2048, 1537, -1, 96, 1, -1) C4, T6 (1536, 1023, -1, 96, 1, -1) C4, T7 (1024, 513, -1, 96, 1, -1) C4, T8 (512, 1, -1, 96, 1, -1) C5/ 6, T1 (4096, 3073, 1, 1, 96, 1) C5/ 6, T2 (3072, 2049, 1, 1, 96, 1) C5/ 6, T3 (2048, 1025, 1, 1, 96, 1) C5/ 6, T4 (1024, 1, 1, 1, 96, 1)
Camera Color <Hybrid, Mono, Pattern, Solid>	Mono
RGB Pattern Size < (T1, Columns*Rows) (T2, Columns*Rows) (T3, Columns*Rows)...>	(T0, 1*1) where 0 is reserved for the default case and individual taps don't need to be articulated
Color Definition (Column, Row, Color) Where color is R,G,B	T0 = (1, 1, M) where 0 is reserved for the default case and individual taps don't need to be defined
Row Color Offset <0,1,2,3...>	0
Column Color Offset <0,1,2,3...>	0
Row Binning Factor <1,2,3 or 1 2 3>	1...8
Column Binning Factor <1,2,3 or 1 2 3>	1 2 4 8
Pretrigger Pixels <0,1,2...or 0..15>	0

Item (when programmable configuration the options are separated with a)	HS-80-04k40
Pretrigger Lines <0,1,2.. or 0..15>	0
Frame Time Minimum <xx μs>	TDI Mode: TBD Area Mode: TBD
Frame Time Maximum <xx μs>	TDI Mode: TBD Area Mode: TBD
Internal Line/ Frame Time Resolution <xx ns> 0 if not applicable	25
Pixel Reset Pulse Minimum Width <xx ns> 0 if not applicable	TDI Mode: NA Area Mode: TBD
Internal Pixel Reset Time Resolution <xx ns> 0 if not applicable	25
Pixel Reset to Exsync Hold time <xx ns>	TBD
BAUD Rate <9600....>	9600, 19200, 57600, 115200
CC1 <Exsync>	EXSYNC
CC2 <Spare>	Spare
CC3 <Forward, Spare>	Forward
CC4 <Spare>	Spare
DVAL out <Strobe Valid, Alternate>	Strobe Valid
Spare out <Spare>	Spare
TDI Stages	TDI Mode: 16 32 48 64 80 96 Area Mode: 16 32 48 64 80 96

Item (when programmable configuration the options are separated with a)	HS-80-08k40/80
Imager Dimension <1,2 or 1 2>	1 2
Imager Columns<number of active columns, X>	8192
Imager Rows<number of active rows, Y> Line Scan/ TDI are defined as 1	16 32 48 64 80 96
Number of CCD Taps <1,2,3.....>	16
Sensor Tap Clock Rate <xx MHz>	40 20
Camera Standard <NTSC, PAL, VS, VW, MW>	VS
Variable Window <Column Start, Column End, Row Start, Row End>	(0,0,0,0) All zeros indicates an unsupported feature
Multiple Window Number of Windows (Column Start 1, Column End 1, Row Start 1, Row End 1) (Column Start 2, Column End 2,...)	0, (0,0,0,0) All zeros indicates an unsupported feature
Number of Camera Configurations<1,2,3,...>	8

Item (when programmable configuration the options are separated with a)	HS-80-08k40/80
Configuration Definition Cx= HDW, Number of Output Taps, Bit Width, Number of Processing Nodes where Cx is the configuration ID x is <1,2,3...> HDW is <Base, Medium, Full> Number of Output Taps is <1,2,3...> Bit width is <8, 10, 12...> Number Processing Nodes is <1 or 2>	TDI Mode C1 = Full, 8, 8, 1 C2 = Full, 8, 12, 1 C3 = Medium, 8, 8, 1 C4 = Medium, 8, 12, 1 Area Mode C5 = Full, 8, 8, 1 C6 = Full, 8, 12, 1 C7 = Medium, 8, 8, 1 C8 = Medium, 8, 12, 1
Tap Reconstruction In some configurations the reconstruction may change. C0 is the default output format and must be listed. Output configurations that don't conform are listed separately. <Cx,Tn (Column Start, Column End, Column Increment, Row Start, Row End, Row Increment)>	Horizontal mirroring is supported. Mirror "on" changes the following unmirrored values to: - The sign of the column increment is inverted. - Column Start becomes the Column End value - Column End becomes the Column Start value TDI Mode Forward Direction left to right readout C1/ 2, T1 (1, 1024, 1, 1, 1, 1) C1/ 2, T2 (1025, 2048, 1, 1, 1, 1) C1/ 2, T3 (2049, 3072, 1, 1, 1, 1) C1/ 2, T4 (3073, 4096, 1, 1, 1, 1) C1/ 2, T5 (4097, 5120, 1, 1, 1, 1) C1/ 2, T6 (5121, 6144, 1, 1, 1, 1) C1/ 2, T7 (6145, 7168, 1, 1, 1, 1) C1/ 2, T8 (7169, 8192, 1, 1, 1, 1) C3/ 4, T1 (1, 2048, 1, 1, 1, 1) C3/ 4, T2 (2049, 4096, 1, 1, 1, 1) C3/ 4, T3 (4097, 6144, 1, 1, 1, 1) C3/ 4, T4 (6145, 8192, 1, 1, 1, 1) TDI Mode Reverse Direction right to left readout C1/ 2, T1 (8192, 7169, -1, 1, 1, 1) C1/ 2, T2 (7168, 6145, -1, 1, 1, 1) C1/ 2, T3 (6144, 5121, -1, 1, 1, 1) C1/ 2, T4 (5120, 4097, -1, 1, 1, 1) C1/ 2, T5 (4096, 3073, -1, 1, 1, 1) C1/ 2, T6 (3072, 2049, -1, 1, 1, 1) C1/ 2, T7 (2048, 1025, -1, 1, 1, 1) C1/ 2, T8 (1024, 1, -1, 1, 1, 1) C3/ 4, T1 (8192, 6144, 1, 1, 1, 1) C3/ 4, T2 (6144, 4097, 1, 1, 1, 1) C3/ 4, T3 (4096, 2049, 1, 1, 1, 1) C3/ 4, T4 (2048, 1, 1, 1, 1, 1) Area Mode Forward Direction left to right readout C5/ 6, T1 (1, 1024, 1, 1, 96, 1) C5/ 6, T2 (1025, 2048, 1, 1, 96, 1) C5/ 6, T3 (2049, 3072, 1, 1, 96, 1)

Item (when programmable configuration the options are separated with a)	HS-80-08k40/80
	C5/ 6, T4 (3073, 4096, 1, 1, 96, 1) C5/ 6, T5 (4097, 5120, 1, 1, 96, 1) C5/ 6, T6 (5121, 6144, 1, 1, 96, 1) C5/ 6, T7 (6145, 7168, 1, 1, 96, 1) C5/ 6, T8 (7169, 8192, 1, 1, 96, 1) C3/ 4, T1 (1, 2048, 1, 1, 96, 1) C3/ 4, T2 (2049, 4096, 1, 1, 96, 1) C3/ 4, T3 (4097, 6144, 1, 1, 96, 1) C3/ 4, T4 (6144, 8192, 1, 1, 96, 1) Area Mode Reverse Direction right to left readout C1/ 2, T1 (8192, 7169, -1, 96, 1, -1) C1/ 2, T2 (7168, 6145, -1, 96, 1, -1) C1/ 2, T3 (6144, 5121, -1, 96, 1, -1) C1/ 2, T4 (5120, 4097, -1, 96, 1, -1) C1/ 2, T5 (4096, 3073, -1, 96, 1, -1) C1/ 2, T6 (3072, 2049, -1, 96, 1, -1) C1/ 2, T7 (2048, 1025, -1, 96, 1, -1) C1/ 2, T8 (1024, 1, -1, 96, 1, -1) C3/ 4, T1 (8192, 6144, 1, 1, 96, 1) C3/ 4, T2 (6144, 4097, 1, 1, 96, 1) C3/ 4, T3 (4096, 2049, 1, 1, 96, 1) C3/ 4, T4 (2048, 1, 1, 1, 96, 1)
Camera Color <Hybrid, Mono, Pattern, Solid>	Mono
RGB Pattern Size < (T1, Columns*Rows) (T2, Columns*Rows) (T3, Columns*Rows....>	(T0, 1*1) where 0 is reserved for the default case and individual taps don't need to be articulated
Color Definition (Column, Row, Color) Where color is R,G,B	T0 = (1, 1, M) where 0 is reserved for the default case and individual taps don't need to be defined
Row Color Offset <0,1,2,3...>	0
Column Color Offset <0,1,2,3...>	0
Row Binning Factor <1,2,3 or 1 2 3>	1...8
Column Binning Factor <1,2,3 or 1 2 3>	1 2 4 8
Pretrigger Pixels <0,1,2...or 0..15>	0
Pretrigger Lines <0,1,2.. or 0..15>	0
Frame Time Minimum <xx µs>	TDI Mode: TBD Area Mode: TBD
Frame Time Maximum <xx µs>	TDI Mode: TBD Area Mode: TBD
Internal Line/ Frame Time Resolution <xx ns> 0 if not applicable	25
Pixel Reset Pulse Minimum Width <xx ns> 0 if not applicable	TDI Mode: NA Area Mode: TBD

Item (when programmable configuration the options are separated with a)	HS-80-08k40/80
Internal Pixel Reset Time Resolution <xx ns> 0 if not applicable	25
Pixel Reset to Exsync Hold time <xx ns>	TBD
BAUD Rate <9600....>	9600, 19200, 57600, 115200
CC1 <Exsync>	EXSYNC
CC2 <Spare>	PRIN (Area Mode only)
CC3 <Forward, Spare>	Forward
CC4 <Spare>	Spare
DVAL out <Strobe Valid, Alternate>	Strobe Valid
Spare out <Spare>	Spare
TDI Stages	TDI Mode: 16 32 48 64 80 96 Area Mode: 16 32 48 64 80 96

Appendix C

EMC Declaration of Conformity

We, **Teledyne DALSA**
605 McMurray Rd.,
Waterloo, ON
CANADA N2V 2E9

declare under sole responsibility, that the product(s):

HS-40-04k40

HS-80-04k40

HS-80-08k40

HS-80-08k80

HS-82-04k80

fulfill(s) the requirements of the standard(s)

EMC:

CISPR-22:1997

EN 50082-1:1997

EN 61000-4-2; +/- 6kV CD, +/-8kV AD:1995

EN 61000-4-3; 3V/m:1996

EN 61000-4-4; 500V, 1100V:1995

This product complies with the requirements of the Low Voltage Directive 73/ 23/ EEC and the EMC Directive 89/ 336/ EEC and carries the CE mark accordingly.

The 8k cameras were equipped with the following lens adapter during testing:
AC-LA-0109 (Adapter, M72 to Nikon lens).

Place of Issue **Waterloo, ON, CANADA**

Date of Issue **April 2005**

Name and Signature of authorized person **Hank Helmond**
Quality Manager, Teledyne DALSA Corp.



This Declaration corresponds to EN 45 014.

Appendix D

Troubleshooting

Common Solutions

The information in this chapter can help you solve problems that may occur during the setup of your camera. Remember that the camera is part of the entire acquisition system. You may have to troubleshoot any or all of the following:

power supplies	cabling
frame grabber hardware & software	host computer
light sources	optics
operating environment	encoder

Your steps in dealing with a technical problem should be:

1. Try the general and specific solutions listed in sections 6.1, 6.2 and 6.3.

If these solutions do not resolve your problem, see section 6.4 on getting product support.

LED

When the camera is first powered up, the LED will glow on the back of the camera. Refer to section 2.2.1 LED Status Indicator for information on the LED.

Connections

The first step in troubleshooting is to verify that your camera has all the correct connections.

Power Supply Voltages

Check for the presence of all voltages at the camera power connector. Verify that all grounds are connected. Issue the command, **vv**, to confirm correct voltages.

EXSYNC

When the camera is received from the factory, it defaults (no external input required) to exposure mode 7 (TBDkHz frame rate, internal Sync to trigger readout, and TDI Mode). After a user has saved settings, the camera powers up with the saved settings.

Data Clocking/Output Signals

To validate cable integrity, have the camera send out a test pattern and verify it is being properly received. Refer to section 4.7.1 Generating a Test Pattern for further information.

Troubleshooting Using the Serial Interface

The following commands can aid in debugging. (The complete command protocol is described in Appendix B and C.)

Communications

To quickly verify serial communications send the help command. The **h** command returns the online help menu. If further problems persist, review Appendix C for more information on communications.

Verify Parameters

To verify the camera parameters, send the **gcp** command. See section 4.7.6 Returning Camera Settings for a complete explanation of the camera parameters screen.

Verify Factory Calibrated Settings

To restore the camera's factory settings and disable the FPN and PRNU coefficients, send the **rfs** command.

After executing this command send the **gcp** command to verify the factory settings.

Verify Timing and Digital Video Path

Use the test pattern feature to verify the proper timing and connections between the camera and the frame grabber and verify the proper output along the digital processing chain. See below.

Generating Test Patterns

The camera can generate a test pattern to aid in system debugging. Use the command **svm 1** to activate the test pattern. The test pattern is a ramp from 0 to 255DN, then starts at 0 again. Use the test pattern to verify the proper timing and connections between the camera and the frame grabber.

- **No test pattern or bad test pattern**— May indicate a problem with the camera (e.g. missing bit) or a system setup problem (e.g. frame grabber or timing). Verify the presence of the LVAL and STROBE signals.
- **Test pattern successful**— Run the **svm 0** command to activate video. Then run the **gl** command under both dark and light conditions to retrieve a line of raw video (no digital processing). Under dark conditions, with factory settings, the analog offset value should be within the specified range (refer to the user specifications). Adjust

the analog offset using the **cao** or **sao** commands. Under light conditions, you should receive a value.

Verify Voltage

Use the **vv** command to display the camera's input voltage.

Verify Temperature

To check the internal temperature of the camera, use the **vt** command. For proper operation, this value should not exceed 75°C.

Note: If the camera reaches 75°C, the camera **will shutdown and the LED will flash red**. If this occurs, the camera **must be rebooted** using the command, **rc** or can be powered down manually. You will have to correct the temperature problem or the camera will shutdown again. If you enter any command other than **vt** or **rc**, the camera responds with:

Error 09: The camera's temperature exceeds the specified operating range>

Verify Pixel Coefficients

Use the **dpc** command to display the pixel coefficients in the order FPN, PRNU, FPN, PRNU... The camera also returns the pixel number for each fifth pixel.

End-of-line Sequence

To further aid debugging, the camera can generate an end-of-line sequence. The end-of-line-sequence outputs "aa", "55", "line counter", "line average", "pixels above threshold", "pixels below threshold". To activate the end-of-line sequence, use the command **els 1**. To disable the end-of-line sequence, use the command **els 0**.

Use the **sut** and **s1t** commands to set threshold values between 0 and 255 for 8 bit data modes, or 0 to 4096 for 12 bit data modes.

Specific Solutions

No Output or Erratic Behavior

If your camera provides no output or behaves erratically, it may be picking up random noise from long cables acting as antennae. Do not attach wires to unused pins. Verify that the camera is not receiving spurious inputs (e.g. EXSYNC if camera is in exposure mode that requires external signals). Unused signals in the Camera Link cable should be terminated in 100Ω

Line Dropout, Bright Lines, or Incorrect Frame Rate

Verify that the frequency of the internal sync is set correctly, or when the camera is set to external sync that the EXSYNC signal supplied to the camera does not exceed the camera's useable frame rate under the current operating conditions.

Noisy Output

Check your power supply voltage outputs for noise. Noise present on these lines can result in poor video quality.

Dark Patches

If dark patches appear in your output the optics path may have become contaminated. Clean your lenses and sensor windows with extreme care.

1. Take standard ESD precautions.
2. Wear latex gloves or finger cots
3. Blow off dust using a filtered blow bottle or dry, filtered compressed air.
4. Fold a piece of optical lens cleaning tissue (approx. 3" x 5") to make a square pad that is approximately one finger-width
5. Moisten the pad on one edge with 2-3 drops of clean solvent—either alcohol or acetone. Do not saturate the entire pad with solvent.
6. Wipe across the length of the window in one direction with the moistened end first, followed by the rest of the pad. The dry part of the pad should follow the moistened end. The goal is to prevent solvent from evaporating from the window surface, as this will end up leaving residue and streaking behind.
7. Repeat steps 2-4 using a clean tissue until the entire window has been cleaned.
8. Blow off any adhering fibers or particles using dry, filtered compressed air.

Stuck Bits

If data bits seem to be stuck or do not change, check that the camera is not saturated by preventing light from entering. To verify the data path integrity, check the levels of the 2 test pattern pixels (first 2 pixels following the last End-of-line pixel. You may need to turn the End-of-line sequence "on" by sending the command **els 1**). Any deviation from a consistent value of these pixels (170DN/ 85DN) could be an indication of the following:

- shorted bits
- stuck bits
- digital noise pickup on the I/ O cable
- open connection.

To activate the test pattern, use the command **svm 1**. Use the test pattern to verify the proper timing and connections between the camera and frame grabber.

To receive a complete line of raw video data (without digital processing or test pattern) through the RS232 port, use the command **g1**. The returned data displays one pixel value after another and the minimum, maximum, and mean value of the sampled line. Use this command to ensure the proper video input range.

Probe the output lines with an oscilloscope. Disconnect the digital cable from the camera and check the digital signals at the output of the camera. Ensure that the correct values are present. Check all cable connections, especially right at the connector; poor connections or broken wires will cause randomly changing bits or stuck bits.

Horizontal Lines or Patterns in Image

A faulty or irregular encoder signal may result in horizontal lines due to exposure time fluctuations; ensure that your exposure time is regular. If you have verified that your exposure time is consistent and patterns of low frequency intensity variations still occur, ensure that you are using a DC or high frequency light source.

Appendix E

CCD Handling Instructions

Electrostatic Discharge and the CCD Sensor

Cameras contain charge-coupled device (CCD) image sensors, which are metal oxide semiconductor (MOS) devices and are susceptible to damage from electrostatic discharge (ESD).

Electrostatic charge introduced to the sensor window surface can induce charge buildup on the underside of the window that cannot be readily dissipated by the dry nitrogen gas in the sensor package cavity. When charge buildup occurs, surface-gated photodiodes (SGPDs) may exhibit higher image lag. Some SGPD sensors, such as the IL-P4 and the IT-P4 used in the Piranha2 cameras, may also exhibit a highly non-uniform response when affected by charge buildup, with some pixels displaying a much higher response when the sensor is exposed to uniform illumination. The charge normally dissipates within 24 hours and the sensor returns to normal operation.



WARNING: Charge buildup will affect the camera's flat-field correction calibration. To avoid an erroneous calibration, ensure that you perform flat-field correction only after a charge buildup has dissipated over 24 hours.

Protecting Against Dust, Oil and Scratches

The CCD window is part of the optical path and should be handled like other optical components, with extreme care.

Dust can obscure pixels, producing dark patches on the sensor response. Dust is most visible when the illumination is collimated. The dark patches shift position as the angle of illumination changes. Dust is normally not visible when the sensor is positioned at the exit port of an integrating sphere, where the illumination is diffuse.

Dust can normally be removed by blowing the window surface using a compressed air blower, unless the dust particles are being held by an electrostatic charge, in which case either an ionized air blower or wet cleaning is necessary.

Oil is usually introduced during handling. Touching the surface of the window barehanded will leave oily residues. Using rubber fingercoats and rubber gloves can prevent oil contamination. However, the friction between the rubber and the window may produce electrostatic charge that may damage the sensor. To avoid ESD damage and

to avoid introducing oily residues, only hold the sensor from the edges of the ceramic package and avoid touching the sensor pins and the window.

Scratches can be caused by improper handling, cleaning or storage of the sensor. Vacuum picking tools should not come in contact with the window surface. CCDs should not be stored in containers where they are not properly secured and can slide against the container.

Scratches diffract incident illumination. When exposed to uniform illumination, a sensor with a scratched window will normally have brighter pixels adjacent to darker pixels. The location of these pixels changes with the angle of illumination.

Cleaning the Sensor Window

1. Use compressed air to blow off loose particles. This step alone is usually sufficient to clean the sensor window.
2. If further cleaning is required, use a lens wiper moistened with alcohol or acetone.
3. We recommend using lint-free ESD-safe cloth wipers that do not contain particles that can scratch the window. The Anticon Gold 9"x 9" wiper made by Milliken is both ESD safe and suitable for class 100 environments. Another ESD acceptable wiper is the TX4025 from Texwipe.
4. An alternative to ESD-safe cloth wipers is Transplex swabs that have desirable ESD properties. There are several varieties available from Texwipe. Do not use regular cotton swabs, since these can introduce charge to the window surface.
5. Wipe the window carefully and slowly.
6. When cleaning long linear sensors, it may be easier to wipe along the width (i.e. as opposed to the length) of the sensor.

Appendix F

Revision History

Revision Number	Change Description
00	RoHS compliant HS camera preliminary release
01	-Help screen examples (gh) revised. -EMC compliance section updated to reflect RoHS designation.
02	-HS-82-04k80 model of camera added to manual. HS-82 specifications added. HS-82 specific additions made to the stage select (stg), set binning vertical (sbv), and set sync frequency (ssf) command descriptions. -“Preliminary” marking removed. -HS-80-04k40 model information added.
03	-Line Rate for HS-80-04K40 model – 68kHz -Updated DALSA logo to Teledyne DALSA logo -8k mechanicals updated showing sensor alignment values measured from the tooling holes.
04	-RoHS compliance updated for HS-40-04k40 model. HS-40-04k40 is not RoHS compliant.

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